Ruukkilabra
UPPER SECONDARY PHYSICS

FY4

Force and motion

KIVIRUUKIN LUKIO

Contents

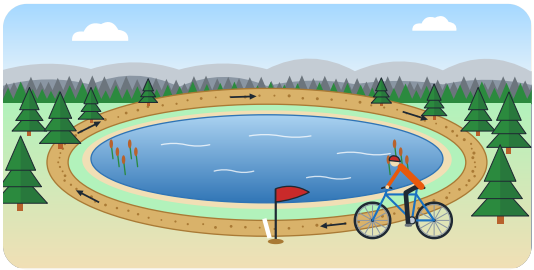
1	Position, velocity and uniform motion as graphs	4
2	Acceleration and uniformly accelerated motion	12
3	Comparing the motions of two bodies	21
4	Interaction and force: Newton's first and third laws	31
5	Newton's second law	40
6	Free-body diagrams and the combined effect of forces	48
7	Equation of motion for connected bodies	57
8	Weight, friction and the inclined plane	66
9	Air resistance and falling through air	77
10	Kinetic energy, potential energy and mechanical energy	87
11	Conservation of mechanical energy and the work–energy principle	96
12	Hydro and wind power as energy converters	106
13	Momentum and impulse	116
14	Conservation of momentum and one-dimensional collisions	126
15	Road safety: braking, seat belts and airbags	137
16	Applied lesson: lift, parachute and sledding hill	148

FY4 · CHAPTER 1

Position, velocity and uniform motion as graphs

Why can the distance a cyclist has travelled be four kilometres, even though her displacement is zero?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy4-1.html`



IN THIS LESSON, YOU WILL LEARN TO

- describe motion as position as a function of time.
- calculate displacement and velocity with their signs.
- read the velocity of uniform motion from the slope of a graph of position against time.

PRIOR KNOWLEDGE

- Velocity from lower secondary school: distance divided by time, $v = s/t$

- Reading a coordinate system: axes, units and the steepness of a line

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Think about it

A cyclist leaves home and rides steadily $2.0\ \mathrm{km}$ along a straight road to the shop. She buys milk and rides straight back home. At the end the odometer shows $4.0\ \mathrm{km}$, but a tracking app that measures the position from home shows $0\ \mathrm{m}$. The app also draws a graph with time on the horizontal axis and the position of the cyclist on the vertical axis.

> **What would such a graph look like, and what could be read from it about the cyclist's velocity and direction of travel?**

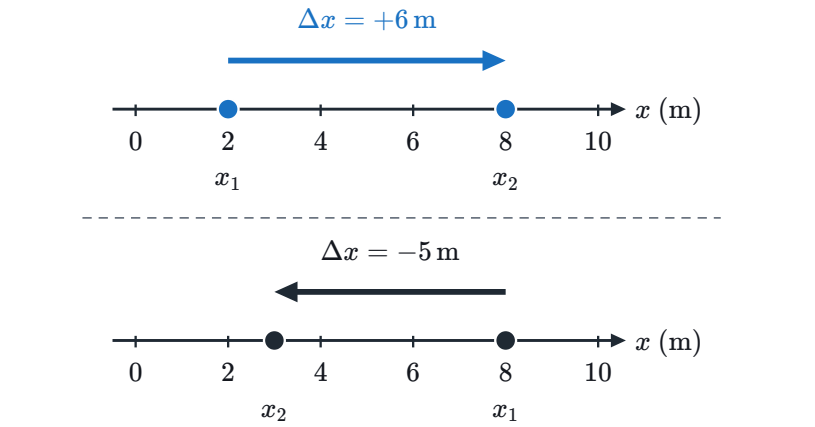
Theory

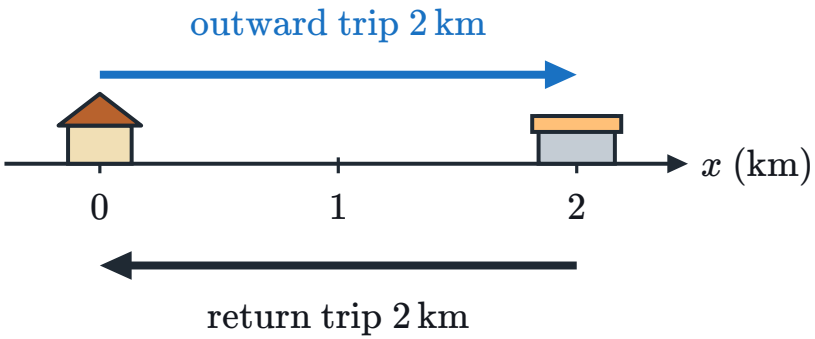
Key idea

Describing motion begins with a convention. We choose the straight line along which the object moves, and an origin and a positive direction on it: this is the x-axis. The x tells where the object is relative to the origin. The sign tells which side: the position $x = -3\ \mathrm{m}$ is three metres from the origin in the negative direction. Time t is measured with a clock, and the motion is described by giving the position at every moment.

When an object moves from the position x_1 to the position x_2, its is the change in position $\Delta x = x_2 - x_1$. Displacement is a vector quantity: it has a magnitude and a direction, and on a line the direction is given by the sign. Plus means the positive direction and minus the opposite one. Displacement is not the same thing as the s. The distance is the length of the route, and it cannot be negative. If a cyclist rides

to the shop and back, the distance is four kilometres, but the displacement is zero, because the final position is the same as the initial position.





$$s = 2\,\mathrm{km} + 2\,\mathrm{km} = 4\,\mathrm{km}$$

$$\Delta x = x_2 - x_1 = 0$$

tells how fast and in which direction the position changes. It is also a vector quantity, and its sign tells the direction. The absolute value of the velocity $|v|$ is the speed. In uniform motion the velocity stays constant: the position changes by the same amount in every time interval of equal length.



In a night photo with a long exposure time, vehicle lights draw streaks. The red streaks are rear lights, the white ones headlights.

Photo: Viorel Horia Surdu, Pexels License

Motion can be shown as a (t, x) **graph**, with time on the horizontal axis and position on the vertical axis. The graph of uniform motion is a straight line. The slope of the line is the velocity. The greater the absolute value of the slope, the greater the speed of the object, and the sign of the slope tells the direction of the motion: a rising line means motion in the positive direction, a falling line motion in the negative direction and a horizontal line being at rest. The graph is not a picture of the path of the object: a rising line does not mean uphill, but that the position increases.

Average velocity and the equation of uniform motion

When the position changes by Δx during the time interval $\Delta t = t_2 - t_1$, the average velocity is the displacement divided by the time:

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

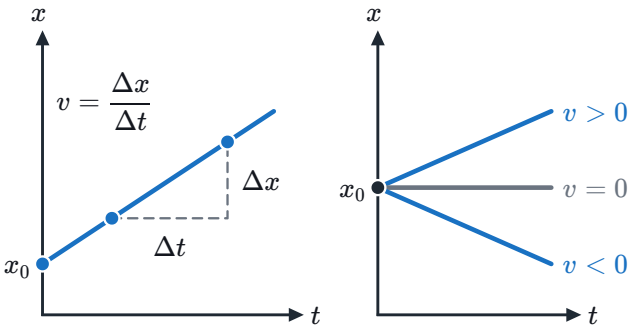
where v_{avg} is the average velocity (m/s), Δx the displacement (m), Δt the elapsed time (s) and x_1 and x_2 the positions (m) at the moments t_1 and t_2. As vectors, the velocity is written $\overline{v}$ and the displacement $\Delta\overline{x}$. On a line we use their components v and Δx, whose sign tells the direction.

In uniform motion the average velocity is the same over any time interval, and it is called the velocity v. In non-uniform motion the average velocity only tells the constant velocity at which the same displacement would be covered in the same

time. If the object is at the position x_0 at the moment $t = 0$, solving the formula for the position gives the position as a function of time:

$$x = x_0 + v\,t$$

where x is the position at the moment t (m), x_0 the initial position (m), v the velocity (m/s) and t the time (s). This is the equation of the straight line in the (t, x) graph: x_0 is where the line crosses the position axis and v is its slope.



The slope is calculated by choosing two points far apart on the line and dividing the change in position by the change in time. In a graph drawn from measured points, a line is fitted through the points so that the points lie evenly on both sides of it. The slope is calculated from the line, not from two neighbouring measured points, because every measurement has a small error.

Common misconception

A (t, x) graph is not a picture of the path of the object: a rising line does not mean uphill, but that the position increases. Another mistake is to think of a negative velocity as a slow velocity. The sign only tells the direction: the velocity $-3\ \mathrm{m/s}$ is greater in magnitude than $+2\ \mathrm{m/s}$. A third mistake is to draw a conclusion about the velocity from the position: an object can be ahead and still move more slowly.

Summary

- The position x tells where the object is relative to the origin. The displacement $\Delta x = x_2 - x_1$ is the change in position, and its sign tells the direction. The distance is the length of the route and is not negative.
- The velocity $v = \Delta x / \Delta t$ is a vector quantity. In uniform motion it is constant, and the position is $x = x_0 + v\,t$.

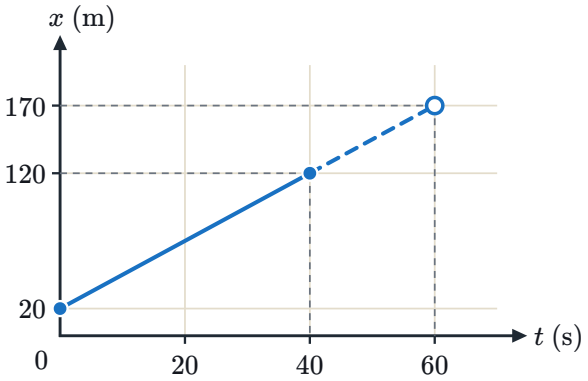
- In a (t, x) graph the line of uniform motion is straight. Its slope is the velocity, and its sign tells the direction.
- A line is fitted to measured data, and the velocity is calculated from points chosen far apart on the line.
- The average velocity of a back-and-forth motion can be zero even though the distance is not zero.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: A jogger's speed from a graph

A jogger runs at a uniform velocity. From the graph we read that at the moment $t = 0$ she is at the position $20\ \mathrm{m}$ and at the moment $t = 40\ \mathrm{s}$ at the position $120\ \mathrm{m}$. What is the jogger's velocity, and where is she at the moment $t = 60\ \mathrm{s}$?



STEP 1/7

Given and find. Two points are read from the graph: $(t_1; x_1) = (0\,\mathrm{s}; 20\,\mathrm{m})$ and $(t_2; x_2) = (40\,\mathrm{s}; 120\,\mathrm{m})$. We are asked for the velocity v and the position x at the moment $t = 60\,\mathrm{s}$.

STEP 2/7

Principle. The jogger's motion is uniform, so the velocity stays constant and the (t, x) graph is a straight line. Then the velocity is the slope of the line, that is the change in position divided by the change in time. Once the velocity is known, the position is obtained by adding the product of the velocity and the time to the initial position.

STEP 3/7

Formula solved. The velocity and the position are solved symbolically before using numbers:

$$v = \frac{x_2 - x_1}{t_2 - t_1}$$
$$x = x_1 + v\,t$$

STEP 4/7

Unit conversions. The positions are in metres and the times in seconds, so the velocity comes directly in the unit $\mathrm{m/s}$. No conversions are needed. The initial position is $x_1 = 20\,\mathrm{m}$ at the moment $t_1 = 0$, so $60\,\mathrm{s}$ can be used directly as the time t.

STEP 5/7

Substitution. Substitute the numerical values with their units:

$$
\begin{aligned}
v &= \frac{x_2 - x_1}{t_2 - t_1} \\
&= \frac{120\,\mathrm{m} - 20\,\mathrm{m}}{40\,\mathrm{s} - 0\,\mathrm{s}} \\
&= 2.5\,\mathrm{m/s}
\end{aligned}
$$

$$
\begin{aligned}
x &= x_1 + v\,t \\
&= 20\,\mathrm{m} + 2.5\,\mathrm{m/s} \cdot 60\,\mathrm{s} \\
&= 20\,\mathrm{m} + 150\,\mathrm{m} \\
&= 170\,\mathrm{m}
\end{aligned}
$$

STEP 6/7

Result. The velocity is $v = 2.5\,\mathrm{m/s}$ and the position at the moment $60\,\mathrm{s}$ is $x = 170\,\mathrm{m}$. The whole numbers read from the graph, $0\,\mathrm{s}$, $40\,\mathrm{s}$, $20\,\mathrm{m}$ and $120\,\mathrm{m}$, are taken as exact values, so the results need no rounding: they are exact values of this model.

STEP 7/7

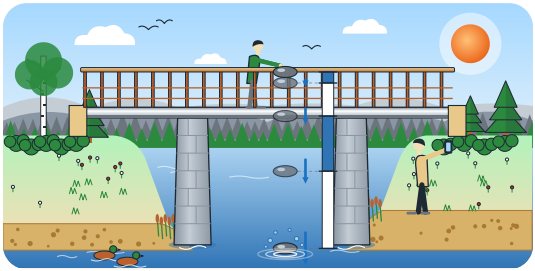
Reasonableness. During the first $40\,\mathrm{s}$ the position grew by $100\,\mathrm{m}$, so during the next $20\,\mathrm{s}$ it grows by half of that, $50\,\mathrm{m}$. The position is $120\,\mathrm{m} + 50\,\mathrm{m} = 170\,\mathrm{m}$, the same result. The velocity $2.5\,\mathrm{m/s}$ is about $9\,\mathrm{km/h}$, which is the speed of jogging. The velocity is positive, because the graph rises.

FY4 · CHAPTER 2

Acceleration and uniformly accelerated motion

Why does a stone dropped from a bridge travel a longer distance during each second than during the second before?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy4-2.html`



IN THIS LESSON, YOU WILL LEARN TO

- explain what acceleration tells you about the change in velocity, and interpret its sign.
- calculate the velocity, position and time of uniformly accelerated motion, and read them from (t, v) and (t, x) graphs.
- determine the acceleration of free fall by fitting a quadratic polynomial to measured points.

PRIOR KNOWLEDGE

- Velocity is the change in position divided by the change in time, and the slope of a (t, x) graph is the velocity
  chapter *Position, velocity and uniform motion as graphs*
- By how much does the velocity of a freely falling object grow in one second when air resistance is small?
  chapter *Acceleration of free fall:* $v = gt$
- The quadratic formula from mathematics

Think about it

You drop a stone from a footbridge into a river, and your friend films the fall with a phone. The video shows that during the first half second the stone falls about $1.2\,\mathrm{m}$. After one whole second it has already fallen almost $5\,\mathrm{m}$. So during the second half second it falls a much longer distance than during the first, even though the time is the same.

> **What changes in the stone's motion so that in every time interval of equal length it travels a longer distance than in the one before?**

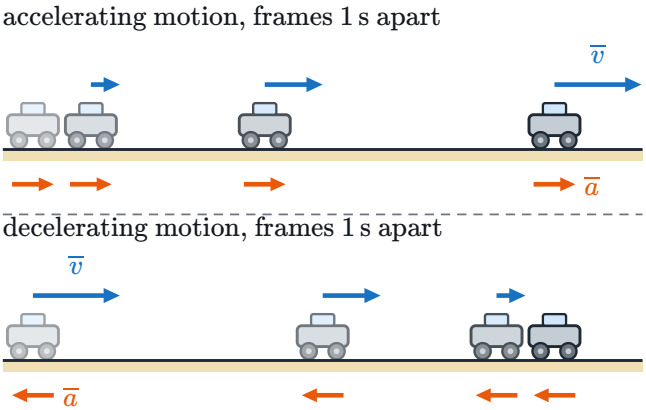
Theory

Key idea

Acceleration tells you how the velocity changes

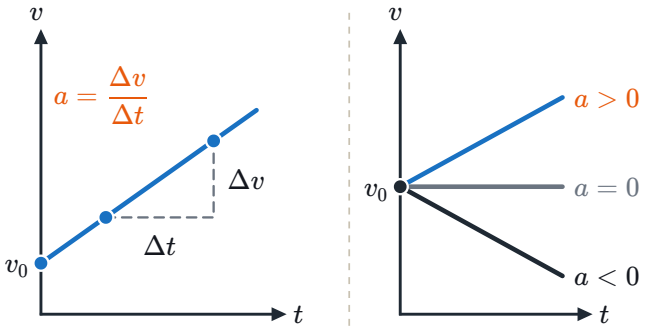
In the chapter *Position, velocity and uniform motion as graphs* the velocity stayed constant. In most motions, however, it changes: a car starts from a traffic light, a cyclist brakes and a dropped stone falls faster and faster. a tells you how much the velocity changes per unit of time. Its unit is $\frac{\mathrm{m/s}}{\mathrm{s}} = \mathrm{m/s^2}$. An acceleration of $2.0\,\mathrm{m/s^2}$ means that the velocity grows by $2.0\,\mathrm{m/s}$ every second.

Acceleration is a vector quantity $\overline{a}$, like velocity $\overline{v}$, and in rectilinear motion its direction is given by its sign. When the acceleration and the velocity point in the same direction, the speed increases. When they point in opposite directions, the speed decreases. So a negative acceleration does not always mean slowing down: if an object moves in the negative direction and $a < 0$, its speed increases. Acceleration can exist even when the velocity is zero for a moment.



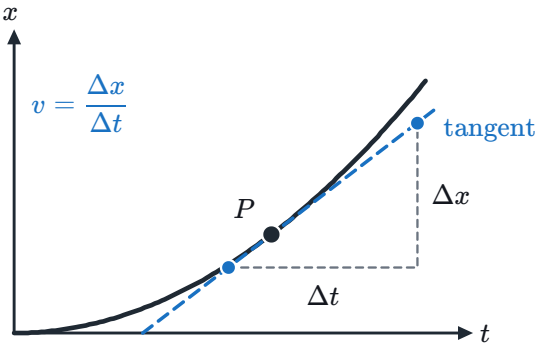
Uniformly accelerated motion and the (t, v) graph

Motion is **uniformly accelerated** when the acceleration stays constant: the velocity changes by the same amount in every time interval of equal length. Then the (t, v) **graph**, with time on the horizontal axis and velocity on the vertical axis, is a straight line. The slope of the line is the acceleration. A rising line means a positive acceleration, a horizontal line uniform motion and a falling line a negative acceleration.



Instantaneous velocity is the slope of the tangent

When the velocity grows, the position changes more in each second than in the one before. That is why the (t, x) graph is not a straight line but a curve that gets steeper. The velocity at a given moment, the , is found from the tangent of the curve: the line that just touches the curve at that point. Two points far apart are chosen on the tangent, and the instantaneous velocity is the change in position between them divided by the change in time. A ratio calculated through two points of the curve would give only the average velocity between them.



Free fall

When air resistance is small, all objects fall with the same acceleration regardless of mass. This is $g = 9.81\ \mathrm{m/s^2}$ at the Earth's surface, directed downwards. The velocity of a dropped stone therefore grows by about $9.8\ \mathrm{m/s}$ every second. That is why the stone travels a longer distance in each second than in the one before:

the distances that an object starting from rest covers in successive intervals of equal length are in the ratio $1 : 3 : 5 : 7$.

Equations of uniformly accelerated motion

Acceleration is the change in velocity divided by the time it took:

$$a = \frac{\Delta v}{\Delta t}$$

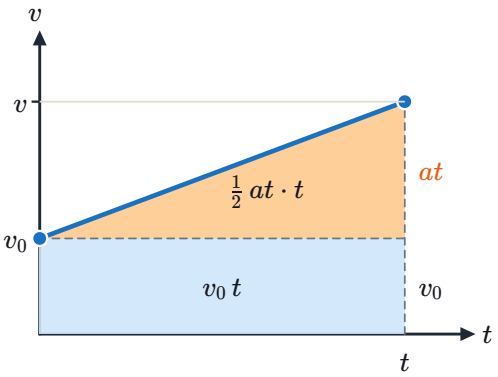
where a is the acceleration ($\mathrm{m/s^2}$), $\Delta v = v_2 - v_1$ the change in velocity ($\mathrm{m/s}$) and Δt the time it took (s). When the acceleration is constant and timing starts at $t = 0$, when the velocity is v_0, the velocity changes by the amount a every second:

$$v = v_0 + a\,t$$

where v is the velocity at time t ($\mathrm{m/s}$), v_0 the initial velocity ($\mathrm{m/s}$), a the acceleration ($\mathrm{m/s^2}$) and t the time (s).

The displacement is the area between the (t, v) graph and the time axis

In uniform motion the displacement $v\,t$ is the area of the rectangle under the (t, v) graph. The same holds when the velocity changes: the displacement is the area between the graph and the time axis, where the part below the time axis ($v < 0$) is counted as negative. In uniformly accelerated motion the region divides into a rectangle whose area is $v_0\,t$ and a triangle whose height is the growth of velocity $a\,t$ and whose width is t. The area of the triangle is $\frac{1}{2} \cdot a\,t \cdot t = \frac{1}{2}\,a\,t^2$.



When the object is at the position x_0 at time $t = 0$, its position is the initial position plus the displacement:

$$x = x_0 + v_0\,t + \tfrac{1}{2}\,a\,t^2$$

where x is the position at time t (m) and x_0 the initial position (m); the other symbols are the same as above. When the time is asked for, this becomes a quadratic equation, and of its solutions you choose the one that fits the situation.

Determining the acceleration of free fall from measurements

The position of a falling object is measured at many moments, for example from a video or with light gates. A quadratic polynomial $s = A\,t^2 + B\,t + C$ is fitted to the measured points with a calculator or a spreadsheet. In real series of measurements the points do not usually lie exactly on one curve because of measurement uncertainty, so the fit uses all the points at once. When the polynomial is compared with the equation of position, the coefficients of t^2 are the same, $A = \tfrac{1}{2}\,a$:

$$a = 2A$$

where A is the coefficient of the t^2 term of the fitted polynomial ($\mathrm{m/s^2}$). The coefficient B is the initial velocity and C the initial position.



A gull taking off, shown as a multiple exposure: the same bird is photographed three times in a row in one picture.

Photo: Bernd Thaller, CC BY 2.0

Common misconception

Many people think that the acceleration is zero whenever the velocity is zero. A ball thrown up is at rest for a moment at the highest point, but its velocity changes all the time: just before the highest point it moves up and just after it moves down. When air resistance is small, the acceleration is $g = 9.81\ \mathrm{m/s^2}$ downwards during the whole flight. Another mistake is to take a negative acceleration to always mean slowing down: the speed decreases only when the acceleration and the velocity point in opposite directions. A third mistake is to think that a heavy object falls faster than a light one. When air resistance is small, both fall with the same acceleration.

Summary

- Acceleration $a = \frac{\Delta v}{\Delta t}$ tells you the change in velocity per unit of time, with the unit $\mathrm{m/s^2}$. It is a vector quantity, and on a straight line its sign tells the direction.
- The speed increases when the acceleration and the velocity point in the same direction, and decreases when they point in opposite directions.
- In uniformly accelerated motion the (t, v) graph is a straight line whose slope is the acceleration and whose signed area between it and the time axis is the displacement; the part below the time axis ($v < 0$) is counted as negative.
- $v = v_0 + a\,t$ and $x = x_0 + v_0\,t + \frac{1}{2}\,a\,t^2$. The (t, x) graph is a curve, and the instantaneous velocity is the slope of its tangent.
- In free fall the acceleration is $g = 9.81\ \mathrm{m/s^2}$ downwards. It can be determined by fitting a quadratic polynomial to the measured points: $a = 2A$.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: A cyclist going downhill

A cyclist reaches the top of a hill at $4.0\ \mathrm{m/s}$ and stops pedalling. On the downhill her acceleration is $0.50\ \mathrm{m/s^2}$, and the hill is $60\ \mathrm{m}$ long. How long does the descent take, and what is her velocity at the bottom?

STEP 1/7

Given and find. The initial velocity is $v_0 = 4.0\ \mathrm{m/s}$, the acceleration $a = 0.50\ \mathrm{m/s^2}$ and the displacement $\Delta x = x - x_0 = 60\ \mathrm{m}$ in the direction of motion, which is chosen as positive. We are asked for the time t and the velocity v at the bottom.

STEP 2/7

Principle. The acceleration is constant, so the motion is uniformly accelerated and the displacement is $\Delta x = v_0\,t + \tfrac{1}{2}\,a\,t^2$. The time is the unknown, and it appears in both the first and the second power, so we get a quadratic equation. Once the time is known, the velocity comes from the equation $v = v_0 + a\,t$.

STEP 3/7

Formula solved. Write the equation in the form $\tfrac{1}{2}\,a\,t^2 + v_0\,t - \Delta x = 0$ and solve it with the quadratic formula. The positive solution is

$$t = \frac{-v_0 + \sqrt{v_0^2 + 2a\,\Delta x}}{a}$$

STEP 4/7

Unit conversions. All quantities are in SI units (m, m/s, m/s²), so no conversions are needed. Under the root the unit is $\mathrm{m^2/s^2}$, so the root is in $\mathrm{m/s}$ and the time comes out in seconds.

STEP 5/7

Substitution. First calculate the expression under the root and its square root, then the time and the velocity:

$$
\begin{aligned}
&v_0^2 + 2a\,\Delta x \\
&= (4.0\,\mathrm{m/s})^2 \\
&\quad + 2 \cdot 0.50\,\mathrm{m/s^2} \cdot 60\,\mathrm{m} \\
&= 76\,\mathrm{m^2/s^2}
\end{aligned}
$$

$$
\sqrt{76\,\mathrm{m^2/s^2}} \approx 8.72\,\mathrm{m/s}
$$

$$
\begin{aligned}
t &= \frac{-4.0\,\mathrm{m/s} + 8.72\,\mathrm{m/s}}{0.50\,\mathrm{m/s^2}} \\
&\approx 9.44\,\mathrm{s}
\end{aligned}
$$

$$
\begin{aligned}
v &= v_0 + a\,t \\
&= 4.0\,\mathrm{m/s} + 0.50\,\mathrm{m/s^2} \cdot 9.44\,\mathrm{s} \\
&\approx 8.72\,\mathrm{m/s}
\end{aligned}
$$

STEP 6/7

Result. The descent takes about $9.4\,\mathrm{s}$, and the velocity at the bottom is about $8.7\,\mathrm{m/s}$. Two significant figures, like the initial velocity $4.0\,\mathrm{m/s}$ and the acceleration $0.50\,\mathrm{m/s^2}$. The distance $60\,\mathrm{m}$ is taken as an exact value. The other solution of the formula, $t \approx -25\,\mathrm{s}$, is rejected because the motion starts at $t = 0$.

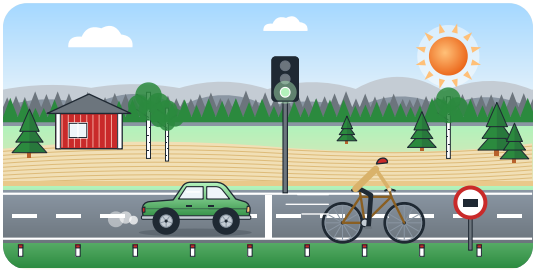
STEP 7/7

Reasonableness. The velocity grows steadily, so the average velocity is the mean of the initial and final velocities, $(4.0\,\mathrm{m/s} + 8.72\,\mathrm{m/s}) : 2 \approx 6.36\,\mathrm{m/s}$, and $6.36\,\mathrm{m/s} \cdot 9.44\,\mathrm{s} \approx 60\,\mathrm{m}$, the length of the hill. The final velocity $8.7\,\mathrm{m/s} \approx 31\,\mathrm{km/h}$ is a usual cycling speed on a downhill.

Comparing the motions of two bodies

Why does a car that starts from rest catch a steadily pedalling cyclist ahead of it only far down the road, even though the car accelerates all the time?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy4-12.html`



IN THIS LESSON, YOU WILL LEARN TO

- calculate the time and place of a meeting of two bodies using position functions.
- read a meeting from the intersection of (t, x) graphs.
- explain how the velocities of the bodies are related to a meeting.

PRIOR KNOWLEDGE

- Position in uniform motion $x = x_0 + v\,t$ and the (t, x) graph
  chapter *Position, velocity and uniform motion as graphs*

- Position in uniformly accelerated motion
  $x = x_0 + v_0 t + \frac{1}{2} a t^2$
  chapter *Acceleration and uniformly accelerated motion*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 5.

Think about it

The traffic light turns green. A car that was standing at the stop line starts moving and accelerates uniformly. At the same moment a cyclist who is $16\ \mathrm{m}$ ahead of the line rides in the same direction at a constant velocity of $6.0\ \mathrm{m/s}$. The car accelerates all the time, but the cyclist is still ahead for a long time.

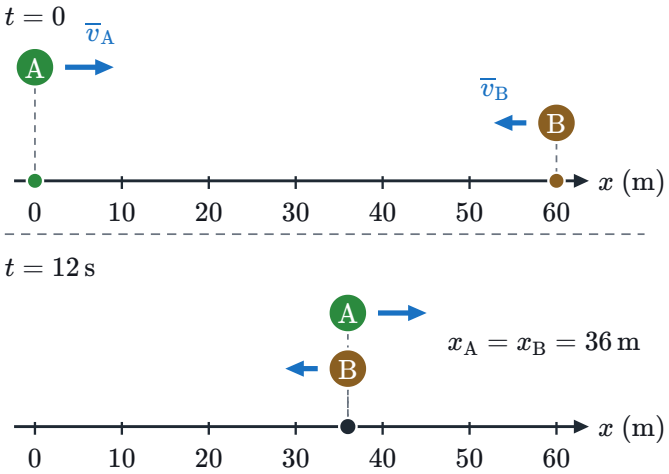
> **When and where does the car catch up with the cyclist, and which of them is faster then?**

Theory

Key idea

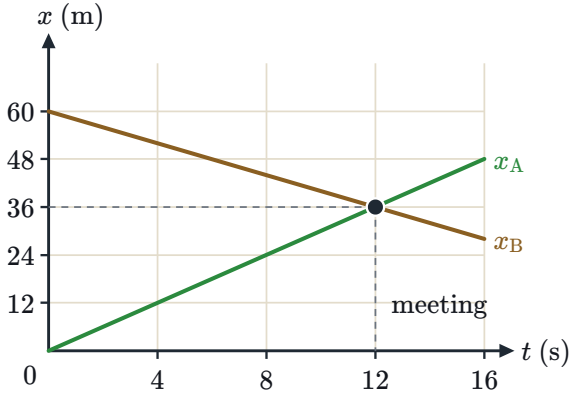
The motions of two bodies can be compared only when both positions are measured in the same coordinate system. We choose one x-axis, one origin, one positive direction and one clock that reads zero when the motions begin. Then each body has its own position as a function of time: $x_{\mathrm{A}}(t)$ and $x_{\mathrm{B}}(t)$. The origin and the direction can be chosen freely, but they are chosen once and kept for the whole problem.

The bodies **meet** when they are at the same place at the same moment, that is, $x_\mathrm{A} = x_\mathrm{B}$. A meeting always raises two questions: when does it happen, and where? If the bodies come from opposite directions, they meet each other. If the one behind catches up with the one ahead and passes it, this is **overtaking**. The physics is the same in both: the positions are equal.



In the figure, body A starts from $0\ \mathrm{m}$ and B from $60\ \mathrm{m}$, and they move toward each other. When the positions of the bodies are marked on the same axis, you can see at once which is where and how the distance between them changes. At the moment of meeting, the distance is zero.

In a (t, x) graph, each motion is its own line in the same coordinate system. The meeting is the intersection of the lines: its horizontal coordinate is the time of the meeting and its vertical coordinate is the place. A line higher in the graph means a larger position coordinate. At the moment of meeting, the bodies' graphs intersect. Which body is ahead also depends on the direction of motion: when the bodies move in the positive x direction, the higher line belongs to the body that is ahead. When they move in the negative direction, the body whose line is lower is ahead.



A meeting does not mean that the bodies have the same velocity. At the intersection the lines have different slopes, because the velocities of the bodies are different. If two uniformly moving bodies had equal velocities, the lines would be parallel and the distance between the bodies would stay the same, so they would not meet if they started from different places. In overtaking, the body coming from behind is faster than the one being overtaken at the moment of meeting.

Meeting condition

The time and place of a meeting are found by writing the position of each body as a function of time and setting the positions equal:

$$x_{\mathrm{A}}(t) = x_{\mathrm{B}}(t)$$

where $x_{\mathrm{A}}(t)$ and $x_{\mathrm{B}}(t)$ are the positions of the bodies in the same coordinate system (m) and t is the time of the meeting (s). Once t is solved, it is substituted into either position function, and the result is the place of the meeting. As a check, the other position function gives the same place.

The position function depends on the kind of motion. In uniform motion it is $x = x_0 + v\,t$ (chapter *Position, velocity and uniform motion as graphs*). In uniformly accelerated motion (chapter *Acceleration and uniformly accelerated motion*) it is

$$x = x_0 + v_0\,t + \tfrac{1}{2}\,a\,t^2$$
$$v = v_0 + a\,t$$

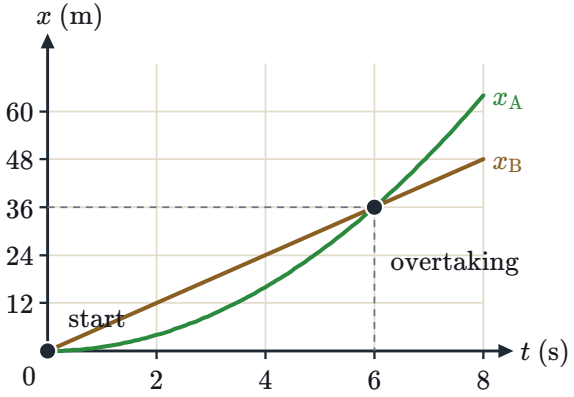
where x_0 is the initial position (m), v_0 the initial velocity ($\mathrm{m/s}$), a the acceleration ($\mathrm{m/s^2}$), t the time (s) and v the velocity at time t ($\mathrm{m/s}$). The formulas hold when the

motion is along a straight line and the acceleration is constant. The equation for two uniform motions is a first-degree equation. If the position functions are different lines, they have at most one point of intersection, so the bodies meet at most once. If the position functions are the same, the bodies are at the same position during the whole time considered. If one of them accelerates, the equation is quadratic, and the lines can have two intersections.

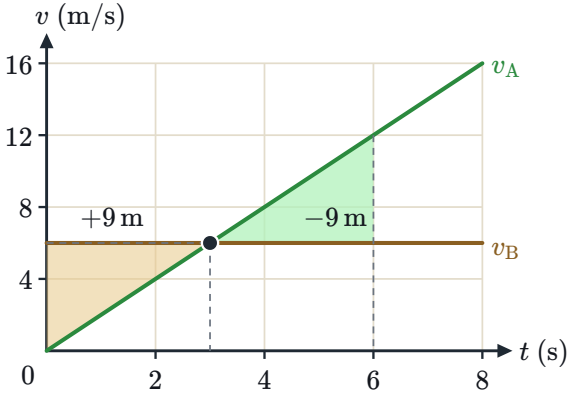
The simplest overtaking: a car starts from rest at the place where a cyclist passes at a constant velocity v. For the car $x_{\mathrm{A}} = \frac{1}{2}at^2$ and for the cyclist $x_{\mathrm{B}} = v\,t$:

$$\begin{aligned} \tfrac{1}{2}\,a\,t^2 &= v\,t \\ t\left(\tfrac{1}{2}\,a\,t - v\right) &= 0 \\ t = 0 \quad &\text{or} \quad t = \frac{2v}{a} \end{aligned}$$

The solution $t = 0$ is the start: the bodies are at the same place then. The overtaking happens at $t = 2v/a$. The velocity of the car is then $a\,t = 2v$, twice the velocity of the cyclist.



Why must the car reach twice the velocity? The reason can be seen in the (t, v) graph. The displacement is the area between the velocity graph and the time axis, and the area between two graphs tells how much the lead changes. As long as the car is slower than the cyclist, the cyclist's lead grows. When the one behind starts slower, the lead is greatest when the velocities are equal, and that is not the moment of meeting. Only after that is the car faster and does it reduce the lead. The overtaking happens when the decrease of the lead equals the earlier growth: in the figure each area corresponds to $9\ \mathrm{m}$. This is why the overtaking comes only far away: the car must first make up the growth of the lead, and in this example the overtaking place is $36\ \mathrm{m}$ from the start, although the lead grew by only $9\ \mathrm{m}$ at most.



If the cyclist is a distance $x_{0\mathrm{B}}$ ahead at the start, the equation is $\frac{1}{2}at^2 = x_{0\mathrm{B}} + v\,t$. It is solved with the quadratic formula, and the time of the meeting is the positive root. A negative root would describe a moment before the start and does not belong to the situation; two positive roots would mean two meetings. If there is no positive root, the bodies do not meet.



Two trams in Helsinki in July 2026. Photo: Pete (comedy_nose), CC0 1.0

Common misconception

The most common mistake is to think that the bodies have the same velocity at the moment of meeting, because they are at the same place. Position and velocity are different quantities: the bodies are at the same place only at that moment, and in overtaking the one coming from behind is moving faster at the moment of overtaking.

A second mistake is to solve the time of the meeting by setting the velocities equal. That finds the moment when the lead is greatest or smallest, not the

moment when the bodies are at the same place. In a meeting, the positions are equal.

A third mistake is to read the intersection of a (t, v) graph as a meeting. There the lines intersect when the velocities are equal. A meeting can be seen only in the (t, x) graph.

Summary

- The positions of two bodies are measured in the same coordinate system: the same origin, the same positive direction and the same clock.
- The bodies meet when $x_{\mathrm{A}}(t) = x_{\mathrm{B}}(t)$. The equation gives the time of the meeting, and substituting it into a position function gives the place.
- In a (t, x) graph the meeting is the intersection of the lines. At a meeting the velocities are usually different.
- When one motion is uniformly accelerated, the equation is quadratic. Every positive root is a meeting (there can be two), and a negative root is before the start and does not belong to the situation.
- When the one behind starts slower, the lead is greatest when the velocities are equal, and that is not the moment of meeting.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: A cyclist and a car at the lights

A cyclist rides along a straight road at a constant velocity of $6.0\ \mathrm{m/s}$. At the moment $t = 0$ she is $16\ \mathrm{m}$ ahead of the stop line at a traffic light. At the same moment a car standing at the line starts moving and accelerates uniformly at $2.0\ \mathrm{m/s^2}$ in the same direction. When does the car overtake the cyclist, and how far from the stop line does the overtaking happen?

STEP 1/7

Given and find. The origin is at the stop line and the positive direction is the direction of travel. Car A: $x_{0\mathrm{A}} = 0, v_{0\mathrm{A}} = 0, a = 2.0\,\mathrm{m/s^2}$. Cyclist B: $x_{0\mathrm{B}} = 16\,\mathrm{m}, v = 6.0\,\mathrm{m/s}$ (constant). We want the time t and the place x of the overtaking.

STEP 2/7

Principle. At the moment of overtaking the bodies are at the same place, so $x_{\mathrm{A}} = x_{\mathrm{B}}$. The motion of the car is uniformly accelerated and that of the cyclist is uniform, so the position of each is known as a function of time. We set the positions equal and solve for the time.

STEP 3/7

Rearranged formula. The position of the car is $x_{\mathrm{A}} = \tfrac{1}{2}at^2$ and that of the cyclist is $x_{\mathrm{B}} = x_{0\mathrm{B}} + v\,t$. The equation is solved with the quadratic formula and the positive root is chosen:

$$\begin{aligned} \tfrac{1}{2}\,a\,t^2 &= x_{0\mathrm{B}} + v\,t \\ \tfrac{1}{2}\,a\,t^2 - v\,t - x_{0\mathrm{B}} &= 0 \\ t &= \frac{v + \sqrt{v^2 + 2\,a\,x_{0\mathrm{B}}}}{a} \end{aligned}$$

The coefficients of the quadratic equation $At^2 + Bt + C = 0$ are here $A = \tfrac{1}{2}a, B = -v$ and $C = -x_{0\mathrm{B}}$, and the quadratic formula

$$t = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

gives the root above.

STEP 4/7

Unit conversions. None are needed: all quantities are in SI units, and under the square root v^2 and $2\,a\,x_{0\mathrm{B}}$ are both in $\mathrm{m^2/s^2}$. The expression under the root is

$$\begin{aligned} v^2 + 2\,a\,x_{0\mathrm{B}} &= 36\,\mathrm{m^2/s^2} + 64\,\mathrm{m^2/s^2} \\ &= 100\,\mathrm{m^2/s^2} \end{aligned}$$

STEP 5/7

Substitution. First we substitute for the time and then for the place:

$$\begin{aligned} t &= \frac{v + \sqrt{v^2 + 2\,a\,x_{0\mathrm{B}}}}{a} \\ &= \frac{6.0\,\mathrm{m/s} + \sqrt{100\,\mathrm{m^2/s^2}}}{2.0\,\mathrm{m/s^2}} \\ &= \frac{6.0\,\mathrm{m/s} + 10\,\mathrm{m/s}}{2.0\,\mathrm{m/s^2}} \\ &= 8.0\,\mathrm{s} \end{aligned}$$

$$\begin{aligned} x &= \tfrac{1}{2}\,a\,t^2 \\ &= \tfrac{1}{2} \cdot 2.0\,\mathrm{m/s^2} \cdot (8.0\,\mathrm{s})^2 \\ &= 64\,\mathrm{m} \end{aligned}$$

STEP 6/7

Result. The car overtakes the cyclist at $t = 8.0\,\mathrm{s}$ at the place $x = 64\,\mathrm{m}$, measured from the stop line. The results have two significant figures, because the data $6.0\,\mathrm{m/s}$, $2.0\,\mathrm{m/s^2}$ and $16\,\mathrm{m}$ have two.

STEP 7/7

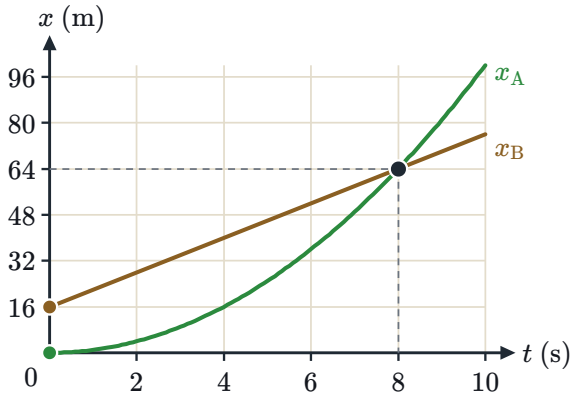
Reasonableness. The position of the cyclist is $16\,\mathrm{m} + 6.0\,\mathrm{m/s} \cdot 8.0\,\mathrm{s} = 64\,\mathrm{m}$, so the other position function gives the same place. The velocity of the car at the moment of overtaking is

$$\begin{aligned} a\,t &= 2.0\,\mathrm{m/s^2} \cdot 8.0\,\mathrm{s} \\ &= 16\,\mathrm{m/s} \end{aligned}$$

and that of the cyclist is $6.0\,\mathrm{m/s}$, so the car is faster as it overtakes, as it should be. The other root

$$\begin{aligned} t &= (6.0 - 10)\,\mathrm{m/s} : 2.0\,\mathrm{m/s^2} \\ &= -2.0\,\mathrm{s} \end{aligned}$$

is before the start and does not belong to the situation.


x (m)
96
80
64
48
32
16
0
2
4
6
8
10
t (s)
x_A
x_B

Interaction and force: Newton's first and third laws

Why does a curling stone slide tens of metres along the ice although nobody is pushing it any more?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy4-3.html`



IN THIS LESSON, YOU WILL LEARN TO

- explain with Newton's first law why motion continues without a forward force
- identify a force and its reaction force and explain why they do not cancel each other
- calculate an unknown force when an object is at rest or moves at constant velocity

PRIOR KNOWLEDGE

- Velocity is a vector, and in uniform motion it stays the same in magnitude and direction
  chapter *Position, velocity and uniform motion as graphs*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 3.

Think about it

A curler slides a few metres with the stone and then lets go. The stone carries on alone along the ice and stops only tens of metres away, in the target area. The sweepers do not touch the stone, and nothing else pushes it after the release. When the stone hits another stone, both stones change their motion.

> **Why does the stone keep moving although nothing pushes it any more, and what happens to each stone in the collision?**

Theory

Key idea

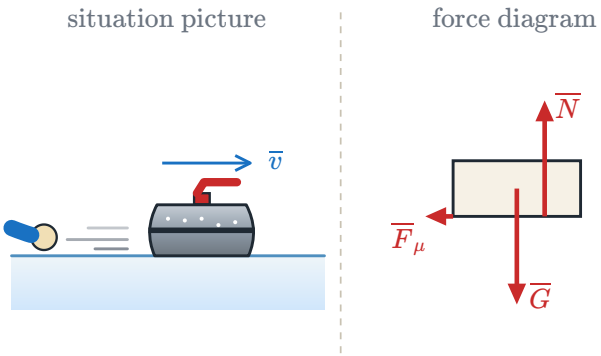
Force is an interaction

When two objects act on each other, there is an between them: a hand pushes a box, a rope pulls a sledge and the Earth pulls an apple. The strength of an

interaction is described by a , whose unit is the newton (N). A force is a vector with a magnitude and a direction, and it is written $\overline{F}$. Every force has a cause, and that cause is always another object. In contact forces the objects touch each other. The gravitational force of the Earth, the $\overline{G}$, and the forces of magnets also act at a distance.

Motion continues without a force

A curler pushes the stone only for a moment. When the hand lets go, nothing pushes the stone forward any more, but the stone still slides tens of metres. During the slide only the Earth and the ice act on the stone (air resistance is very small). In the vertical direction the weight and the of the ice, the force that supports the stone upwards and is perpendicular to the surface, cancel each other. In the horizontal direction there is only the small of the ice, a force between the surfaces that opposes motion, points backwards and slows the stone down very gradually.



If there were no friction at all, the speed of the stone would not change, and it would slide in the same direction for ever. This is , also called the **law of inertia**: if the sum of the forces acting on an object is zero, the object stays at rest or moves in a straight line at constant velocity. A force is needed to *change* motion, not to keep it going. The tendency of an object to keep its state of motion is called .



Tug of war players lean back and press their feet against the ground.

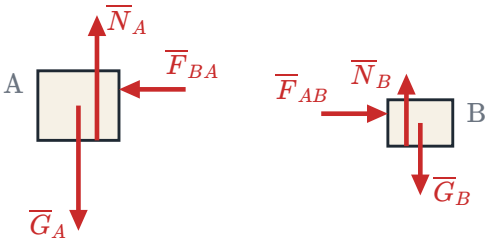
Photo: Trougnouf (Benoit Brummer), CC BY 4.0

Everyday experience seems to say otherwise, because on Earth friction and air resistance slow down almost everything that moves. A bicycle stops when you stop pedalling because the road and the air brake it, not because the "force of motion" has been used up. The smaller the friction, the longer the motion continues: further on ice than on asphalt, and in space almost without limit.

Action and reaction

When skater A pushes skater B, both start to glide away from each other. B does not just receive the push, but acts on A at the same time. An interaction always works in both directions. This is , the **law of action and reaction**: when object A exerts a force on object B, B exerts an equally large but oppositely directed force on A. The two forces form an action–reaction pair. A force and its reaction force arise in the same interaction at the same time, and they are of the same kind: push and counter-push, attraction and counter-attraction.







A Falcon 9 rocket lifts off in Florida in September 2024. The rocket pushes the exhaust gases downwards, and the gases push the rocket upwards with a force of equal magnitude. The rocket does not need air or ground to push against.

Photo: NASA/Keegan Barber, public domain (NASA)

An adult and a child receive an equally large force from each other, although the adult is much heavier. The child still starts moving with a greater speed, because the same force changes the motion of the smaller mass more; we return to this in the chapter *Newton's second law*. The most important point is that a force and its reaction force act on **different objects**. That is why they never cancel each other, and why they are not drawn in the force diagram of the same object.

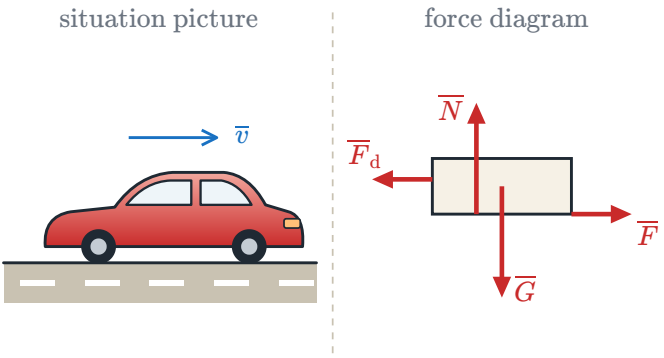
Newton's first and third laws

Newton's first law is a condition for the sum of the forces acting on an object:

$$\Sigma \overline{F} = \overline{0} \;\Leftrightarrow\; \overline{v} = \text{constant}$$

where $\Sigma \overline{F}$ is the vector sum of the forces acting on the object, the (N), and $\overline{v}$ is the velocity of the object (m/s). Constant velocity means that both the magnitude and the direction of the velocity stay the same; rest, $\overline{v} = \overline{0}$, is also constant velocity.

A net force of zero does not mean that there are no forces, but that they cancel each other. On a car driving along a straight road at constant velocity, the weight and the normal force of the road act in the vertical direction, and the force of the road on the tyres forwards and the air resistance backwards. Because the velocity does not change, the forward force is as large as the air resistance when rolling resistance is ignored.



Newton's third law is written for the forces between two objects:

$$\overline{F}_{AB} = -\overline{F}_{BA}$$

where $\overline{F}_{AB}$ is the force with which object A acts on object B (N), and $\overline{F}_{BA}$ is the force with which B acts on A (N). The minus sign tells that the forces are opposite in direction; the magnitudes are the same, $F_{AB} = F_{BA}$.

In calculations the magnitude of the weight is often needed:

$$G = mg$$

where G is the weight (N), m is the mass of the object (kg) and $g = 9.81\ \mathrm{m/s^2}$ is the acceleration due to gravity. The reaction force of the weight is the equally large gravitational force of the object on the Earth.

Common misconception

Many people think that a force must act on a moving object in the direction of motion and that the motion ends when this force "wears out". According to the first law this is not so: no force acts forwards on a sliding curling stone, and it slows down only because of friction. Another common error is to think that a force and its reaction force cancel each other. They act on different objects: a horse pulls a cart and the cart pulls the horse, but the motion of the cart is decided only by the forces acting on the cart. A third error is to take the weight of a book resting on a table and the normal force of the table as a force and its reaction force. They are equally large because of the first law. The reaction force of the normal force is the force of the book on the table, and the reaction force of the weight is the gravitational force of the book on the Earth.

Summary

- A force describes the interaction of two objects. Every force has a cause, another object, and the unit of force is the newton.
- Newton's first law: if the net force is zero, the object stays at rest or moves in a straight line at constant velocity.
- A force is needed to change motion, not to keep it going. In everyday life motion slows down because of friction and air resistance.
- Newton's third law: $\overline{F}_{AB} = -\overline{F}_{BA}$. A force and its reaction force are equally large, opposite in direction and of the same kind, and they act on different objects, so they do not cancel each other.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Forces on an airliner

A passenger aircraft flies horizontally in a straight line at a constant velocity of $850\ \mathrm{km/h}$. Each of its two jet engines pushes air and exhaust gases backwards with a force of $48\ \mathrm{kN}$. How large is the air resistance on the aircraft?

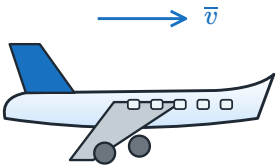
STEP 1/7

Given and asked. The force of one engine on the gases is $F_1 = 48\ \mathrm{kN}$ backwards, there are two engines, and the velocity $850\ \mathrm{km/h}$ stays the same in magnitude and direction. We are asked for the air resistance F_d.

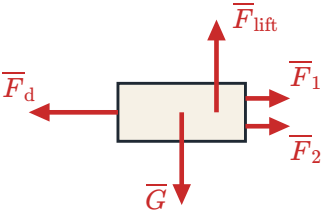
STEP 2/7

Principle. By Newton's third law the gases push each engine forwards with an equally large force F_1; this is the thrust of the engine. The aircraft flies straight at constant velocity, so by Newton's first law its net force is zero. In the horizontal direction only two thrusts $\overline{F}_1$ and $\overline{F}_2$ act forwards on the aircraft (both $48\ \mathrm{kN}$), and the air resistance $\overline{F}_\mathrm{d}$ acts backwards, so they cancel each other. In the vertical direction the lift of the wings and the weight cancel each other. In the force diagram the aircraft is a rectangle and the forces acting on it are arrows:

situation picture



force diagram



STEP 3/7

Formula solved. Choose the forward direction as positive and solve the air resistance:

$$\Sigma F_x = 2F_1 - F_\mathrm{d} = 0$$
$$F_\mathrm{d} = 2F_1$$

STEP 4/7

Unit conversions. The velocity $850\,\mathrm{km/h}$ is not needed: it is enough to know that it does not change. The forces are in kilonewtons, so the result is obtained directly in kilonewtons: $48\,\mathrm{kN} = 48\,000\,\mathrm{N}$.

STEP 5/7

Substitution. Substitute the force of the engine with its unit:

$$\begin{aligned} F_\mathrm{d} &= 2F_1 \\ &= 2 \cdot 48\,\mathrm{kN} \\ &= 96\,\mathrm{kN} \end{aligned}$$

STEP 6/7

Result. The air resistance is $96\,\mathrm{kN}$, or $9.6 \cdot 10^4\,\mathrm{N}$, and it points backwards. The engine force $48\,\mathrm{kN}$ is taken as an exact value, so the air resistance $96\,\mathrm{kN}$ is an exact value of this model and needs no rounding.

STEP 7/7

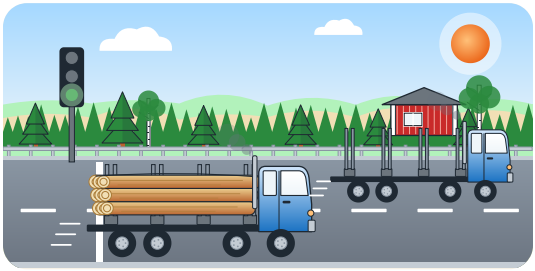
Reasonableness. If the air resistance were smaller than the thrusts, the net force would point forwards and the aircraft would accelerate; if it were larger, the aircraft would slow down. Constant velocity is possible only when the air resistance is as large as the sum of the thrusts $2 \cdot 48\,\mathrm{kN}$. The same reasoning holds for all constant velocities: the sum of the thrust and the air resistance is always zero. The magnitudes of the forces can still differ at different speeds, because air resistance grows as the speed grows, so a greater constant speed also needs a greater thrust.

FY4 · CHAPTER 5

Newton's second law

Why does a fully loaded truck accelerate away from the traffic lights more slowly than an empty one, even though the engine is the same?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy4-4.html`



IN THIS LESSON, YOU WILL LEARN TO

- explain how the acceleration depends on the net force and the mass.
- work out the direction of the net force from the acceleration, also in slowing-down motion.
- calculate an acceleration or a force using the free-body diagram and the equation of motion $\Sigma F = ma$.

PRIOR KNOWLEDGE

- Acceleration is the change in velocity per unit of time, $a = \frac{\Delta v}{\Delta t}$
  chapter *Acceleration and uniformly accelerated motion*
- When the forces acting on an object cancel each other, the object is at rest or moves at constant velocity
  chapter *Interaction and force: Newton's first and third laws*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 4.

Think about it

Two identical timber trucks stand side by side at the traffic lights. One is empty, the other is full of logs. The light turns green, and both drivers accelerate with equal effort. Within a few seconds the empty truck is clearly ahead.

> **What decides how quickly the velocity of an object changes, and why does the same engine not get the loaded truck up to speed as quickly?**

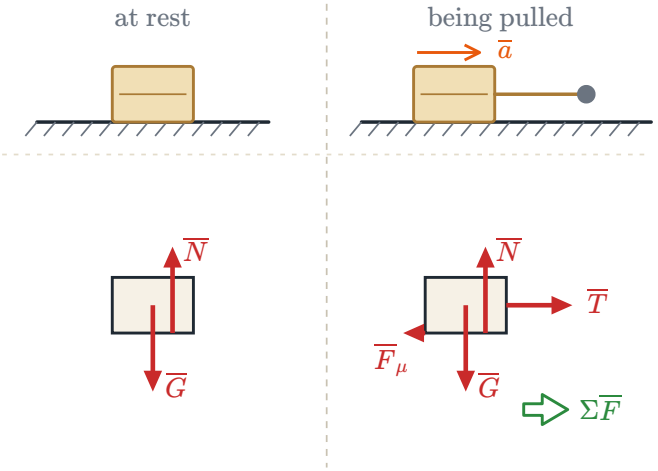
Theory

Key idea

The net force changes the velocity

In the chapter *Interaction and force: Newton's first and third laws* you learned that an object's velocity stays unchanged when the forces acting on it cancel each other. The object is then at rest or moves in a straight line at constant velocity. When the forces do not cancel, the velocity changes: the object speeds up, slows down or turns. What matters is not any single force but the combined effect of all the forces acting on the object, the $\Sigma\overline{F}$. It is the vector sum of the forces, not just one force added to another.

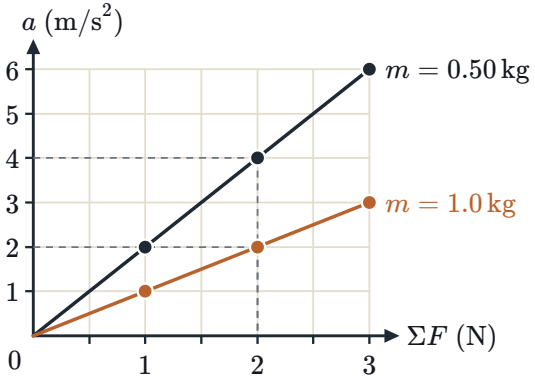
On the left of the figure a box rests on the floor. The weight $\overline{G}$ and the normal force $\overline{N}$ from the floor are equal and opposite, so the net force is zero. On the right the box is pulled by a rope. The vertical forces still cancel, but horizontally the tension $\overline{T}$ in the rope is greater than the friction force $\overline{F}_\mu$ from the floor. The net force points to the right, and the box accelerates to the right.



Mass resists a change in velocity

Measurements with a cart reveal two regularities. When the net force doubles, the acceleration doubles: the acceleration is directly proportional to the net force. When the same net force acts on a cart with twice the mass, the acceleration is halved. Mass therefore tells us about the inertia of an object, that is, how hard it is

to change its velocity. In the graph each mass gives a straight line through the origin, and the line of the heavier cart is half as steep.



This gives the answer to the opening question. The engine turns the driving wheels, and the road pushes the tyres forward. When this force is equally large for the loaded and the empty truck, the much greater mass of the loaded truck makes its acceleration small. The weight of the load does not pull the truck backward as such, because the normal force from the road cancels it. The heavy load does increase the rolling resistance of the tyres, but the main reason for the slow acceleration is the large mass.

Newton's second law

Newton's second law, also called the fundamental law of dynamics, gathers the observations into one equation:

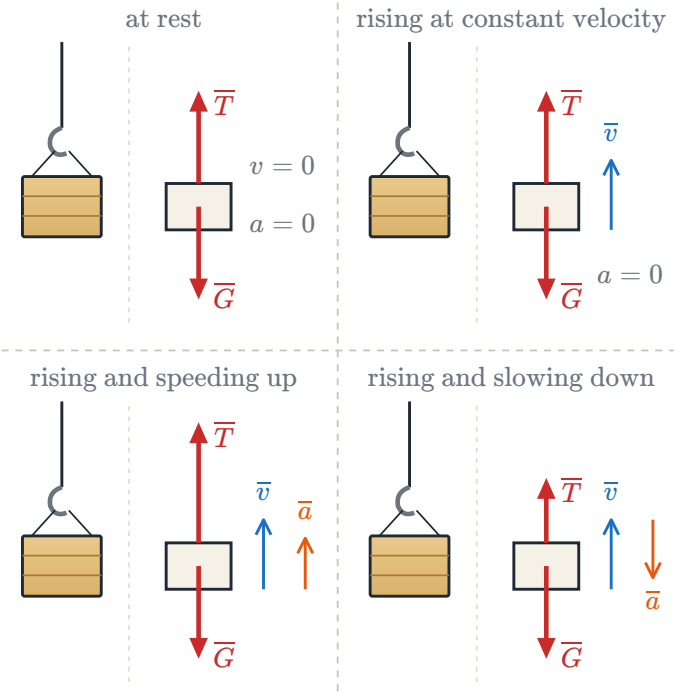
$$\Sigma \overline{F} = m\overline{a}$$

where $\Sigma \overline{F}$ is the vector sum of the forces acting on the object, the net force (N), m is the mass of the object (kg) and $\overline{a}$ is its acceleration ($\mathrm{m/s^2}$). The unit of force is defined by this law: one newton is the force that gives a mass of one kilogram an acceleration of $1\ \mathrm{m/s^2}$. So $1\ \mathrm{N} = 1\ \mathrm{kg} \cdot \mathrm{m/s^2}$.

The net force points in the direction of the acceleration

The law is a vector equation: the net force and the acceleration always point in the same direction. The direction of the velocity can be another one. In the figure a crane holds a crate on a rope. When the crate rises while speeding up, the force of the rope is greater than the weight and the net force points up. When the crate rises while slowing down, the force of the rope is smaller than the weight: the net

force and the acceleration point down, even though the crate still moves upward. At rest and when rising at constant velocity, the force of the rope and the weight are equal.



First the forces, then the equation

When the law is used in a calculation, the steps are always taken in the same order. Choose the object and draw its free-body diagram, in which every force has a cause: the Earth, a rope, a surface or a puller. Choose a positive direction, usually the direction of the acceleration. Only then write the net force from the forces, with signs, and set it equal to ma. For the crate of the crane, when up is positive, we get $T - G = ma$. In the chapter *Free-body diagrams and the combined effect of forces* the same method extends to slanted forces.

Common misconception

Many people think that a force always acts in the direction of motion and that the net force points in the direction of motion. The net force, however, points in the direction of the acceleration. When air resistance is ignored, a ball thrown upward keeps rising even though only its weight, which points down, acts on it, and that is why it slows down. In a slowing motion the net force points against the velocity. A

second mistake is to think that a constant force keeps up a constant velocity. When the net force is constant and not zero, the velocity changes steadily. A third mistake is to confuse mass and weight. On a horizontal track the weight does not resist the motion, because the normal force cancels it: it is the mass that makes the acceleration smaller.

Summary

- When the net force is zero, the velocity does not change. When it is not zero, the object accelerates in the direction of the net force.
- Newton's second law is $\Sigma \overline{F} = m\overline{a}$. The acceleration is directly proportional to the net force and inversely proportional to the mass.
- The net force points in the direction of the acceleration, which is not always the direction of the velocity: in a slowing motion it points against the velocity.
- First the free-body diagram is drawn and a positive direction is chosen, then $\Sigma F = ma$ is written with signs.
- The unit of force is $1\,\mathrm{N} = 1\,\mathrm{kg}\cdot\mathrm{m/s^2}$.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: A crane lifting a panel

A construction crane lifts a concrete slab of mass $480\,\mathrm{kg}$. The tension in the rope is $5.20\,\mathrm{kN}$. What is the acceleration of the slab, and in which direction does it point? Air resistance is ignored.

STEP 1/8

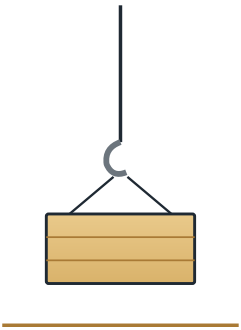
Given and find. $m = 480\,\mathrm{kg}$, $T = 5.20\,\mathrm{kN}$ and $g = 9.81\,\mathrm{m/s^2}$. We want the acceleration a and its direction.

STEP 2/8

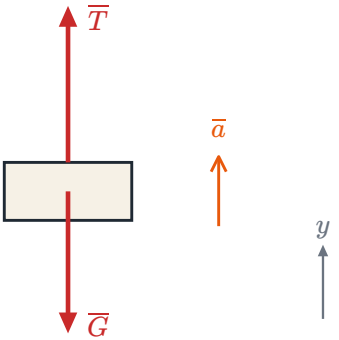
Principle. Choose the concrete slab as the object. The Earth (weight) and the rope (tension) act on it. By Newton's second law, the acceleration is determined by the sum of these forces. The force of the rope alone does not accelerate the slab, because most of it is used to hold up the weight.

STEP 3/8

Free-body diagram. The slab is drawn as a rectangle. The weight $\overline{G}$ starts from the centre and points down, and the tension $\overline{T}$ starts from the top surface and points up. The positive direction is chosen to be up.



situation picture



free-body diagram

STEP 4/8

Rearranged formula. From the diagram we write the net force $T - mg$ and solve for the acceleration:

$$T - mg = ma$$
$$a = \frac{T - mg}{m}$$

STEP 5/8

Unit conversions. The tension is converted to newtons:

$$5.20\,\mathrm{kN} = 5.20 \cdot 1000\,\mathrm{N}$$
$$= 5200\,\mathrm{N}$$

STEP 6/8

Substitution. First we calculate the weight and then the acceleration:

$$\begin{aligned} a &= \frac{T - mg}{m} \\ mg &= 480\,\mathrm{kg} \cdot 9.81\,\tfrac{\mathrm{m}}{\mathrm{s}^2} \\ &= 4708.8\,\mathrm{N} \\ a &= \frac{5200\,\mathrm{N} - 4708.8\,\mathrm{N}}{480\,\mathrm{kg}} \\ &\approx 1.02\,\mathrm{m/s^2} \end{aligned}$$

STEP 7/8

Result. The acceleration is $a \approx 1.02\,\mathrm{m/s^2}$ upward. It has three significant figures, like the tension $5.20\,\mathrm{kN}$ and the free-fall acceleration $9.81\,\mathrm{m/s^2}$. The whole number $480\,\mathrm{kg}$ is taken as an exact value, so it does not limit the precision.

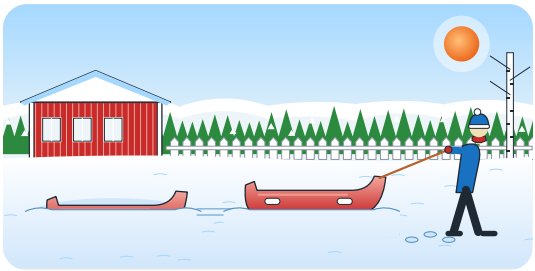
STEP 8/8

Reasonableness. The force of the rope is only about ten percent greater than the weight $4.71\,\mathrm{kN}$, so the acceleration should also be about a tenth of the free-fall acceleration: $0.1 \cdot 9.81\,\mathrm{m/s^2} \approx 1\,\mathrm{m/s^2}$. If the weight were forgotten, the result would be $5200\,\mathrm{N} : 480\,\mathrm{kg} \approx 10.8\,\mathrm{m/s^2}$. That would be wrong, because besides the force of the rope the weight acts on the element downward, and the acceleration is set by the sum of these forces. For example, when the element rises at a constant velocity, the rope pulls with a force equal to the weight and the forces cancel, but a calculation that forgets the weight would still give an acceleration of about $9.8\,\mathrm{m/s^2}$.

Free-body diagrams and the combined effect of forces

Why can a sled press on the ground beneath it differently when it is pulled upward at an angle by a rope than when it simply stands still?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy4-5.html`



IN THIS LESSON, YOU WILL LEARN TO

- identify the forces acting on an object and draw them in a free-body diagram.
- tell forces apart from their components and reason out the combined effect of forces.
- form the equations of motion with the help of a free-body diagram.

PRIOR KNOWLEDGE

- Newton's second law $\Sigma\overline{F} = m\overline{a}$ from the chapter *Newton's second law*
- The sine and cosine of a right triangle

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 4.

Think about it

Two children take turns moving a sled on ice. One pulls on a short rope obliquely upward, and the other pushes the sled from behind with a handle that points obliquely downward. Both use an equal force. Friction on ice is small, but the runners of the sled press on the ice in different ways.

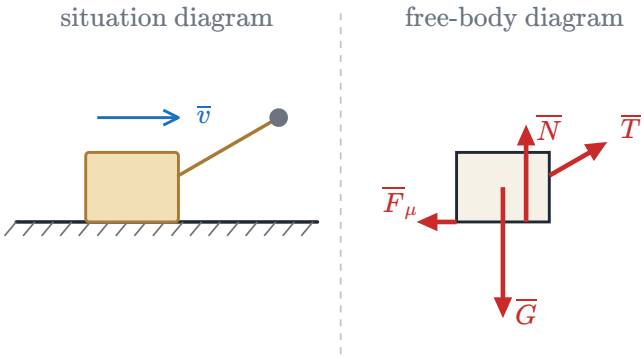
> **Which forces act on the sled in each case, and does the sled press on the ice with the same force as when it stands still?**

Theory

Key idea

Usually several forces act on an object at the same time. The Earth pulls it down, the surface supports it, a rope pulls and the ground resists the motion. Each force describes one interaction between the object and some other object. The motion is decided by the **combined effect** of these forces, the $\Sigma\overline{F}$, which is the vector sum of the forces. In the chapter *Newton's second law* you learn that the net force determines the acceleration: $\Sigma\overline{F} = m\overline{a}$.

To include every force and to keep anything extra out of the calculation, the forces are first drawn in a . It is a different picture from the **situation diagram**. The situation diagram shows the whole situation: the surface, the rope, the person pulling and the direction of motion. The free-body diagram shows only the chosen object, simplified to a rectangle, and the forces acting on it, as arrows. The direction of an arrow is the direction of the force, and the lengths of the arrows tell which force is larger.



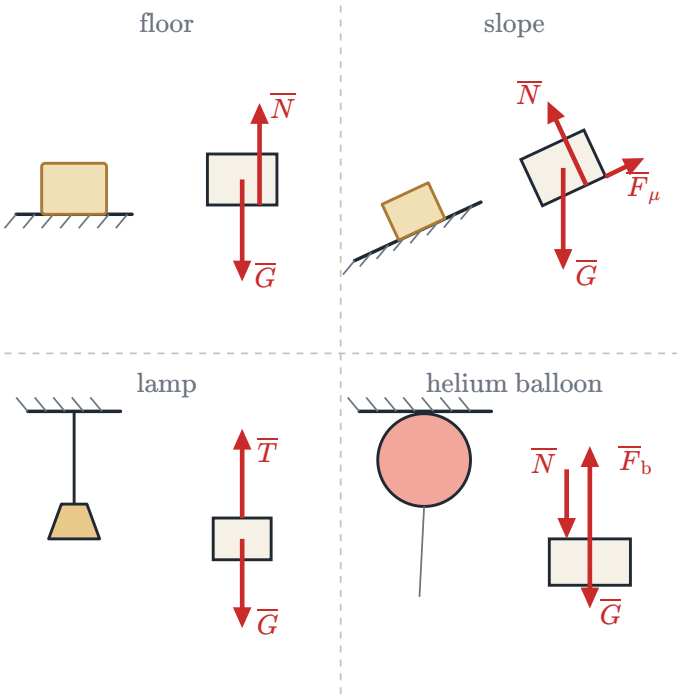
In the figure, a box is pulled along by a rope that points obliquely upward. The force of the rope points both forward and upward. The upward part helps to hold the box up, so the floor does not have to support it as much as when the box stands still. That is why the normal force N is smaller than the weight G. The velocity is not drawn in the free-body diagram, because velocity is not a force.

Drawing a free-body diagram

A free-body diagram is always drawn with the same eight steps. They are worth going through every time, even when the situation looks easy.

1. **Choose the object.** Decide whose forces you draw. Only forces that act **on this object** go into the diagram. Forces that the object itself exerts on others do not belong in it.
2. **Find the interactions.** List everything that acts on the object: the Earth (weight), every surface the object touches (normal force and friction), every rope (tension), a push or a pull, and air or water (drag and buoyancy). Every force has a cause outside the object. Motion alone does not cause a force.

3. **Draw the object and the forces.** Draw the object simplified to a rectangle. Draw one arrow for each force, starting at the point where the force acts. The arrow of a pushing force may also end at that point.
    - The weight $\overline{G}$ points straight down.
    - The normal force $\overline{N}$ is perpendicular to the contact surface and pushes the object away from the surface. It does not always point up: a ceiling pushes down and a wall pushes sideways.
    - The friction force $\overline{F}_{\mu}$ is parallel to the contact surface. Its direction comes from the direction in which the object slides, or tends to slide, relative to the surface. It does not always oppose the velocity. The friction arrow starts at a corner of the contact surface, outside the object, or ends there. It is not drawn along a side of the object from the middle of the side.
    - The tension $\overline{T}$ acts along the rope and pulls the object toward the rope.
    - Air drag is drawn only when the situation and its assumptions call for it.
4. **Name the forces.** Give each force its own symbol. Two similar forces are told apart with subscripts, for example $\overline{T}_1$ and $\overline{T}_2$.
5. **Think about the magnitudes.** At rest and in uniform motion the net force is zero, so the arrows cancel each other. In accelerating motion the net force points in the direction of the acceleration, which is not always the direction of the velocity. Do not invent exact lengths if the data does not fix them.
6. **Only forces.** Velocity, acceleration and displacement are not forces. If they are needed, they are drawn beside the object as thin arrows in different colours: in this course forces are red, velocity is blue and acceleration is orange. The net force is not one more force.
7. **Components are not new forces.** When a force is split into components, the components together replace the force in the calculation. They are drawn with a dashed line so that they are not mistaken for forces.
8. Only now are the axes chosen and Newton's second law written component by component from the forces and signs of the diagram.



The figure shows four situations in which the directions of the forces often surprise people. On a slope the normal force is perpendicular to the slope, not vertical, and friction holds the box in place along the slope. On a helium balloon pressed against the ceiling the ceiling pushes downward.

In a free-body diagram the object is always a simplified rectangle, and each force starts from its point of action: the weight from the centre, the normal force and friction from the contact surface, and the tension from where the rope is attached. The arrow of a pushing force may also end at the point where the push acts. In the situation diagram objects are drawn as they look, but in the free-body diagram only the rectangle and the forces are drawn.

The marking instructions of the matriculation examinations give task-specific points for the free-body diagram. For example, in the spring 2025 helium-balloon task a missing, extra or wrongly directed force cost all the points of the diagram, and so did drawing velocity or acceleration as a force on the balloon. In the autumn 2025 examination a missing name cost one point. The exact scoring varies from task to task, but every examination requires the same things: all the forces, no invented forces, correct directions and unambiguous names.

Equation of motion by components

Newton's second law concerns the net force:

$$\Sigma \overline{F} = m\overline{a}$$

where $\Sigma \overline{F}$ is the vector sum of the forces acting on the object (N), m is the mass of the object (kg) and $\overline{a}$ is its acceleration (m/s²). When the forces are not along one line, two perpendicular axes are chosen and the law is written separately for each axis direction:

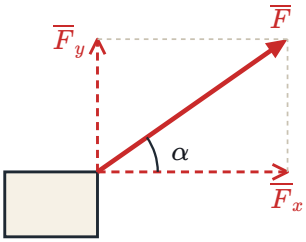
$$\Sigma F_x = ma_x$$
$$\Sigma F_y = ma_y$$

where ΣF_x and ΣF_y are the sums of the components of the forces (N) and a_x and a_y are the components of the acceleration (m/s²). A component pointing along the axis is positive and one pointing against it is negative. It is wise to choose the axes so that one of them points in the direction of the acceleration.

The components of an oblique force F come from a right triangle, when α is the angle between the force and the x-axis:

$$F_x = F \cos \alpha$$
$$F_y = F \sin \alpha$$

where F is the magnitude of the force (N), F_x and F_y are its components (N) and α is the angle. The components are not new forces. They replace the force F in the calculation.



$$F_x = F \cos \alpha$$

$$F_y = F \sin \alpha$$

Common misconception

A moving object does not need a force in the direction of its motion. Only the weight, the normal force and friction act on a sliding box, and there is no "force of motion": there is no object that would cause it. Another common mistake is to

think that the normal force is always equal to the weight. The normal force is exactly as large as the situation requires: in an oblique pull it is smaller than the weight, and in an oblique push or in an elevator accelerating upward it is larger. A third mistake is to draw the components as forces in addition to the force itself, which puts the same force into the calculation twice.

Summary

- A free-body diagram contains only the chosen object, as a rectangle, and the forces acting on it. The situation diagram and the free-body diagram are different pictures.
- Every force has a cause: the Earth, a surface, a rope, a pusher or the air. Velocity and acceleration are not forces, and they are drawn in different colours from the forces.
- The normal force is perpendicular to the surface, friction is parallel to the surface and tension is along the rope.
- The net force is the vector sum of the forces, and $\Sigma \overline{F} = m\overline{a}$ is written by components: $\Sigma F_x = ma_x$ and $\Sigma F_y = ma_y$.
- The components of an oblique force are $F\cos\alpha$ and $F\sin\alpha$. They replace the force in the calculation and are not new forces.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Pulling a sled with a rope

A sled of mass $12\ \mathrm{kg}$ is pulled on horizontal snow by a rope that makes an angle of 35° with the horizontal. The tension in the rope is $80\ \mathrm{N}$ and the friction force from the snow is $30\ \mathrm{N}$. What is the acceleration of the sled, and what is the normal force from the snow?

STEP 1/8

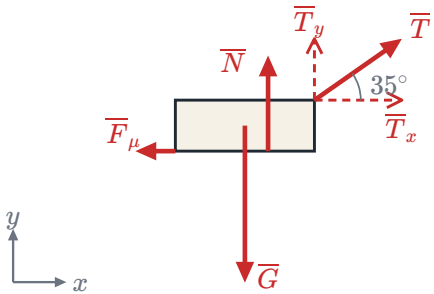
Given and find. $m = 12\,\mathrm{kg}, T = 80\,\mathrm{N}, \alpha = 35^\circ, F_\mu = 30\,\mathrm{N}$ and $g = 9.81\,\mathrm{m/s^2}$. We want the acceleration a and the normal force N.

STEP 2/8

Principle. Choose the sled as the object. The Earth (weight), the snow (normal force and friction) and the rope (tension) act on it. Newton's second law is written in two directions: horizontally the sled accelerates, vertically it stays on the surface of the snow, so $a_y = 0$.

STEP 3/8

Free-body diagram. The sled is drawn as a rectangle, and the forces start from their points of action. The axes are chosen so that x points in the direction of motion and y points up. The force of the rope is split into the components $T_x = T\cos\alpha$ and $T_y = T\sin\alpha$, drawn with dashed lines.



STEP 4/8

Rearranged formula. The equations of motion are read from the diagram, and the asked quantities are solved:

$$x\colon\ T\cos\alpha - F_\mu = ma$$
$$y\colon\ N + T\sin\alpha - mg = 0$$
$$a = \frac{T\cos\alpha - F_\mu}{m}$$
$$N = mg - T\sin\alpha$$

STEP 5/8

Unit conversions. None are needed: all quantities are in SI units, and $\frac{\mathrm{N}}{\mathrm{kg}} = \frac{\mathrm{m}}{\mathrm{s}^2}$.

STEP 6/8

Substitution. Substitute the values with their units:

$$\begin{aligned} a &= \frac{T\cos\alpha - F_\mu}{m} \\ &= \frac{80\,\mathrm{N}\cdot\cos 35^\circ - 30\,\mathrm{N}}{12\,\mathrm{kg}} \\ &\approx 2.96\,\mathrm{m/s^2} \end{aligned}$$

$$\begin{aligned} N &= mg - T\sin\alpha \\ &= 12\,\mathrm{kg}\cdot 9.81\,\mathrm{m/s^2} \\ &\quad - 80\,\mathrm{N}\cdot\sin 35^\circ \\ &\approx 71.8\,\mathrm{N} \end{aligned}$$

STEP 7/8

Result. The acceleration is $a \approx 2.96\,\mathrm{m/s^2}$ and the normal force is $N \approx 71.8\,\mathrm{N}$. The whole numbers of the data, $12\,\mathrm{kg}$, $80\,\mathrm{N}$, $30\,\mathrm{N}$ and 35°, are taken as exact values. The acceleration is calculated from exact values only and is given with three significant figures. The precision of the normal force is limited by the free-fall acceleration $9.81\,\mathrm{m/s^2}$, so it is also given with three significant figures.

STEP 8/8

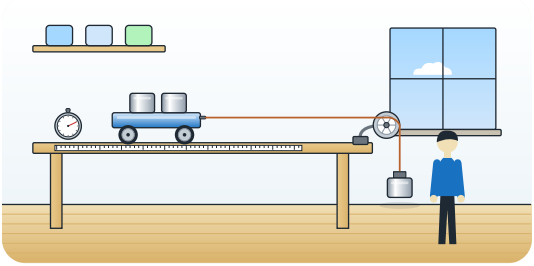
Reasonableness. The weight of the sled is $mg \approx 118\,\mathrm{N}$. The normal force $71.8\,\mathrm{N}$ is smaller, because the vertical component of the rope force, $T_y \approx 45.9\,\mathrm{N}$, supports part of the weight, and $71.8\,\mathrm{N} + 45.9\,\mathrm{N} \approx 118\,\mathrm{N}$. Without friction the acceleration would be $65.5\,\mathrm{N} : 12\,\mathrm{kg} \approx 5.5\,\mathrm{m/s^2}$, so friction reduces it, as it should.

FY4 · CHAPTER 7

Equation of motion for connected bodies

Why does a mass hanging over the edge of a table fall more slowly than a freely falling object when it pulls a cart along the table with a string?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy4-13.html`



IN THIS LESSON, YOU WILL LEARN TO

- calculate the acceleration of connected bodies and the tension in the string from the equation of motion of each body.
- draw a separate free-body diagram for each body.
- explain why the tension for an accelerating hanging body is not its weight.

PRIOR KNOWLEDGE

- Newton's second law $\Sigma\overline{F} = m\overline{a}$
  chapter *Newton's second law*

- Drawing a free-body diagram and writing the equation of motion by components
  chapter *Free-body diagrams and the combined effect of forces*
- A force and its reaction force act on different bodies
  chapter *Interaction and force: Newton's first and third laws*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 5.

Think about it

A cart stands on a table, and a string tied to it passes over a pulley at the edge of the table. A mass hangs from the other end of the string. When the cart is released, the mass starts to descend and pulls the cart along. The mass does not fall as fast as it would in free fall, though.

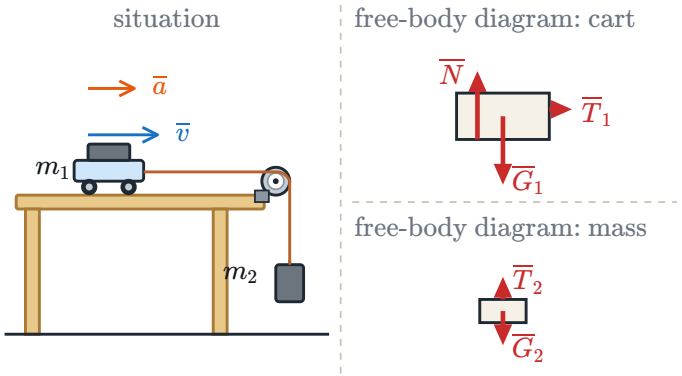
> **What pulls the cart, and how large is the force of the string compared with the weight of the mass?**

Theory

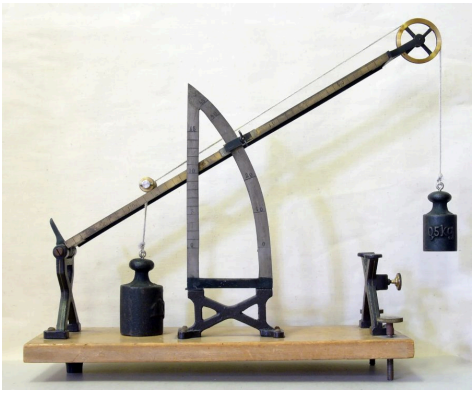
Key idea

Often two bodies move together because a string joins them. When the string is taut and does not stretch, the bodies travel the same distance in the same time, so their accelerations have the same magnitude a. The string passes a force from one body to the other: it pulls each body toward itself. In this chapter the string is light and inextensible, the pulley is light and frictionless, and friction on the table and air resistance are neglected.

Connected bodies are treated one at a time. Each gets its own free-body diagram that shows only the forces acting on it, and each gets its own equation of motion. In the figure, the cart is acted on by its weight $\overline{G}_1$, the normal force $\overline{N}$ from the table and the tension $\overline{T}_1$ in the string. The hanging mass is acted on by its weight $\overline{G}_2$ and the tension $\overline{T}_2$ in the string.



The string pulls the cart with the force $\overline{T}_1$ and the hanging mass with the force $\overline{T}_2$. They act on different bodies, so they appear in different free-body diagrams. A light string and a frictionless pulley only pass the force on and turn its direction, so the magnitudes are equal: $T_1 = T_2 = T$. The forces $\overline{T}_1$ and $\overline{T}_2$ are nevertheless not an action–reaction pair. When the string pulls the cart, the cart pulls the string back with an equally large force, and that force acts on the string and is not in the cart's free-body diagram.



An inclined plane teaching apparatus from the early 20th century in the school museum of Bremerhaven, Germany. A roller on

the incline and a hanging weight are connected by a string that runs over a pulley.

Photo: Hannes Grobe, CC BY 3.0

When the hanging mass accelerates downward, the tension in the string is smaller than its weight. The acceleration is downward, so the net force on the mass points down and $G_2 > T$. If T were equal to G_2, the net force on the mass would be zero and it would not start to move. The mass therefore descends more slowly than a freely falling object: the acceleration $a = \frac{m_2 g}{m_1 + m_2}$ is smaller than g, because the weight of the hanging mass also accelerates the cart.

Equations of motion for connected bodies

Each body gets its own equation of motion $\Sigma F = ma$. The positive direction is chosen separately for each body along its acceleration: to the right for the cart and down for the hanging mass. Because the string does not stretch, both bodies have the same acceleration magnitude a. The equations of the cart and the hanging mass are

$$T = m_1 a$$
$$m_2 g - T = m_2 a$$

where m_1 is the mass of the cart (kg), m_2 the mass of the hanging mass (kg), T the tension in the string (N), a the acceleration (m/s²) and $g = 9.81\ \mathrm{m/s^2}$. There are two unknowns, a and T, and two equations.

When the equations are added side by side, T cancels, because the string is an internal force between the bodies. This gives the acceleration and then the tension:

$$a = \frac{m_2}{m_1 + m_2}\, g$$
$$T = m_1 a = \frac{m_1 m_2}{m_1 + m_2}\, g$$

The tension is always smaller than the weight $m_2 g$ of the hanging mass, because $\frac{m_1}{m_1 + m_2} < 1$. The heavier the cart is compared with the hanging mass, the less the bodies accelerate and the closer T is to the weight $m_2 g$.

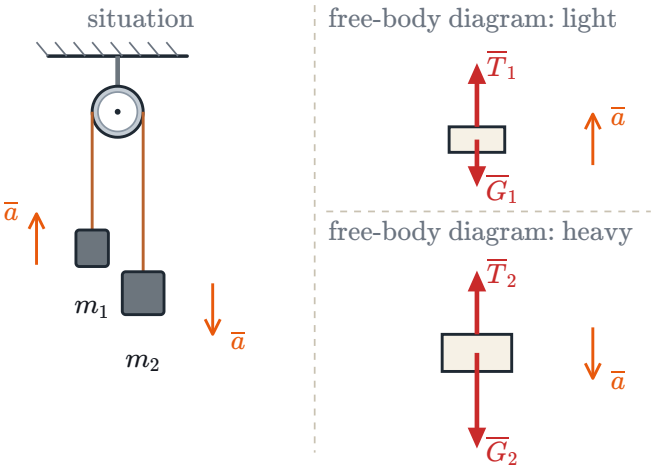
Two hanging bodies

When a body hangs at each end of the string, the heavier one (m_2) goes down and the lighter one (m_1) goes up. The positive direction is chosen along the motion: up for the lighter body and down for the heavier one. Then

$$T - m_1 g = m_1 a$$
$$m_2 g - T = m_2 a$$

$$a = \frac{m_2 - m_1}{m_1 + m_2}\, g$$
$$T = \frac{2 m_1 m_2}{m_1 + m_2}\, g$$

where $m_2 > m_1$. The tension lies between the weights $m_1 g$ and $m_2 g$ of the bodies. If the masses are equal, $a = 0$ and $T = mg$.



Method for connected bodies

1. **Draw a free-body diagram** for each body. The string pulls each body toward itself, and each body has its own tension arrow.

2. **Choose the positive direction** for each body along its acceleration. If the actual direction of the acceleration is not known, the positive direction may be assumed, as long as the directions of the bodies fit together along the string. If the solution gives a negative acceleration, the actual acceleration is in the opposite direction. The direction of friction depends on the direction of motion, so with friction the assumed direction must be checked.
3. **Write the equation of motion** $\Sigma F = ma$ for each body separately. The acceleration a and the tension magnitude T are the same in both. If a body is also pulled by an external force $\overline{F}$, that force belongs only in the free-body diagram and equation of motion of the body being pulled.
4. **Solve** the two equations: add them, so that T cancels, or substitute the T from one into the other.
5. **Check** the signs and compare: the tension is not equal to the weight of a hanging body when the bodies accelerate.

Common misconception

People often think that the cart is pulled by the weight of the hanging mass, so that the tension would be $m_2 g$. The cart is pulled by the string, however, and the force of the string is T. Only when the bodies do not accelerate, for example when the cart is held still, is $T = m_2 g$. When the hanging mass descends with increasing speed, its net force $G_2 - T = m_2 a$ must point down, so $T < m_2 g$.

Another mistake is to treat the forces $\overline{T}_1$ and $\overline{T}_2$ as an action–reaction pair. They act on different bodies but belong to different interactions: one to the string–cart pair and the other to the string–mass pair. They are equal because the string is light and the pulley is light and frictionless, not because they would be the forces of one interaction pair. A third mistake is to think that the heavier body accelerates more. Bodies at the ends of an inextensible string have the same acceleration magnitude.

Summary

- Bodies connected by a string or over a pulley each have their own free-body diagram and their own equation of motion $\Sigma F = ma$, and the positive direction is chosen for each body along its acceleration, or assumed if the direction is not known: a negative solution means that the acceleration is in the opposite direction.
- A light, inextensible string and a light, frictionless pulley give the bodies the same acceleration magnitude a and equal tensions $T_1 = T_2 = T$.

- For a cart and a hanging mass, $a = \frac{m_2}{m_1+m_2}\,g$ and $T = \frac{m_1 m_2}{m_1+m_2}\,g$. For two hanging bodies, $a = \frac{m_2-m_1}{m_1+m_2}\,g$ and $T = \frac{2m_1 m_2}{m_1+m_2}\,g$.
- The tension for an accelerating hanging body is not its weight: for the cart and the hanging mass $T < m_2 g$, and for two hanging bodies T lies between the weights.
- The tensions $\overline{T}_1$ and $\overline{T}_2$ act on different bodies and are not an action–reaction pair.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: A cart and a hanging mass

A cart of mass $1.5\,\mathrm{kg}$ stands on a table. A string tied to it passes over a light pulley at the edge of the table, and a mass of $0.60\,\mathrm{kg}$ hangs from the other end of the string. The cart is released from rest. Friction and air resistance are neglected, and the string is light and inextensible. What is the acceleration of the cart and what is the tension in the string?

STEP 1/8

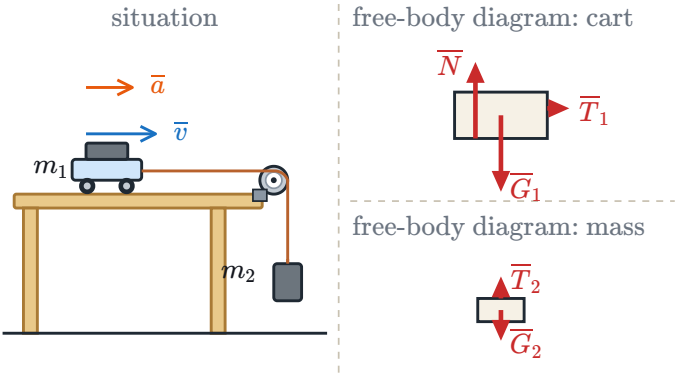
Given and find. $m_1 = 1.5\,\mathrm{kg}$, $m_2 = 0.60\,\mathrm{kg}$ and $g = 9.81\,\mathrm{m/s^2}$. We want the acceleration a and the tension T in the string.

STEP 2/8

Principle. The cart and the hanging mass are two bodies, so each gets its own equation of motion. The string does not stretch, so the accelerations of the bodies have the same magnitude a. The string is light and the pulley is light and frictionless, so the string pulls both bodies with the same force T. There is no friction, so in the horizontal direction only the string acts on the cart.

STEP 3/8

Free-body diagram. Each body has its own free-body diagram. The positive direction is to the right for the cart and down for the hanging mass. The cart does not accelerate vertically, so $N = G_1$. In the horizontal direction the cart is pulled only by $\overline{T}_1$. The hanging mass is acted on by $\overline{G}_2$ downward and $\overline{T}_2$ upward.



STEP 4/8

Rearranged formula. The two equations of motion are read from the diagrams and solved:

$$
\begin{aligned}
T &= m_1 a \\
m_2 g - T &= m_2 a \\
m_2 g &= (m_1 + m_2) a \\
a &= \frac{m_2 g}{m_1 + m_2} \\
T &= m_1 a
\end{aligned}
$$

STEP 5/8

Unit conversions. None are needed: the masses are in kilograms and $\frac{\mathrm{kg \cdot m/s^2}}{\mathrm{kg}} = \mathrm{m/s^2}$.

STEP 6/8

Substitution. The numerical values are substituted with their units:

$$\begin{aligned} a &= \frac{m_2 g}{m_1 + m_2} \\ &= \frac{0.60\,\mathrm{kg} \cdot 9.81\,\mathrm{m/s^2}}{1.5\,\mathrm{kg} + 0.60\,\mathrm{kg}} \\ &\approx 2.80\,\mathrm{m/s^2} \end{aligned}$$

$$\begin{aligned} T &= m_1 a = 1.5\,\mathrm{kg} \cdot 2.803\,\mathrm{m/s^2} \\ &\approx 4.20\,\mathrm{N} \end{aligned}$$

STEP 7/8

Result. The acceleration of the cart is $a \approx 2.8\,\mathrm{m/s^2}$ and the tension in the string is $T \approx 4.2\,\mathrm{N}$, with two significant figures like the masses $1.5\,\mathrm{kg}$ and $0.60\,\mathrm{kg}$.

STEP 8/8

Reasonableness. The weight of the hanging mass is $m_2 g \approx 5.9\,\mathrm{N}$, and the tension $4.2\,\mathrm{N}$ is smaller than that, as it should be. The net force on the hanging mass is $5.886\,\mathrm{N} - 4.204\,\mathrm{N} \approx 1.68\,\mathrm{N}$, and this equals

$$\begin{aligned} m_2 a &= 0.60\,\mathrm{kg} \cdot 2.803\,\mathrm{m/s^2} \\ &\approx 1.68\,\mathrm{N} \end{aligned}$$

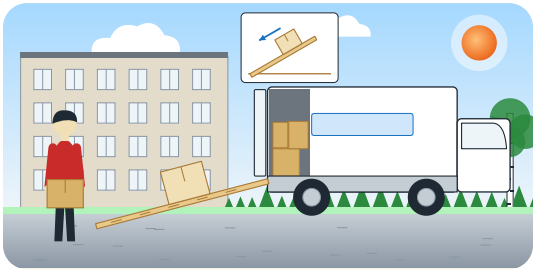
The acceleration is smaller than g, because the hanging mass also accelerates the cart. Without the cart, the mass would fall with acceleration g.

FY4 · CHAPTER 8

Weight, friction and the inclined plane

Why does a moving box stay in place on a slanted loading ramp, but start to slide when the ramp is made a little steeper?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy4-6.html`



IN THIS LESSON, YOU WILL LEARN TO

- distinguish static friction from kinetic friction and calculate the friction force using the coefficient of friction.
- draw the free-body diagram of an object on an inclined plane and write its equations of motion.
- determine the coefficient of friction experimentally.

PRIOR KNOWLEDGE

- Drawing a free-body diagram and the equations of motion $\Sigma F_x = ma_x$ and $\Sigma F_y = ma_y$
  chapter *Free-body diagrams and the combined effect of forces*
- The components $F\cos\alpha$ and $F\sin\alpha$ of an oblique force
  chapter *Free-body diagrams and the combined effect of forces*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 3.

Think about it

A mover lifts a box onto the loading ramp of a van. The box may be empty or packed full. It stays in place even though the ramp is slanted. When the lower end of the ramp is moved closer to the van, the ramp gets steeper. At some point the box suddenly starts to slide down.

> **What holds the box in place on a gentle ramp, and why is it no longer enough on a steep ramp?**

Theory

Key idea

Weight

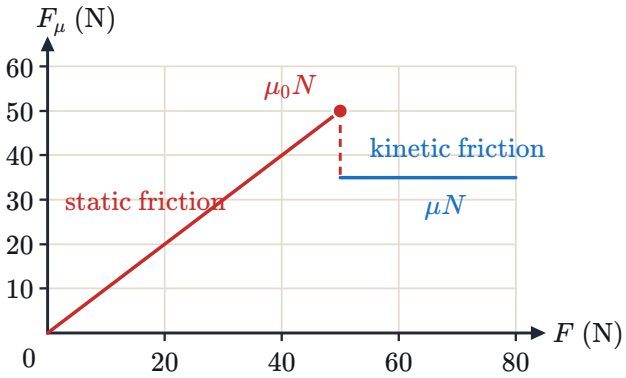
The $\overline{G}$ of an object is the force with which the Earth pulls the object. It always points straight down, also on a slope, and in a free-body diagram it is drawn from

the center of the object. The magnitude of the weight is the product of the mass and the acceleration of free fall, mg. Mass describes the inertia of an object, that is how hard it is to change its state of motion, and it is the same everywhere. Weight, on the other hand, depends on the place: on the surface of the Moon $g = 1.62\,\mathrm{m/s^2}$, so the weight of a $60\,\mathrm{kg}$ person there is about $97\,\mathrm{N}$, whereas on Earth it is about $590\,\mathrm{N}$.

Static friction and kinetic friction

When two surfaces touch each other, they resist sliding relative to each other. This force parallel to the surface is $\overline{F}_\mu$. The surfaces are not perfectly smooth: tiny bumps catch on each other, and there are attractive forces between the particles of the surfaces. The harder the surfaces are pressed against each other, that is the greater the normal force N is, the greater the friction can be.

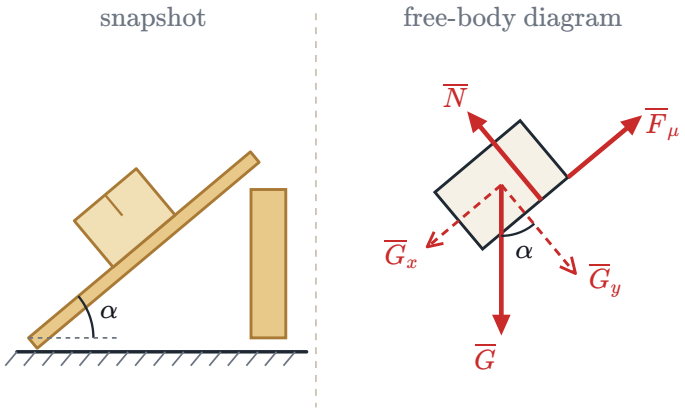
When a box on a floor is pulled only weakly, it does not move. Then acts on it, equal to the pull and in the opposite direction, so the net force is zero. Static friction is therefore not constant but adapts to the situation. It does have an upper limit, though. When the pull exceeds it, the box starts to slide, and the friction changes into . Kinetic friction is usually smaller than the greatest static friction, and at ordinary speeds it is approximately constant.



In the graph the normal force of the floor is $100\,\mathrm{N}$. Static friction grows with the pull up to $50\,\mathrm{N}$, where the box starts to move. A sliding box is acted on by a kinetic friction of $35\,\mathrm{N}$, even if the pull is increased. This is why a heavy cabinet is hardest to get moving, and once it is moving it goes more easily.

The inclined plane

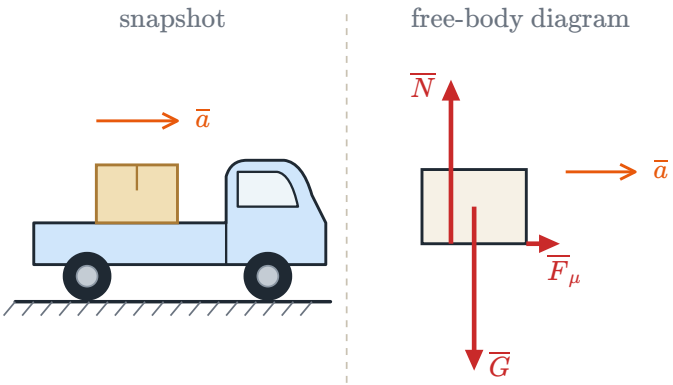
On an inclined plane, that is a slope, the weight still points straight down, but the surface is slanted. The normal force is perpendicular to the surface, and the friction is parallel to the surface. For calculation the weight is split into two **components**: the component along the slope tends to take the object down the slope, and the component perpendicular to the surface presses the object against the surface. The components are not new forces. They replace the weight in the calculation, and in a free-body diagram they are drawn as dashed lines.



The steeper the slope, the greater the component along the slope and the smaller the perpendicular component. When only the weight and the normal force act perpendicular to the surface and the object does not accelerate in that direction, the normal force decreases together with the perpendicular component, and with it the upper limit of static friction. This is why a box starts to slide on a steep slope: the component pulling it down the slope grows greater than the greatest static friction that the ramp can give.

Friction does not always oppose motion

Friction opposes the sliding of surfaces **relative to each other**, not necessarily the motion of the object. When a pickup truck starts to move with an acceleration, a box on its bed accelerates along with it. When air drag is ignored, the only horizontal force on the box is the friction of the bed, so it must point forward, in the direction of motion: the bed tends to slide forward from under the box, and static friction prevents this. Likewise, when we walk, the foot pushes the ground backward, and the static friction of the ground pushes the foot forward. On mirror-smooth ice, where there is very little friction, getting started is therefore difficult.



Besides friction, other contact forces can act on an object. is the force with which a liquid or a gas pushes upward an object immersed in it, and the is the force with which air or water opposes the motion of an object. Air drag is treated in more detail in the chapter *Air resistance and falling through air*. In this chapter air drag and buoyancy are ignored.

The friction force and the inclined plane

The magnitude of the weight is

$$G = mg$$

where G is the weight (N), m the mass (kg) and g the acceleration of free fall, $9.81\ \mathrm{m/s^2}$ on the surface of the Earth.

Kinetic friction is proportional to the normal force:

$$F_\mu = \mu N$$

where F_μ is the magnitude of the kinetic friction (N), μ the **coefficient of kinetic friction** (no unit) and N the magnitude of the normal force (N). Static friction is at most the product of the normal force and the **coefficient of static friction** μ_0:

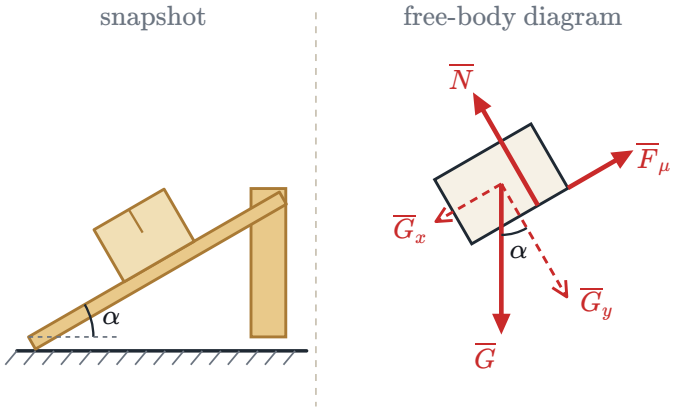
$$F_{\mu 0} \leq \mu_0 N$$

where $F_{\mu 0}$ is the magnitude of the static friction (N). The coefficients of friction depend on the materials and the condition of the surfaces, and usually $\mu < \mu_0$. The formulas are empirical models that describe dry surfaces well. In them the friction does not depend on the area of contact or on the sliding speed.

On an inclined plane whose angle of inclination is α, the x-axis is chosen along the slope and the y-axis perpendicular to the surface. The components of the weight are

$$G_x = G \sin \alpha$$
$$G_y = G \cos \alpha$$

where G_x is the component along the slope (N) and G_y the component perpendicular to the surface (N). The angle between the weight and the component G_y is the same α as the angle of inclination of the slope, because each is perpendicular to its counterpart: the weight to the horizontal and G_y to the slope.



If only the weight and the normal force act perpendicular to the surface and the object does not accelerate in that direction, the normal force equals G_y: $N = mg \cos \alpha$, which is less than the weight. (A pulling or pushing force directed toward or away from the surface would change the normal force.) The static friction acting on an object at rest equals G_x. The object stays in place as long as $mg \sin \alpha \leq \mu_0 \, mg \cos \alpha$, that is $\tan \alpha \leq \mu_0$. The mass cancels, so according to this model a heavy and a light box start to slide on the same surface at the same angle.



A child slides down a snowy slope on a plastic sled.

Photo: Michał Frąckowiak, CC BY-SA 2.0

Determining the coefficient of friction

In the diagrams and in the simulation, the static friction acting on an object at rest is denoted $\overline{F}_\mu$; its magnitude is $F_{\mu 0} \leq \mu_0 N$, and while sliding $F_\mu = \mu N$.

The coefficient of static friction is obtained from an experiment in which a board is tilted slowly until the object starts to slide. At this limiting angle α_0 the static friction is at its greatest, so $\mu_0 = \tan \alpha_0$. The coefficient of kinetic friction is obtained for example by pulling the object on a horizontal table with a spring balance, horizontally and at a constant velocity. Then the pull equals the kinetic friction and the normal force equals the weight, so $\mu = F : mg$. When the measurement is repeated with different masses, the coefficient of friction is the slope of the straight line through the origin fitted to the (N, F_μ) graph.

Common misconception

Many people think that the friction force is always $\mu_0 N$ or μN and always against the motion. Static friction, however, is only as large as the situation requires: if a box is pushed with a force of $20\ \mathrm{N}$ and it does not move, the static friction is $20\ \mathrm{N}$, even if its upper limit is greater. Friction opposes the sliding of surfaces relative to each other, so it can also point in the direction of motion, as on the bed of an accelerating truck. Another common mistake is to take the normal force to be always equal to the weight. On a slope $N = mg\cos\alpha$ is less than the weight,

when only the weight and the normal force act perpendicular to the surface and the object does not accelerate in that direction, and the weight still points straight down and not perpendicular to the slope.

Summary

- The magnitude of the weight is $G = mg$, and the weight always points straight down. Mass is the same everywhere, but weight depends on the acceleration of free fall.
- Friction is parallel to the surface and opposes the sliding of surfaces relative to each other, not always the motion of the object.
- Static friction adapts to the situation: $F_{\mu 0} \leq \mu_0 N$. A sliding object is acted on by kinetic friction $F_\mu = \mu N$, and usually $\mu < \mu_0$.
- On a slope the weight is split into the components $G_x = mg \sin \alpha$ and $G_y = mg \cos \alpha$, and the normal force is $N = mg \cos \alpha$, when only the weight and the normal force act perpendicular to the surface and the object does not accelerate in that direction. The components are drawn as dashed lines, and they are not new forces.
- The coefficient of static friction is obtained from the limiting angle at which the object starts to slide: $\mu_0 = \tan \alpha_0$.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: A box sliding down a ramp

A moving box of mass $18\ \mathrm{kg}$ slides down a loading ramp. The ramp is at an angle of 25° to the horizontal, and the coefficient of kinetic friction between the box and the ramp is 0.30. How large is the acceleration of the box, and how large is the friction force? Air drag is ignored.

STEP 1/8

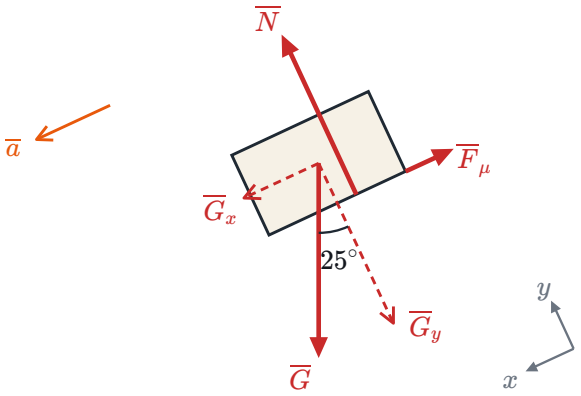
Given and asked. $m = 18\,\mathrm{kg}, \alpha = 25°, \mu = 0.30$ and $g = 9.81\,\mathrm{m/s^2}$. The acceleration a and the friction force F_μ are asked.

STEP 2/8

Principle. The box is chosen as the object. The Earth (weight) and the ramp (normal force and kinetic friction) act on it. The box slides down, so the kinetic friction points up the slope. Newton's second law is written along the slope, where the box accelerates, and perpendicular to the surface, where it does not accelerate: $a_y = 0$.

STEP 3/8

Free-body diagram. The box is drawn as a rectangle along the slope, and the forces start from their points of application. The axes are chosen so that x points down the slope, that is in the direction of the acceleration, and y perpendicular to the surface, outward. The weight is split into the components $G_x = mg\sin\alpha$ and $G_y = mg\cos\alpha$, drawn with a dashed line.



STEP 4/8

The formula solved. The equations of motion are read from the diagram and the asked quantities are solved:

$$\begin{aligned} x&:\ mg\sin\alpha - \mu N = ma \\ y&:\ N - mg\cos\alpha = 0 \\ F_\mu &= \mu mg\cos\alpha \\ a &= g(\sin\alpha - \mu\cos\alpha) \end{aligned}$$

STEP 5/8

Unit conversions. None are needed: all quantities are in SI units, and the coefficient of friction has no unit.

STEP 6/8

Substitution. The numerical values are substituted with their units:

$$\begin{aligned} F_\mu &= \mu mg\cos\alpha \\ &= 0.30 \cdot 18\,\mathrm{kg} \\ &\quad \cdot 9.81\,\mathrm{m/s^2} \cdot \cos 25^\circ \\ &\approx 48.0\,\mathrm{N} \end{aligned}$$

$$\begin{aligned} a &= g(\sin\alpha - \mu\cos\alpha) \\ &= 9.81\,\mathrm{m/s^2} \\ &\quad \cdot (\sin 25^\circ - 0.30 \cdot \cos 25^\circ) \\ &\approx 1.48\,\mathrm{m/s^2} \end{aligned}$$

STEP 7/8

Result. The acceleration of the box is $a \approx 1.5\,\mathrm{m/s^2}$ down the slope and the friction force is $F_\mu \approx 48\,\mathrm{N}$, to two significant figures like the data $\mu = 0.30$ and $m = 18\,\mathrm{kg}$.

STEP 8/8

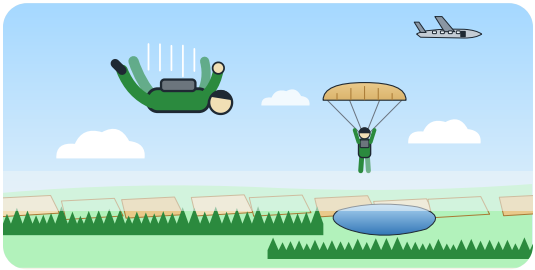
Reasonableness. Without friction the acceleration would be $g \sin 25^\circ \approx 4.1\,\mathrm{m/s^2}$, so friction reduces it, as it should. The friction force $48\,\mathrm{N}$ is less than the component along the slope $G_x \approx 75\,\mathrm{N}$, so the net force points down the slope and the box accelerates down. The mass does not affect the acceleration, because both G_x and F_μ are proportional to the mass.

FY4 · CHAPTER 9

Air resistance and falling through air

Why does a skydiver's speed stop increasing, even though the weight pulls the skydiver down throughout the whole fall?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy4-14.html`



IN THIS LESSON, YOU WILL LEARN TO

- explain why the speed of a falling object settles at a terminal velocity and how mass and shape affect it.
- draw the free-body diagram of a falling and a rising object when air resistance is not ignored.
- calculate the acceleration or the air resistance from the equation of motion, also when the net force points against the motion.

PRIOR KNOWLEDGE

- The net force points in the direction of the acceleration, and $\Sigma \overline{F} = m\overline{a}$ (the chapter *Newton's second law*)
- The weight is $G = mg$, and friction opposes sliding along a surface (the chapter *Weight, friction and the inclined plane*)

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 4.

Think about it

A skydiver steps out of an aeroplane at a height of about four kilometres. During the first seconds the speed grows quickly, but after about ten seconds the reading of the wrist computer almost stops growing and settles at about $200\ \mathrm{km/h}$, although the skydiver still falls for tens of seconds. When the parachute opens, the speed decreases in a few seconds to below $20\ \mathrm{km/h}$.

> **Why does the speed stop growing, although the weight pulls the skydiver down all the time, and what happens to the forces when the parachute opens?**

Theory

Key idea

Air resists motion

When an object moves through air or water, it has to push the medium out of its way. The medium pushes back on the object: the object experiences **resistance of the medium**, in air (also called **drag**), which this book writes as $\overline{F}_{\mathrm{v}}$. It points against the velocity of the object when the air is still. In the chapter *Weight, friction and the inclined plane*, friction resisted sliding along a surface. Air resistance is a similar resistive force, but it is caused by the surrounding air, and the object does not need to touch any solid surface. A bicycle and a car also have **rolling resistance** $\overline{F}_{\mathrm{r}}$, the resistance caused by the tyres and the road, which is a different force from air resistance.



NASA's EPIC test in California in June 2025: a drone dropped a parachute fitted with instruments and its payload. There is no person in the photo, only a test body.

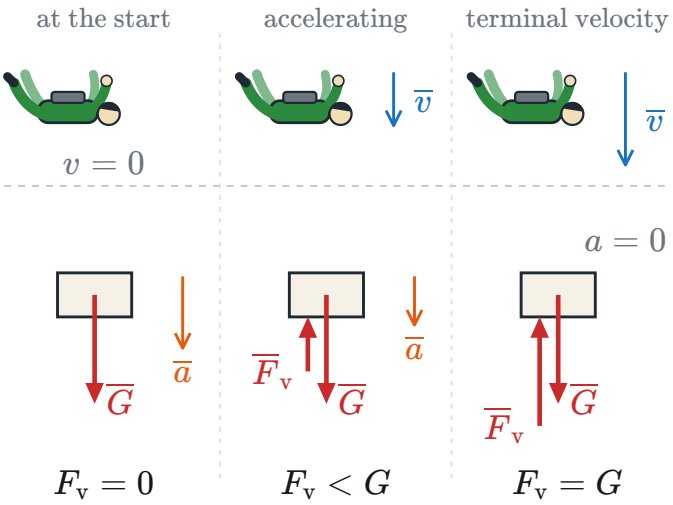
Photo: NASA/Christopher LC Clark, public domain (NASA)

Air resistance differs from sliding friction in that it depends on speed. Air resistance does not act at all on an object that is at rest relative to the air, and it

grows quickly when the speed of the object relative to the air grows. At the same speed, air resistance is greater the larger the cross-sectional area of the object and the worse the air can flow past it. At the same speed, the resistance in water is much greater than in air, because water is denser.

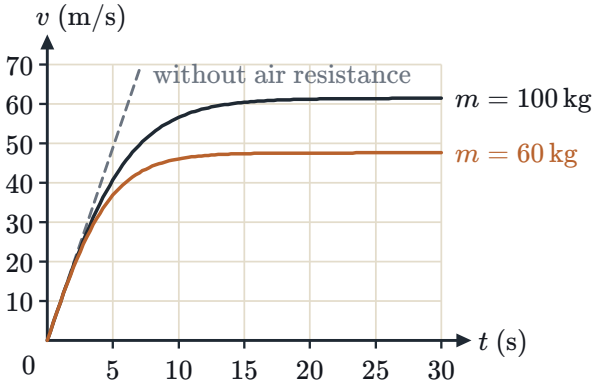
The three stages of a fall

The skydiver starts from rest relative to the air (the speed of the plane is ignored), so right after the jump there is no air resistance yet. The only force is the weight, and the skydiver accelerates downward with the free-fall acceleration g. As the speed grows, air resistance grows. The weight stays the same, so the net force becomes smaller, and the acceleration becomes smaller with it: the speed grows more and more slowly. Finally the air resistance is equal to the weight. Then the net force is zero, the acceleration is zero and the speed stays constant, as Newton's first law says. This speed is called the .



Terminal velocity depends on mass and shape

At terminal velocity, air resistance is equal to the weight. A heavier skydiver has a greater weight, so the speed has time to grow greater before air resistance has grown to the size of the weight. In the graph, the curves of a $60\ \mathrm{kg}$ and a $100\ \mathrm{kg}$ skydiver falling in the same position start equally steep, but they level off at different speeds, in the values of the model at about $48\ \mathrm{m/s}$ and $61\ \mathrm{m/s}$. A skydiver diving head first has a smaller cross-sectional area and a more streamlined shape, so the terminal velocity is greater.



This gives the answer to the question at the start. The weight pulls the skydiver down all the time, but as the speed grows, air resistance grows until it cancels the weight. After that the speed no longer changes. Without air resistance, all objects would fall with the same acceleration g. In air, a light and wide object, such as a sheet of paper or a feather, reaches a small terminal velocity very soon and therefore falls slowly.

Equation of motion with air resistance

Air resistance is one of the forces acting on an object, so the motion follows Newton's second law. An object falling or rising through air is acted on by its weight and by air resistance:

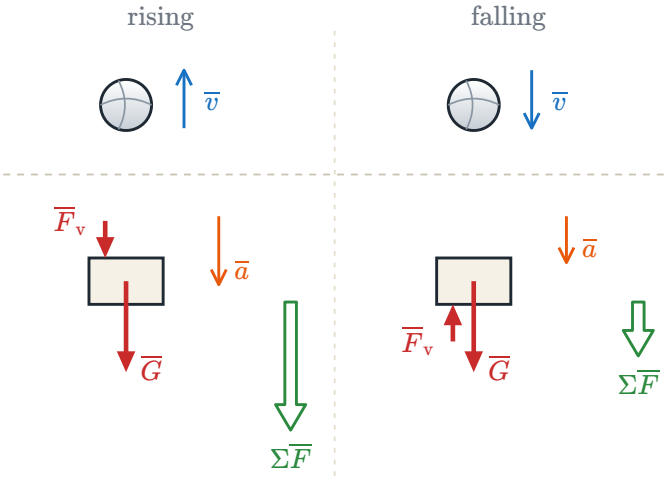
$$\overline{G} + \overline{F}_{\mathrm{v}} = m\overline{a}$$

where $\overline{G}$ is the weight of the object (N), $\overline{F}_{\mathrm{v}}$ is the air resistance (N), m is the mass of the object (kg) and $\overline{a}$ is its acceleration ($\mathrm{m/s^2}$). In this chapter the magnitude of air resistance is not calculated from the speed, but it is given or deduced from the motion: at terminal velocity $F_{\mathrm{v}} = G$.

A falling and a rising object

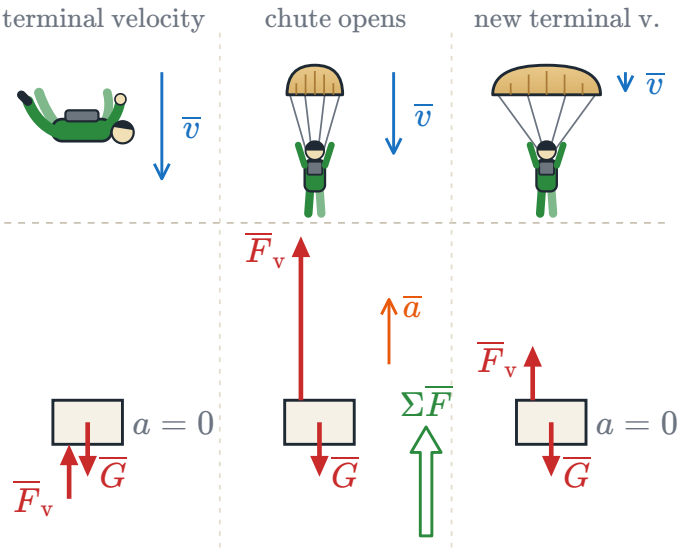
As always, first choose the object, draw its free-body diagram and choose the positive direction, and only then write the equation. The air resistance of a falling object points up, against the weight. When the positive direction is down, $mg - F_{\mathrm{v}} = ma$, so the acceleration is smaller than g. The air resistance of a ball thrown upward points down, in the same direction as the weight. When the positive direction is up, $-mg - F_{\mathrm{v}} = ma$. The magnitude of the acceleration is

then greater than g, and the rising ball slows down faster than without air resistance.



When the parachute opens

The large area of the parachute makes air resistance many times greater. When the parachute opens, the skydiver is still moving downward at terminal velocity, so for a moment the air resistance is many times the weight. The net force and the acceleration point up, even though the skydiver is moving down: the skydiver slows down but does not rise. As the speed decreases, air resistance decreases until, at a new, small terminal velocity, it is again equal to the weight. How many times the weight the air resistance is depends on the parachute and the moment: in the diagram it is three times, in the example almost three times, and in the model of the simulation at most about five times. The direction of the net force can therefore differ from the direction of motion. It always tells the direction of the acceleration, not of the velocity.



Common misconception

In videos, the skydiver seems to rise upward when the parachute opens. The camera operator is still falling without a parachute, and the skydiver only slows down relative to the camera operator. Relative to the ground, the skydiver moves downward all the time: the net force points up for a moment, but it decreases the speed and does not turn the motion around. Another mistake is to think that at terminal velocity air resistance is greater than the weight or that there are no forces any more. At terminal velocity the forces are equal and opposite. A third mistake is to claim that a heavy object always falls faster. Without air resistance, everything falls with the same acceleration. In air, of objects of the same size and shape the heavier one reaches the greater terminal velocity.

Summary

- Resistance of the medium, in air air resistance $\overline{F}_\mathrm{v}$, points against the velocity and grows when the speed grows. It is greater with a large cross-sectional area and a poorly streamlined shape.
- The acceleration of a falling object decreases as air resistance grows. At terminal velocity $F_\mathrm{v} = G$, the net force is zero and the speed stays constant.
- Of objects of the same size and shape, the heavier one has the greater terminal velocity. A parachute lowers the terminal velocity to a safe one.
- The equation of motion is $\overline{G} + \overline{F}_\mathrm{v} = m\overline{a}$. For a rising object, air resistance points down together with the weight, so the deceleration is greater than g.

- When the parachute opens, the net force and the acceleration point up even though the skydiver moves down: the direction of the net force can differ from the direction of motion.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: A parachute opening

A skydiver whose mass with equipment is $85.0\,\mathrm{kg}$ opens the parachute. At one moment while the parachute is opening, the air resistance is $2.40\,\mathrm{kN}$, and the skydiver is still moving downward. What is the acceleration of the skydiver at this moment, and in which direction does it point?

STEP 1/7

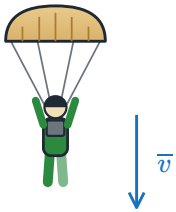
Given and find. $m = 85.0\,\mathrm{kg}$, $F_{\mathrm{v}} = 2.40\,\mathrm{kN}$ and $g = 9.81\,\mathrm{m/s^2}$. We want the acceleration a and its direction.

STEP 2/7

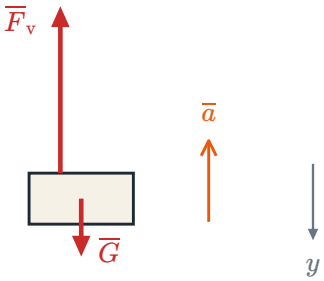
Principle. Choose the skydiver with the equipment and the parachute as the object. The Earth (weight) and the air (air resistance) act on it. By Newton's second law, the acceleration is determined by the sum of these forces; the direction of the velocity does not decide the direction of the acceleration.

STEP 3/7

Rearranged formula. In the free-body diagram the skydiver is a rectangle. The weight $\overline{G}$ starts from the centre and points down. Air resistance $\overline{F}_{\mathrm{v}}$ acts mainly on the parachute, so it starts from the top surface and points up. The positive direction is chosen down, the direction of motion.



situation



free-body diagram

From the diagram we write the net force $mg - F_{\mathrm{v}}$ and solve the acceleration:

$$mg - F_{\mathrm{v}} = ma$$
$$a = \frac{mg - F_{\mathrm{v}}}{m}$$

STEP 4/7

Unit conversions. Air resistance is converted to newtons:

$$\begin{aligned} 2.40\,\mathrm{kN} &= 2.40 \cdot 1000\,\mathrm{N} \\ &= 2400\,\mathrm{N} \end{aligned}$$

STEP 5/7

Substitution. First calculate the weight and then the acceleration:

$$a = \frac{mg - F_{\mathrm{v}}}{m}$$
$$\begin{aligned} mg &= 85.0\,\mathrm{kg} \cdot 9.81\,\tfrac{\mathrm{m}}{\mathrm{s}^2} \\ &= 833.85\,\mathrm{N} \end{aligned}$$
$$\begin{aligned} a &= \frac{833.85\,\mathrm{N} - 2400\,\mathrm{N}}{85.0\,\mathrm{kg}} \\ &\approx -18.4\,\mathrm{m/s^2} \end{aligned}$$

STEP 6/7

Result. The acceleration is negative, so it points up, opposite to the direction chosen as positive. The magnitude of the acceleration is $18.4\ \mathrm{m/s^2}$ upward, given with three significant figures like the mass $85.0\ \mathrm{kg}$ and the air resistance $2.40\ \mathrm{kN}$. The skydiver moves down, but the speed decreases.

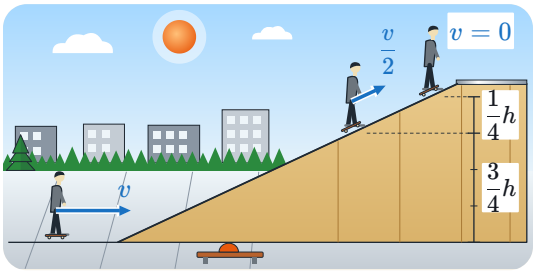
STEP 7/7

Reasonableness. Air resistance is almost three times the weight $834\ \mathrm{N}$, so about two weights of net force remain upward. The acceleration must therefore be about two times g, or about $20\ \mathrm{m/s^2}$, and the result $18.4\ \mathrm{m/s^2}$ fits this. To the skydiver, the opening of the parachute feels like a strong jerk, but it lasts only a few seconds.

Kinetic energy, potential energy and mechanical energy

Why does a skateboarder who is twice as fast rise four times as high on a ramp?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy4-7.html`



IN THIS LESSON, YOU WILL LEARN TO

- calculate the kinetic energy, potential energy and mechanical energy of an object relative to a chosen zero level.
- explain why kinetic energy grows with the square of the speed and why only the change in potential energy matters.
- identify conservative forces and explain why no potential energy is associated with friction.

PRIOR KNOWLEDGE

- An object thrown straight up at speed v rises to a height $\frac{v^2}{2g}$ when air resistance is small
  chapter *Acceleration and uniformly accelerated motion*
- The weight of an object is $G = mg$
  chapter *Weight, friction and the inclined plane*
- Energy changes from one form to another and its unit is the joule
  chapter *Energy transformations and the joule*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 1.

Think about it

A skateboarder rolls along level ground and goes up a straight ramp until she stops for a moment and starts back down. Friction and air resistance are small. When her speed at the foot of the ramp is $3.0\ \mathrm{m/s}$, she rises to a height of about $0.46\ \mathrm{m}$. When the speed is doubled, $6.0\ \mathrm{m/s}$, she rises to about $1.8\ \mathrm{m}$, that is, four times as high.

> **What grows so much in the skateboarder's motion that doubling the speed is enough for four times the rise?**

Theory

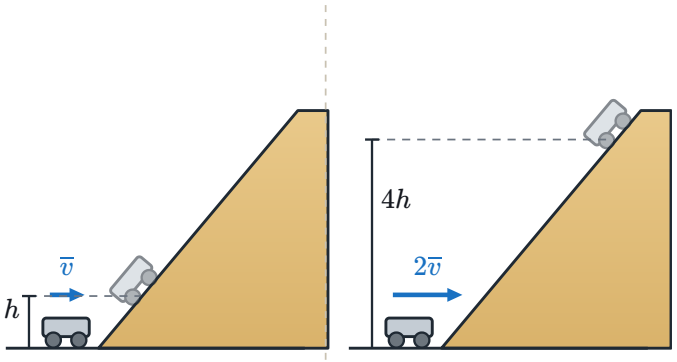
Key idea

Height stores energy

In the chapter *Forms of energy and conservation of energy*, potential energy and kinetic energy were forms of energy. Now we calculate how much of them there is. When an object is lifted higher, energy is transferred to it: a book lifted onto a shelf can gain speed when it falls, and water pumped uphill can turn a turbine. This energy tied to the position of an object is called . Gravitational potential energy is greater the greater the mass of the object and the higher it is. Lifting two identical books onto the same shelf takes twice as much energy as lifting one, and lifting by two metres takes twice as much as lifting by one metre. The height is measured from a chosen **zero level**, for example the floor.

Kinetic energy grows with the square of the speed

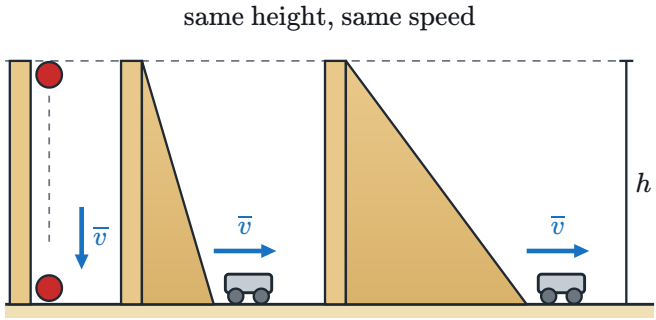
A moving object has . When a cart goes up a frictionless ramp, it slows down, and its kinetic energy changes into potential energy. The cart rises as high as a ball thrown straight up at the same speed in the chapter *Acceleration and uniformly accelerated motion*, that is, to the height $h = \frac{v^2}{2g}$: on the ramp the acceleration is smaller, but the distance is correspondingly longer. At double the speed the cart therefore rises four times as high, and that takes four times as much energy. **Kinetic energy is proportional to the square of the speed.** The mass does not change the rise height, but lifting a cart that is twice as heavy to the same height takes twice the energy. Kinetic energy is therefore also proportional to the mass.



The square of the speed shows in traffic. When the speed of a car grows from $50\ \mathrm{km/h}$ to $70\ \mathrm{km/h}$, the kinetic energy grows by the factor $\left(\frac{70}{50}\right)^2 \approx 2.0$, that is, almost doubles. This is exactly the energy that braking or a collision has to change into other energy.

A conservative force returns the energy

When a frictionless cart has risen to its highest point on the ramp, it starts back down and is at the foot of the ramp as fast as when it started. Gravity returns as kinetic energy all the energy that changed into potential energy in the rise. In addition, a frictionless object that starts from rest at a height h gets the same speed on the way down, whether it falls straight down or slides down a steep or a gentle ramp: the energy depends only on height, not on the path. A force with these properties is called a . Potential energy is always associated with a conservative force.



Gravitational potential energy is not the only kind of potential energy. A stretched spring, a drawn bow and a compressed trampoline return the energy as kinetic energy: the spring force is conservative, and a stretched spring has **spring potential energy**. The force between electric charges is also conservative. Friction and air resistance, on the other hand, are not conservative. They change kinetic energy into the internal energy of the objects and the air, and this energy does not return as motion even if the object returns to its starting place. That is why no potential energy is associated with friction.

Mechanical energy

The kinetic energy and the potential energy of an object together are its . When a cart goes up a frictionless ramp, the kinetic energy decreases and the potential

energy grows by the same amount, so the mechanical energy stays the same. When friction acts, part of the mechanical energy changes into internal energy, and the mechanical energy decreases. In the chapter *Conservation of mechanical energy and the work–energy principle* this becomes a tool for calculation.

Kinetic energy and potential energy

Gravitational potential energy is

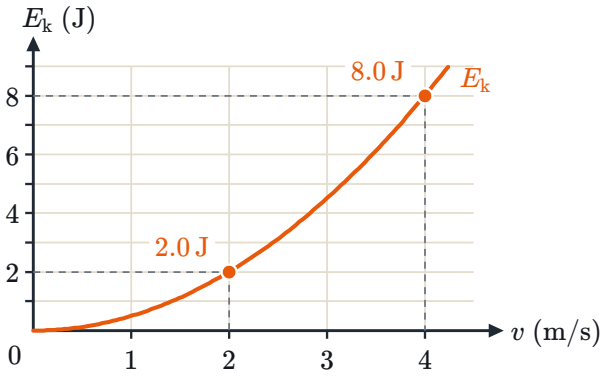
$$E_{\mathrm{p}} = mgh$$

where E_{p} is the potential energy (J), m the mass (kg), $g = 9.81\ \mathrm{m/s^2}$ the acceleration of free fall and h the height from the chosen zero level (m). The expression holds near the surface of the Earth, where g is constant. Below the zero level $h < 0$, so the potential energy is negative. The unit is $\mathrm{kg \cdot m/s^2 \cdot m = N \cdot m = J}$.

Kinetic energy is

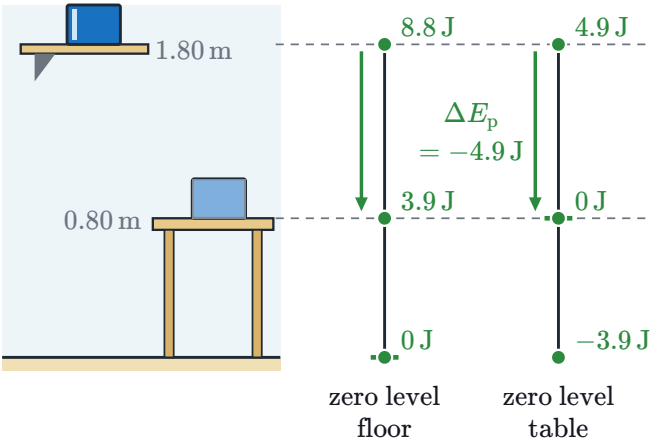
$$E_{\mathrm{k}} = \tfrac{1}{2}\, mv^2$$

where E_{k} is the kinetic energy (J), m the mass (kg) and v the speed $(\mathrm{m/s})$. The expression is obtained from the rise height on a ramp. A frictionless cart rises to the height $h = \frac{v^2}{2g}$, and its potential energy then grows by $mgh = mg \cdot \frac{v^2}{2g} = \frac{1}{2}\, mv^2$. That is exactly how much kinetic energy the cart had at the foot of the ramp. Kinetic energy is never negative, and it does not depend on the direction of motion.



The zero level is chosen

You may choose the zero level freely, as long as it stays the same throughout the calculation. When a $0.50\ \mathrm{kg}$ book is on a shelf $1.80\ \mathrm{m}$ above the floor, its potential energy is $8.8\ \mathrm{J}$ relative to the floor but only $4.9\ \mathrm{J}$ relative to the surface of a table $0.80\ \mathrm{m}$ high. When the book is moved from the shelf to the table, the potential energy decreases by the same amount in both cases, $4.9\ \mathrm{J}$. **Only the change in potential energy has physical significance**, and it does not depend on the zero level.



Mechanical energy is the sum of kinetic and potential energy:

$$E_{\mathrm{mech}} = E_{\mathrm{k}} + E_{\mathrm{p}}$$

where E_{mech} is the mechanical energy (J). Because potential energy depends on the zero level, the value of the mechanical energy depends on it too. Always state in a calculation what the zero level is.



A roller coaster car at the top of a loop on a sunny day.

Photo: Tony Rojas, Pexels License

Common misconception

Many people think that double the speed means double the kinetic energy. Kinetic energy, however, grows with the square of the speed: double the speed gives four times and triple the speed nine times the kinetic energy. Another mistake is to treat potential energy as a property of the object itself with one correct value. Its value depends on the chosen zero level, and only the change in potential energy is the same whatever the zero level. A third mistake is to think that potential energy is associated only with gravity. Potential energy is also associated with the spring force, but not with friction, because friction does not return the energy as motion.

Summary

- Kinetic energy $E_\mathrm{k} = \frac{1}{2}\,mv^2$ is proportional to the mass and the square of the speed: double the speed gives four times the kinetic energy.
- Gravitational potential energy $E_\mathrm{p} = mgh$ depends on the height above the chosen zero level. Below the zero level it is negative, and only its change has physical significance.
- Potential energy is associated with a conservative force, such as gravity or the spring force, which returns the energy as kinetic energy and whose energy depends only on position, not on the path. Friction and air resistance are not conservative.
- Mechanical energy is $E_\mathrm{mech} = E_\mathrm{k} + E_\mathrm{p}$. When there is no friction or air resistance and the object is not pushed or pulled, it stays the same; friction and air resistance change part of it into internal energy.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: A snowboarder's energy

A snowboarder whose mass with equipment is $68\ \mathrm{kg}$ goes down a slope. At a point $25\ \mathrm{m}$ above the bottom of the slope her speed is $12\ \mathrm{m/s}$. How great is the snowboarder's mechanical energy at this point when the bottom of the slope is chosen as the zero level?

STEP 1/7

Given and find. Mass $m = 68\ \mathrm{kg}$, speed $v = 12\ \mathrm{m/s}$, height above the zero level $h = 25\ \mathrm{m}$ and acceleration of free fall $g = 9.81\ \mathrm{m/s^2}$. We are asked for the mechanical energy E_{mech}.

STEP 2/7

Principle. Mechanical energy is the sum of kinetic energy and potential energy. Kinetic energy is calculated from the speed and potential energy from the height above the zero level, that is, the bottom of the slope. With such a small difference in height the acceleration of free fall is constant, so $E_{\mathrm{p}} = mgh$ holds.

STEP 3/7

Formula solved.

$$E_{\mathrm{mech}} = E_{\mathrm{k}} + E_{\mathrm{p}}$$
$$E_{\mathrm{mech}} = \tfrac{1}{2}mv^2 + mgh$$

STEP 4/7

Unit conversions. All quantities are in SI units, so no conversions are needed. Both terms come in the unit $\mathrm{kg \cdot m^2/s^2} = \mathrm{J}$.

STEP 5/7

Substitution. First we calculate each energy and then their sum:

$$
\begin{aligned}
E_{\mathrm{k}} &= \tfrac{1}{2}\, m v^{2} \\
&= \tfrac{1}{2} \cdot 68\,\mathrm{kg} \cdot (12\,\mathrm{m/s})^{2} \\
&= 4896\,\mathrm{J}
\end{aligned}
$$

$$
\begin{aligned}
E_{\mathrm{p}} &= m g h \\
&= 68\,\mathrm{kg} \cdot 9.81\,\mathrm{m/s^{2}} \cdot 25\,\mathrm{m} \\
&\approx 16\,677\,\mathrm{J}
\end{aligned}
$$

$$
\begin{aligned}
E_{\mathrm{mech}} &= 4896\,\mathrm{J} + 16\,677\,\mathrm{J} \\
&\approx 21\,573\,\mathrm{J}
\end{aligned}
$$

STEP 6/7

Result. The snowboarder's mechanical energy is about $22\,\mathrm{kJ}$ relative to the bottom of the slope. Two significant figures, like the mass $68\,\mathrm{kg}$, the speed $12\,\mathrm{m/s}$ and the height $25\,\mathrm{m}$.

STEP 7/7

Reasonableness. With rounded numbers

$$
\begin{aligned}
E_{\mathrm{k}} &\approx \tfrac{1}{2} \cdot 70\,\mathrm{kg} \cdot 150\,\mathrm{m^{2}/s^{2}} \\
&\approx 5\,\mathrm{kJ}
\end{aligned}
$$

$$
\begin{aligned}
E_{\mathrm{p}} &\approx 70\,\mathrm{kg} \cdot 10\,\mathrm{m/s^{2}} \cdot 25\,\mathrm{m} \\
&= 17.5\,\mathrm{kJ}
\end{aligned}
$$

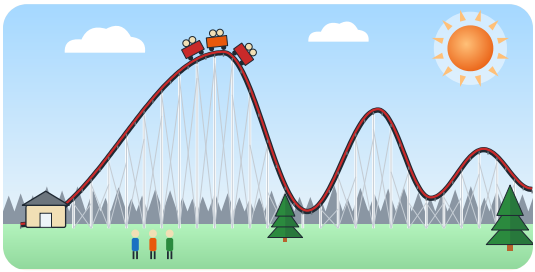
together about $23\,\mathrm{kJ}$, close to the result. Both energies are positive, because the snowboarder is moving and is above the zero level.

FY4 · CHAPTER 11

Conservation of mechanical energy and the work–energy principle

Why is the first hill of a roller coaster always the highest of the track, even though the train is no longer pulled up after it?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy4-8.html`



IN THIS LESSON, YOU WILL LEARN TO

- explain when mechanical energy is conserved and what it turns into when friction or air resistance does work.
- calculate a speed or a height using the law of conservation of mechanical energy.
- calculate the work done by forces other than weight using the work–energy principle.

PRIOR KNOWLEDGE

- Mechanical energy is the sum of kinetic energy and potential energy, $E_\mathrm{k} = \frac{1}{2}mv^2$ and $E_\mathrm{p} = mgh$
  chapter *Kinetic energy, potential energy and mechanical energy*
- A force does work when an object moves in the direction of the force, and work transfers energy
  chapter *Work and energy*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 5.

Think about it

A roller coaster train leaves the station, and a chain lift slowly pulls it to the top of the first hill, at a height of $40\ \mathrm{m}$. After that nothing pulls the train any more. The train plunges into the valley, speeds over the next hill and the one after that, and finally rolls to the brakes at the station. In the drawings of the track each hill is lower than the one before.

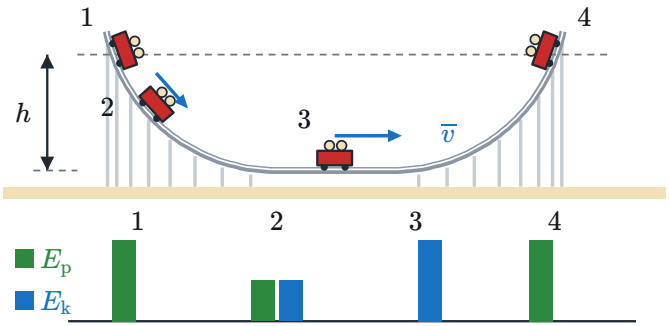
> **Where does the train get its speed without an engine, and why can no later hill be as high as the first?**

Theory

Key idea

Mechanical energy changes its form

In the chapter *Kinetic energy, potential energy and mechanical energy* you learned that the mechanical energy of an object is the sum of its kinetic energy and potential energy, $E_{\text{mech}} = E_{\text{k}} + E_{\text{p}}$. When a roller coaster car rolls into a valley, its height decreases and its speed increases: potential energy turns into kinetic energy. When the car climbs the opposite slope, kinetic energy turns back into potential energy. If there is no friction or air resistance, the car rises exactly to its starting height and stops there for a moment. The mechanical energy is then equal at every moment, even though its form changes. This is called .

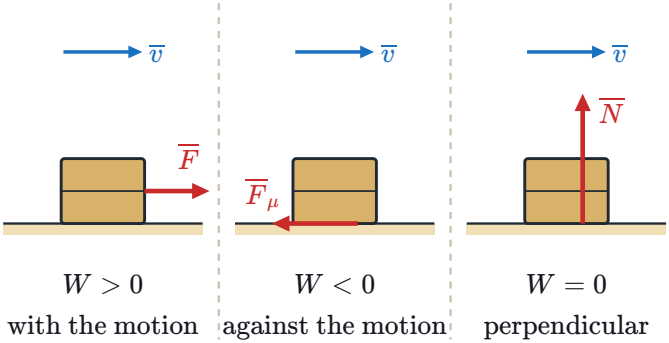


Not all the forces acting on the car change its energy. The weight does work when the car goes down or up, and it is exactly this work that turns potential energy into kinetic energy and back. The normal force of the track is perpendicular to the surface of the track and therefore also to the motion, so it does no work. That is why the mechanical energy is conserved when only the weight does work, however twisting the track is. The speed of the car at any point then depends only on how much lower it has descended from the starting point, not on the shape of the track or the mass of the car.

Work transfers energy

In the chapter *Work and energy* you learned that a force does work when an object moves in the direction of the force. Now we also need the sign of the work. A force

does positive work when it has a component in the direction of the motion: it increases the energy of the object. A force that opposes the motion, such as friction, does negative work: it decreases the energy of the object. A force perpendicular to the motion, such as the normal force, does no work at all. The unit of work is the same as that of energy, the joule.



Friction turns mechanical energy into internal energy

In a real roller coaster the rolling resistance of the wheels and the air resistance oppose the motion the whole way. They do negative work, and the mechanical energy of the car decreases by exactly this work. Energy does not disappear but turns into internal energy of the wheels, the rails and the air, which warm up slightly. The total energy is always conserved. Mechanical energy is conserved only when there is no friction or air resistance, or when they are so small that they can be ignored.

You remember the difference between the weight, that is gravity, and friction from the chapter *Kinetic energy, potential energy and mechanical energy*: the weight is a . What is new is what follows from this for calculating energy: the work done by the weight depends only on the difference in height, so when an object returns to the same height, it gets back the same energy, whatever route it took. That is why a potential energy can be associated with the weight. The negative work done by friction and air resistance grows with every metre travelled, so it depends on the route. For such a **non-conservative force** a potential energy cannot be defined.

Answer to the opening question

A chain lift lifts the train to the top of the first hill. After that the train has only the mechanical energy it got during the lift, and nothing adds to it. The resistive forces decrease it with every metre. If the next hill were as high as the first, the train would not reach its top. Every later hill must be lower than the first and lower than the

height that the remaining energy would reach, so that the train still has some kinetic energy at the top.

the work–energy principle

Work and kinetic energy

When an object moves in a straight line in one direction a distance s and a constant force $\overline{F}$ acts on it, the work done by the force is

$$W = F_{\parallel}\, s$$

where $F_{\parallel}$ is the component of the force in the direction of the motion (N), negative when the component points against the motion, and s is the distance travelled (m). The unit of work is $1\,\mathrm{N}\cdot\mathrm{m} = 1\,\mathrm{J}$.

When an object moves along the x-axis and a constant net force ΣF_x along the axis acts on it, it accelerates uniformly, and its acceleration is $a = \frac{\Sigma F_x}{m}$. In the chapter *Acceleration and uniformly accelerated motion* you learned that in uniformly accelerated motion the acceleration is $a = \frac{v_2 - v_1}{t}$ and the displacement is the average velocity times the time, $\Delta x = \frac{v_1 + v_2}{2}\, t$. Here the velocities v_1 and v_2 are signed: a negative velocity means motion in the negative direction of the axis. When these equations are multiplied together, the time cancels:

$$\begin{aligned} a\,\Delta x &= \frac{v_2 - v_1}{t} \cdot \frac{v_1 + v_2}{2}\, t \\ &= \tfrac{1}{2}\left(v_2^2 - v_1^2\right) \end{aligned}$$

When the equation is multiplied on both sides by the mass m and $ma = \Sigma F_x$ is substituted, we get $\Sigma F_x\,\Delta x = \tfrac{1}{2}mv_2^2 - \tfrac{1}{2}mv_1^2$, that is, $\Sigma F_x\,\Delta x = \Delta E_{\mathrm{k}}$.

The displacement Δx is not the same as the distance travelled s. If the object moves in one direction all the time, the magnitude of the displacement equals the distance travelled, and the work of the net force is $\Sigma F_{\parallel}\, s$. If the object turns back, like a ball thrown upwards, the distance travelled is longer than the displacement. The work of a constant force is then calculated from the displacement, $W = F_x\,\Delta x$, or in parts so that on each part the motion is in one direction. Friction and air resistance, on the other hand, turn with the motion, so their work is always negative and its magnitude grows with the distance travelled. In both cases the work done by the net force equals the change in kinetic energy:

$$W_{\text{net}} = \Delta E_{\text{k}}$$

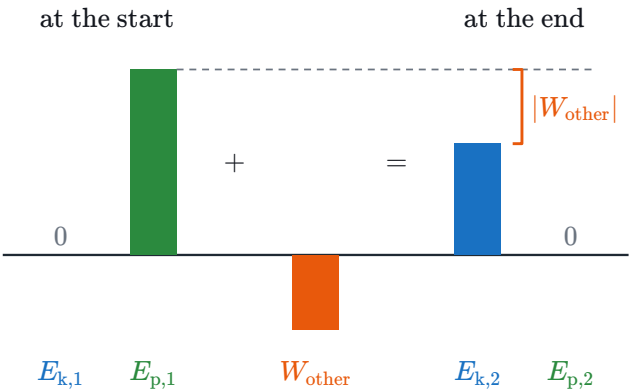
where W_{net} is the work done together by all the forces acting on the object (J) and $\Delta E_{\text{k}} = E_{\text{k},2} - E_{\text{k},1}$ is the change in kinetic energy (J). For example, when the mass of a cart is $2.0\ \text{kg}$ and a net force of $12\ \text{N}$ accelerates it from rest over a distance of $3.0\ \text{m}$, the force does the work $36\ \text{J}$, and the speed of the cart at the end is $6.0\ \text{m/s}$.

The work–energy principle

The total work is divided into two parts: the work done by the weight W_G and the work done by all the other forces W_{other}. Other forces include, for example, friction, air resistance, a pulling rope and a pushing hand. When an object descends a height h, the weight does the work mgh and the potential energy decreases by the same amount, so $W_G = -\Delta E_{\text{p}}$. Substituting this into the equation $W_G + W_{\text{other}} = \Delta E_{\text{k}}$ gives the :

$$\begin{aligned} &E_{\text{k},1} + E_{\text{p},1} + W_{\text{other}} \\ &= E_{\text{k},2} + E_{\text{p},2} \end{aligned}$$

where the index 1 means the initial situation and 2 the final situation, $E_{\text{k}} = \tfrac{1}{2}mv^2$ is the kinetic energy, $E_{\text{p}} = mgh$ is the potential energy at a height h from the chosen zero level, and W_{other} is the work done by forces other than the weight (J). In words: the change in mechanical energy equals the work done by forces other than the weight, $\Delta E_{\text{mech}} = W_{\text{other}}$. The zero level can be chosen freely, for example at the lowest point of the situation, because only the change in height affects the result.



When an object moves against a stationary surface or air, friction and air resistance oppose the motion and do negative work, so they decrease the mechanical energy. (A moving surface or the wind can also do positive work, for example a conveyor belt carrying a box.) A pulling rope or a motor can do positive work and increase it, like the chain lift that lifts the train. If the other forces do no work, $W_{\text{other}} = 0$, and the work–energy principle becomes the **law of conservation of mechanical energy**:

$$E_{\text{k},1} + E_{\text{p},1} = E_{\text{k},2} + E_{\text{p},2}$$

When an object starts from rest at a height h and descends to the zero level, the conservation law $mgh = \frac{1}{2}mv^2$ gives $v = \sqrt{2gh}$. The mass cancels, so the speed does not depend on the mass.

Common misconception

Many people think that a heavier object is faster at the bottom of a hill. When friction and air resistance can be ignored, the mass cancels from the conservation law: a full and an empty car are equally fast at the bottom of the valley. The heavier car has more kinetic energy at the bottom, but it also had more potential energy at the start.

Another mistake is to say that friction destroys energy. Friction decreases the mechanical energy, but an equal amount of energy turns into internal energy, and the total energy is conserved. A third mistake is to calculate the work of friction from the difference in height. The work of friction is the frictional force times the distance travelled along the track. When the resistive force is the same, on a long gentle track more mechanical energy turns into internal energy than on a short steep one, even if the difference in height is the same.

Summary

- The mechanical energy $E_{\text{mech}} = E_{\text{k}} + E_{\text{p}}$ is conserved when only the weight does work, that is, when friction and air resistance can be ignored and nothing pulls or pushes. The normal force does no work, because it is perpendicular to the motion.
- A force does work with its component in the direction of the motion, $W = F_{\parallel} s$ when the object moves in one direction: positive in the direction of the motion and negative against it.
- The work done by the net force equals the change in kinetic energy, $W_{\text{net}} = \Delta E_{\text{k}}$.
- The work–energy principle: the change in mechanical energy is the work done by forces other than the weight, $\Delta E_{\text{mech}} = W_{\text{other}}$. Friction and air resistance turn mechanical energy into internal energy, and the total energy is conserved.
- The weight is a conservative force, because its work depends only on the difference in height. Friction and air resistance are non-conservative forces, because their work grows with the distance travelled.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Energy on a sledging hill

A child slides down a hill on a sled. The combined mass of the child and the sled is $32\ \text{kg}$. The sled starts from rest at the top of the hill, which is $5.0\ \text{m}$ higher than the foot of the hill. At the foot of the hill the speed of the sled is measured to be $7.6\ \text{m/s}$. How much work do friction and air resistance together do on the sled during the descent?

STEP 1/7

Given and find. $m = 32\,\mathrm{kg}$, difference in height $h = 5.0\,\mathrm{m}$, initial speed $v_1 = 0$, final speed $v_2 = 7.6\,\mathrm{m/s}$ and $g = 9.81\,\mathrm{m/s^2}$. We want the work W_{other} done together by friction and air resistance.

STEP 2/7

Principle. The weight, the normal force of the snow, friction and air resistance act on the sled. The normal force is perpendicular to the motion, so it does no work. The work of the weight is taken into account as the change in potential energy. Friction and air resistance are other forces, so we use the work–energy principle. The zero level is chosen at the foot of the hill.

STEP 3/7

Formula solved. At the start $E_{\mathrm{k},1} = 0$ and $E_{\mathrm{p},1} = mgh$, at the end $E_{\mathrm{k},2} = \frac{1}{2}mv_2^2$ and $E_{\mathrm{p},2} = 0$:

$$mgh + W_{\mathrm{other}} = \tfrac{1}{2}mv_2^2$$
$$W_{\mathrm{other}} = \tfrac{1}{2}mv_2^2 - mgh$$

STEP 4/7

Unit conversions. Not needed: all quantities are in SI units, and $1\,\mathrm{kg}\cdot\frac{\mathrm{m}^2}{\mathrm{s}^2} = 1\,\mathrm{J}$.

STEP 5/7

Substitution. First we calculate the kinetic energy at the end and the potential energy at the start, and then their difference:

$$\begin{aligned} &\tfrac{1}{2} m v_2^2 \\ &= \tfrac{1}{2} \cdot 32\,\mathrm{kg} \cdot \left(7.6\,\tfrac{\mathrm{m}}{\mathrm{s}}\right)^2 \\ &= 924.16\,\mathrm{J} \end{aligned}$$

$$\begin{aligned} &mgh \\ &= 32\,\mathrm{kg} \cdot 9.81\,\tfrac{\mathrm{m}}{\mathrm{s}^2} \cdot 5.0\,\mathrm{m} \\ &= 1569.6\,\mathrm{J} \end{aligned}$$

$$\begin{aligned} &W_{\text{other}} \\ &= 924.16\,\mathrm{J} - 1569.6\,\mathrm{J} \\ &= -645.44\,\mathrm{J} \end{aligned}$$

STEP 6/7

Result. Friction and air resistance together do the work $W_{\text{other}} \approx -6.5 \cdot 10^2\,\mathrm{J}$. The result is given with two significant figures like the height $5.0\,\mathrm{m}$ and the speed $7.6\,\mathrm{m/s}$. The work is negative, because friction and air resistance oppose the motion: about $650\,\mathrm{J}$ of mechanical energy turned into internal energy of the snow, the runners and the air.

STEP 7/7

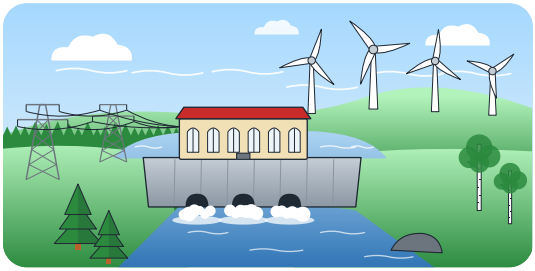
Reasonableness. Without friction and air resistance the speed of the sled at the foot of the hill would be $\sqrt{2gh} \approx 9.9\,\mathrm{m/s}$, so the measured $7.6\,\mathrm{m/s}$ is smaller, as it should be. About $41\,\%$ of the initial potential energy turned into internal energy. If the slope is $24\,\mathrm{m}$ long, the average resistive force is $645\,\mathrm{J} : 24\,\mathrm{m} \approx 27\,\mathrm{N}$, less than a tenth of the weight $314\,\mathrm{N}$, which is believable for a sliding sled.

FY4 · CHAPTER 12

Hydro and wind power as energy converters

Why does the same water of the Vuoksi river give three times as much power at Imatra as at Tainionkoski? And why does the power of a wind turbine grow so much more than the wind speed?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy4-15.html`



IN THIS LESSON, YOU WILL LEARN TO

- explain how hydro and wind power plants convert the potential energy of water and the kinetic energy of wind into electricity.
- calculate the power, efficiency and produced energy of hydro and wind power plants.
- justify why the power of a wind turbine grows with the third power of the wind speed.

PRIOR KNOWLEDGE

- Potential energy is $E_\mathrm{p} = mgh$ and kinetic energy $E_\mathrm{k} = \frac{1}{2}mv^2$
  chapter *Kinetic energy, potential energy and mechanical energy*
- When only gravity does work, potential energy turns into kinetic energy and mechanical energy is conserved
  chapter *Conservation of mechanical energy and the work–energy principle*
- Power tells how fast energy is converted: $P = \frac{\Delta E}{\Delta t}$
  chapter *Power*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 3.

Think about it

The Vuoksi river, flowing out of Lake Saimaa, runs first through the Tainionkoski power plant and a few kilometres downstream through the Imatra power plant. The same water turns the turbines of both, but the power of Imatra is 192 MW and that of Tainionkoski 62 MW. The power of a wind turbine, on the other hand, varies with the wind: in a weak wind it is small, but when the wind strengthens only a little, the power can grow many times over.

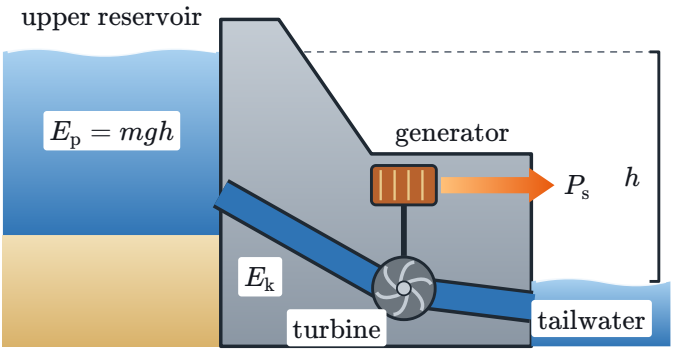
> **What do the powers of a hydro and a wind power plant depend on, and why does the wind speed affect**

Theory

Key idea

A power plant converts energy

Hydro and wind power plants do not produce energy out of nothing. They convert the mechanical energy of water or air into electricity. Water behind a dam has potential energy $E_\mathrm{p} = mgh$ relative to the water below. When the water flows down a pipe, the potential energy turns into kinetic energy, as in the chapter *Conservation of mechanical energy and the work–energy principle*. The fast stream of water pushes on the blades of the turbine and makes it rotate (in many turbines the pressure of the water also does work, but the energy chain is the same), and the shaft of the turbine turns a generator, which converts the energy of rotation into electricity. After the turbine, the water flows slowly on to the tailwater. The difference in height between the upper and the lower water surface is called the **head** h.



Power tells how much energy per second

A power plant is described by its power: how much energy it converts in one second. When one cubic metre of water, that is $1000\ \mathrm{kg}$, falls the $24\ \mathrm{m}$ head of Imatra, it releases potential energy $mgh \approx 235\ \mathrm{kJ}$. If hundreds of cubic metres

flow through the turbines every second, the power is hundreds of megawatts. So power grows for two reasons: when more water falls each second, and when it falls from higher up. The water of the Vuoksi flows first through the Tainionkoski power plant and then through the Imatra power plant, so the flow rate is the same at both. The head at Imatra is about three times as large, and so is its power.

Efficiency

Not all the energy the water loses becomes electricity. The water rubs against the walls of the pipe, the turbine and the generator warm up, and the water leaving into the tailwater still has a little kinetic energy. The **efficiency** of a power plant is the ratio of the electric power it delivers P_{s} to the power taken in P_{o}:

$$\eta = \frac{P_{\mathrm{s}}}{P_{\mathrm{o}}}$$

where η (eta) is the efficiency, given as a decimal number or as a percentage. Efficiency is at most $100\ \%$. In real devices it is usually below $100\ \%$, because part of the energy taken in does not end up in the intended useful form, here electricity. The efficiency of a large hydropower plant is about $90\ \%$.

The kinetic energy of the wind

In a wind turbine the source of energy is the kinetic energy of moving air, $E_{\mathrm{k}} = \frac{1}{2}mv^2$. The wind pushes on the rotor blades and makes the rotor turn, and the air slows down as it passes the rotor. The rotor cannot stop the air completely, because the air must be able to flow away behind the rotor. So even an ideal rotor can capture only about $59\ \%$ of the power that the undisturbed wind carries through the area swept by the rotor, and a real wind turbine captures less than half. Air is light: a cubic metre has a mass of only about $1.25\ \mathrm{kg}$. This is why the rotor of a wind turbine has to be large, so that a lot of air flows through it every second.



The Middelgrunden offshore wind farm off Copenhagen, Denmark, photographed from a boat.
Photo: Kritzolina, CC BY-SA 4.0

Hydro and wind power

Hydropower

The flow rate Q tells how many cubic metres of water flow through the turbines each second. In time t a volume $V = Qt$ flows through the turbines, and its mass is $m = \rho V = \rho Q t$. Its potential energy over the head h is $mgh = \rho Q t g h$, so the power the water brings is $P_\mathrm{o} = \rho Q g h$. By the definition of efficiency, the electric power is η times this:

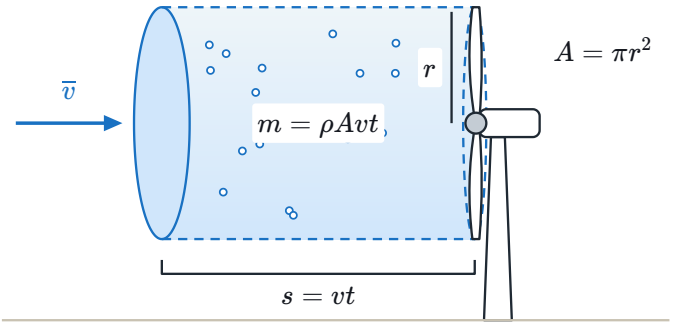
$$P = \eta \rho Q g h$$

where P is the electric power of the plant (W), η the efficiency, ρ the density of water ($1000\ \mathrm{kg/m^3}$), Q the flow rate ($\mathrm{m^3/s}$), g the acceleration due to gravity ($9.81\ \mathrm{m/s^2}$) and h the head (m). The formula assumes that the kinetic energy of the water in the upper and lower water is small, so the energy that is converted is the change in potential energy.

Wind power

Consider first the undisturbed wind, that is, the wind before the rotor has slowed it down. In time t the undisturbed wind of speed v would carry through the area swept by the rotor, $A = \pi r^2$, the air that fills a cylinder of length vt. The mass of this air is $m = \rho A v t$, and its kinetic energy is $\tfrac{1}{2} m v^2 = \tfrac{1}{2} \rho A v^3 t$. So the power flow of the undisturbed wind through the swept area is $P_\mathrm{wind} = \tfrac{1}{2} \rho A v^3$.

In reality the rotor slows the air down already before the blades, so at the rotor the air flows more slowly than v, and its mass flow is not $\rho A v$. Therefore P_{wind} is a reference power: it tells how much energy per second would pass through an area of the same size without the turbine. The efficiency η of the turbine is defined as a fraction of this reference power, $\eta = P : P_{\text{wind}}$.



So the electric power is:

$$P = \frac{1}{2}\eta\rho A v^3$$

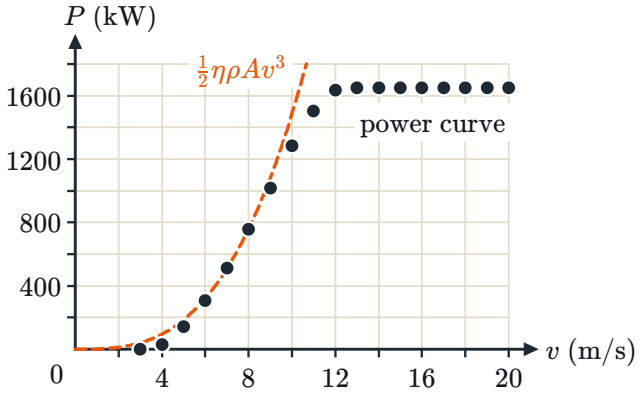
where P is the electric power (W), η the efficiency of the turbine, that is, the electric power as a fraction of the reference power of the undisturbed wind, ρ the density of air (about $1.25\ \mathrm{kg/m^3}$), A the area swept by the rotor ($\mathrm{m^2}$) and v the speed of the undisturbed wind ($\mathrm{m/s}$).

Why is the speed raised to the third power?

When the wind speed doubles, twice as much air passes through the swept area each second, and each kilogram of air has four times as much kinetic energy. So the reference power grows by a factor of $2 \cdot 4 = 2^3 = 8$. The electric power grows in the same ratio only in a model where the efficiency stays roughly constant. In the power curve of a real turbine, the power grows about 8.9 times when the wind speed goes from $5\ \mathrm{m/s}$ to $10\ \mathrm{m/s}$, because the efficiency is slightly better at the higher speed.

A real turbine also has three ranges of wind speed that the cubic model does not describe. In a wind weaker than the cut-in speed, for this turbine about 3–$4\ \mathrm{m/s}$, the turbine produces no electricity. From a wind of about $13\ \mathrm{m/s}$ on, the power

stays at $1650\ \mathrm{kW}$. This is the rated power of the turbine, which the generator cannot exceed, so the blades are turned to let part of the wind pass by. Then the formula with a constant efficiency no longer describes the turbine. In a storm wind stronger than the cut-out speed the turbine is stopped to protect its structure, and its power drops to zero.



The points of the graph are the manufacturer's power curve of the Vestas V82 wind turbine (rotor diameter $82\ \mathrm{m}$, rated power $1650\ \mathrm{kW}$). A manufacturer's power curve is given for standard conditions, usually an air density of $1.225\ \mathrm{kg/m^3}$ and ten-minute average winds at hub height. The dashed model curve is calculated with the density $1.25\ \mathrm{kg/m^3}$, so the comparison is approximate. In too strong a wind the turbine is stopped, and its power drops to zero; the graph does not show this region. Source: Vestas Wind Systems, technical data of the V82 turbine; values from the NREL turbine archive (turbine-models, Vestas V82).

Common misconception

People often think that a power plant produces energy. A power plant converts energy: the electricity of a hydropower plant comes from the potential energy of water and the electricity of a wind turbine from the kinetic energy of the wind, and some of the energy always turns into something other than electricity. A second mistake is to assume that double the wind gives double the power. The power grows with the third power of the speed, until the turbine reaches its rated power. A third mistake is to mix up power and energy: power is given in watts, and energy is obtained by multiplying power by time. A power plant with a power of $3\ \mathrm{MW}$ produces $3\ \mathrm{MWh}$ of energy in an hour.

Summary

- Hydro and wind power plants convert mechanical energy into electricity: a hydropower plant the potential energy of water, a wind turbine the kinetic energy of air.

- The efficiency $\eta = \frac{P_s}{P_o}$ is at most $100\ \%$; for a large hydropower plant it is about $90\ \%$ and for a wind turbine below $50\ \%$.
- The power of a hydropower plant is $P = \eta \rho Q g h$: it is proportional to the flow rate and the head.
- The power of a wind turbine is $P = \frac{1}{2}\eta \rho A v^3$: when the efficiency stays roughly constant, double the wind speed gives about eight times the power, until the turbine reaches its rated power. In a very strong wind the turbine is stopped.
- Energy is power times time, $E = Pt$.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Power of a wind turbine

The rotor of a wind turbine has a diameter of $120\ \mathrm{m}$. The wind speed is $9.0\ \mathrm{m/s}$ and the density of air is $1.25\ \mathrm{kg/m^3}$. The turbine converts into electricity $45\ \%$ of the power that the undisturbed wind carries through the area swept by the rotor. What is the electric power of the turbine?

STEP 1/7

Given and find. The rotor diameter is $d = 120\ \mathrm{m}$, the wind speed $v = 9.0\ \mathrm{m/s}$, the density of air $\rho = 1.25\ \mathrm{kg/m^3}$ and the efficiency $\eta = 45\ \%$. We are asked for the electric power P.

STEP 2/7

Principle. The undisturbed wind carries the power $\frac{1}{2}\rho A v^3$ through the area swept by the rotor. The turbine converts the fraction η of this reference power into electricity, so the electric power comes from the formula for the power of a wind turbine. The area A is the circle swept by the rotor.

STEP 3/7

Formula solved. The formula gives the asked power directly, once the area is calculated from the radius $r = \frac{d}{2}$:

$$P = \tfrac{1}{2}\eta\rho A v^3$$
$$A = \pi r^2$$

STEP 4/7

Unit conversions. The radius is $r = 120\,\mathrm{m} : 2 = 60\,\mathrm{m}$, and the efficiency as a decimal number is $\eta = 0.45$. The other quantities are in SI units, and

$$\begin{aligned}\tfrac{\mathrm{kg}}{\mathrm{m}^3} \cdot \mathrm{m}^2 \cdot \tfrac{\mathrm{m}^3}{\mathrm{s}^3} &= \tfrac{\mathrm{kg}\cdot\mathrm{m}^2}{\mathrm{s}^3} \\ &= \mathrm{W}\end{aligned}$$

STEP 5/7

Substitution. First we calculate the area and then the power:

$$\begin{aligned}A &= \pi r^2 \\ &= \pi \cdot (60\,\mathrm{m})^2 \\ &= 11\,309.7\,\mathrm{m}^2 \\ P &= \tfrac{1}{2}\eta\rho A v^3 \\ &= 0.5 \cdot 0.45 \cdot 1.25\ \tfrac{\mathrm{kg}}{\mathrm{m}^3} \\ &\quad \cdot 11\,309.7\,\mathrm{m}^2 \cdot \left(9.0\ \tfrac{\mathrm{m}}{\mathrm{s}}\right)^3 \\ &= 2\,318\,849\,\mathrm{W}\end{aligned}$$

STEP 6/7

Answer. The electric power is about $2.3\,\mathrm{MW}$, to two significant figures like the wind speed $9.0\,\mathrm{m/s}$ and the efficiency 0.45.

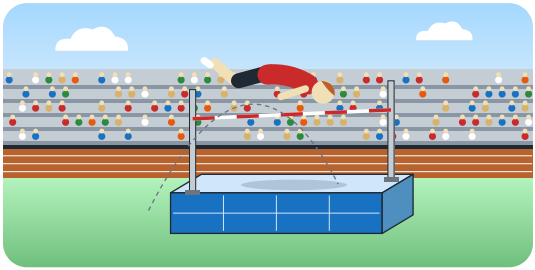
STEP 7/7

Reasonableness. The turbine of the power curve, with a rotor diameter of $82\,\mathrm{m}$, gives a power of $1017\,\mathrm{kW}$ at a wind speed of $9\,\mathrm{m/s}$. The area of this rotor is $(120:82)^2 \approx 2.1$ times as large, so the power should be about $2.1 \cdot 1.0\,\mathrm{MW} \approx 2.2\,\mathrm{MW}$. The result $2.3\,\mathrm{MW}$ is of the same order of magnitude. The reference power of the undisturbed wind is $5.2\,\mathrm{MW}$, and less than half of it becomes electricity, as it should.

Momentum and impulse

Why can a high jumper fall backwards over the bar onto a thick mat without getting hurt, even though a similar fall onto hard ground would be dangerous?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy4-9.html`



IN THIS LESSON, YOU WILL LEARN TO

- calculate momentum, impulse and the change in momentum, with their signs.
- use the impulse–momentum principle and read the impulse from the area of a (t, F) graph.
- explain why a longer stopping time lowers the force of a collision.

PRIOR KNOWLEDGE

- The net force determines the acceleration, $\Sigma \overline{F} = m\overline{a}$
  chapter *Newton's second law*
- Acceleration is the change in velocity divided by the time, $a = \frac{\Delta v}{\Delta t}$
  chapter *Acceleration and uniformly accelerated motion*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 4.

Think about it

A high jumper takes off and arches backwards over a bar more than two metres high. Behind the bar the jumper lands on a thick mat and gets straight up. Without the mat, the same fall onto grass could break bones. The jumper hits both surfaces at the same velocity and stops on both.

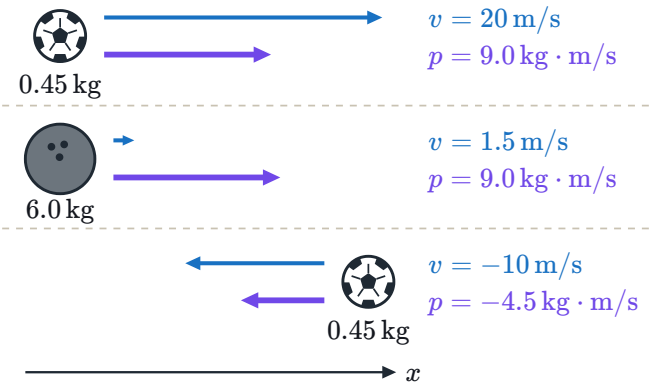
> **What is it about the mat that protects the jumper, when the jumper's velocity changes by the same amount in both cases?**

Theory

Key idea

Momentum: mass and velocity together

In the chapter *Newton's second law* you learned that a force changes the velocity of an object, and that changing the velocity of a heavy object takes a greater force than for a light one. So both the mass and the velocity affect how hard it is to stop or speed up a moving object. Their product is called . A light, fast football can have as much momentum as a heavy, slow bowling ball. Momentum is a vector with the same direction as the velocity. In straight-line motion a positive direction is chosen, and the momentum of an object moving in the opposite direction is negative.

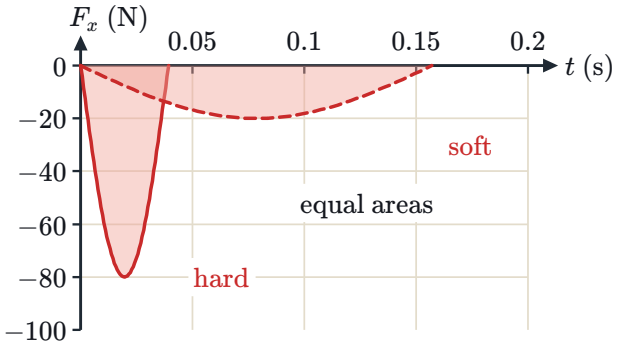


Force and time change momentum

When a net force acts on an object, its velocity changes, and so does its momentum. The size of the change depends on two things: how large the force is and how long it acts. The product of a force and the time it acts is called . A small force for a long time can change the momentum as much as a large force for a short time. A javelin thrower pulls the javelin over a long distance so that the force has time to act for as long as possible and the momentum of the javelin grows large.

The same applies to stopping. When a moving object is stopped, the change in its momentum is fixed: all of the original momentum has to be removed. The stopping force can be large and short or small and long, but the product of force and time is

the same in both cases. When the x-component of the force is drawn as a function of time, the x-component of the impulse is the signed area between the curve and the time axis: an area below the time axis counts as negative. If the x-component of the force is negative, its impulse is negative too, as for the bumpers in the figure below. A hard bumper stops a cart quickly with a large force and a soft bumper slowly with a small force, but the areas are equal.

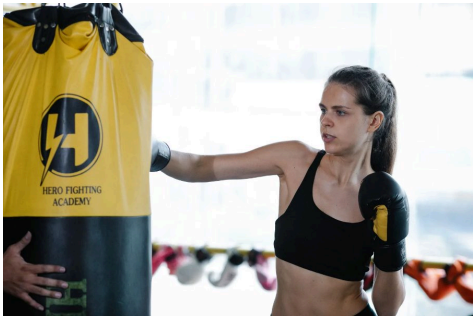


Rebounding changes momentum more than stopping

If an object does not just stop but bounces back, its momentum changes more. When a ball hits a wall and rebounds straight back at the same speed, its momentum changes from mv to $-mv$. The change is $-2mv$, and its magnitude is twice as large as when stopping. So the wall must give a rebounding ball twice the impulse. When calculating a change in momentum the signs are essential, since even a change in the direction of motion alone changes the momentum.

The answer to the opening question

The jumper hits the surface at the same velocity whether it is hard or soft, and stops in both cases. So the change in the jumper's momentum is the same, and so is the impulse of the net force on the jumper. The force of the surface is larger than this net force by the weight, but it too becomes smaller when the stopping time grows. A thick mat is compressed over tens of centimetres, so stopping takes many times longer than on hard ground, where the body stops within a few centimetres. Since impulse is force times time, a longer stopping time means a smaller average net force, and so a smaller average force from the surface too. A smaller force puts less strain on the body, so the risk of injury is lower. This is also why we bend our knees when landing, boxing gloves are padded and a cycling helmet has compressible foam inside.



A boxer punches a heavy punching bag.

Photo: Annushka Ahuja, Pexels License

the impulse–momentum principle

Momentum and impulse

The momentum of an object is

$$\overline{p} = m\overline{v}$$

where m is the mass of the object (kg) and $\overline{v}$ its velocity ($\mathrm{m/s}$). The unit of momentum is $\mathrm{kg \cdot m/s}$. In straight-line motion the vectors are replaced by numbers with signs, $p = mv$, and v is negative when the object moves in the negative direction.

When a constant force $\overline{F}$ acts on an object for a time interval Δt, the impulse of the force is

$$\overline{I} = \overline{F}\,\Delta t$$

where $\overline{F}$ is the force (N) and Δt the time it acts (s). The impulse has the direction of the force, and its unit is $\mathrm{N \cdot s}$. Since $1\,\mathrm{N} = 1\,\mathrm{kg \cdot m/s^2}$, we have $1\,\mathrm{N \cdot s} = 1\,\mathrm{kg \cdot m/s}$: impulse and momentum have the same unit.

The impulse–momentum principle from Newton's second law

When the net force $\Sigma\overline{F}$ is constant, the object accelerates uniformly. In the chapter *Acceleration and uniformly accelerated motion* you learned that then the acceleration is the change in velocity divided by the time, $\overline{a} = \frac{\overline{v}-\overline{v}_0}{\Delta t}$. Substituting this into Newton's second law $\Sigma\overline{F} = m\overline{a}$ and multiplying both sides by the time Δt gives

$$\Sigma\overline{F}\,\Delta t = m\overline{v} - m\overline{v}_0$$

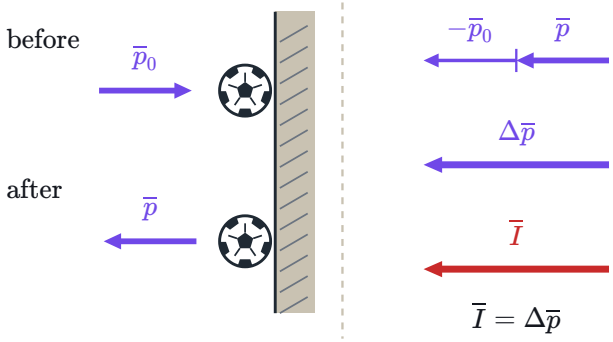
On the left is the impulse of the net force and on the right the change in momentum. This is the :

$$\Delta\overline{p} = \overline{I}$$

where $\Delta\overline{p} = \overline{p} - \overline{p}_0$ is the change in momentum ($\mathrm{kg}\cdot\mathrm{m/s}$) and $\overline{I}$ the impulse of the net force acting on the object ($\mathrm{N}\cdot\mathrm{s}$). In words: the momentum of an object changes by as much as the net force gives it impulse. For example, when a $2.0\,\mathrm{kg}$ cart is pushed in the direction of motion with a net force of $4.0\,\mathrm{N}$ for $0.50\,\mathrm{s}$, the impulse is $2.0\,\mathrm{N}\cdot\mathrm{s}$, and the velocity of the cart increases by $1.0\,\mathrm{m/s}$.

Signs in one dimension

Before calculating, a positive direction is chosen. Velocities, momenta and forces that point in the opposite direction get a minus sign. When a ball rebounds straight back from a wall at the same speed and the positive direction is towards the wall, $p_0 = mv$ and $p = -mv$, so $\Delta p = -mv - mv = -2mv$. The change points away from the wall, in the same direction as the force of the wall on the ball.

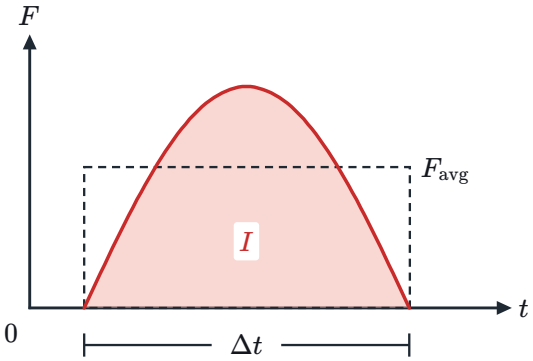


A changing force

In a collision the force is not constant: it grows from zero to its greatest value and falls back. The impulse–momentum principle holds then too, when the impulse is calculated as the signed area between the (t, F) graph of the force and the time axis. The effect of a changing force is often described with an average force, which would give the same impulse in the same time:

$$F_{\text{avg}} = \frac{|\Delta p|}{\Delta t}$$

where F_{avg} is the average force (N), $|\Delta p|$ the magnitude of the change in momentum ($\text{kg} \cdot \text{m/s}$) and Δt the duration of the collision (s). The formula gives the magnitude of the average force. The x-component of the average force is signed, $F_{\text{avg},x} = \frac{\Delta p_x}{\Delta t}$; for example, the component of the force of a stopping bumper is negative when the cart moves in the positive direction. In the graph the average force is the height of the rectangle whose area is the same as the area under the curve. The greatest magnitude of a changing force is always larger than the magnitude of the average force.



Common misconception

Many people think that a soft mat or an airbag lowers the impulse of a collision or "absorbs the momentum". But the change in momentum is the same whether the object stops against a hard or a soft surface, so the impulse of the net force is the same too. A soft surface lengthens the stopping time, and that is why a smaller average net force is enough for the same impulse.

Another mistake is to forget the signs when an object rebounds. If a ball hits a wall at $10\ \text{m/s}$ and rebounds at $10\ \text{m/s}$, the magnitude of its velocity does not change, but its momentum changes from mv to $-mv$. The change is $-2mv$, not zero: its

magnitude is $2mv$. A third mistake is to confuse momentum and kinetic energy: momentum is a vector and proportional to the velocity, kinetic energy is a scalar and proportional to the square of the velocity.

Summary

- Momentum is the product of mass and velocity, $\overline{p} = m\overline{v}$. It is a vector with the direction of the velocity; the unit is $\mathrm{kg} \cdot \mathrm{m/s}$.
- The impulse of a constant force is $\overline{I} = \overline{F}\,\Delta t$. The impulse of a changing force is the signed area between the (t, F) graph and the time axis; the unit is $\mathrm{N} \cdot \mathrm{s} = \mathrm{kg} \cdot \mathrm{m/s}$.
- The impulse–momentum principle: the impulse of the net force equals the change in momentum, $\Delta\overline{p} = \overline{I}$.
- The same change in momentum is obtained with a large force in a short time or a small force in a long time. A soft surface, bent knees and the foam of a helmet lengthen the stopping time and lower the force.
- In one dimension a positive direction is chosen and signs are used. For an object rebounding straight back at the same speed the magnitude of the change in momentum is $2mv$, twice as large as for one that stops.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Racket's impulse on a tennis ball

A tennis ball of mass $58\ \mathrm{g}$ flies horizontally at $25\ \mathrm{m/s}$ towards a racket. The racket hits the ball straight back, and the ball leaves the racket horizontally at $35\ \mathrm{m/s}$. The ball is in contact with the racket for $4.0\ \mathrm{ms}$. How large is the average force with which the racket pushes the ball?

STEP 1/7

Given and find. $m = 58\,\mathrm{g}$, incoming velocity $25\,\mathrm{m/s}$, outgoing velocity $35\,\mathrm{m/s}$ in the opposite direction and contact time $\Delta t = 4.0\,\mathrm{ms}$. We are asked for the average force of the racket F_{avg}.

STEP 2/7

Principle. The momentum of the ball changes because the racket pushes it. The motion is horizontal, and during contact the force of the racket is so large that the effect of the ball's weight is neglected. Then the force of the racket is the net force, and we use the impulse–momentum principle. The positive direction is chosen as the outgoing direction of the ball, so the incoming velocity is $v_0 = -25\,\mathrm{m/s}$ and the outgoing velocity $v = +35\,\mathrm{m/s}$.

STEP 3/7

Formula solved. From the impulse–momentum principle $F_{\mathrm{avg}}\,\Delta t = mv - mv_0$, so

$$F_{\mathrm{avg}} = \frac{m(v - v_0)}{\Delta t}$$

STEP 4/7

Unit conversions. The mass to kilograms and the time to seconds: $58\,\mathrm{g} = 0.058\,\mathrm{kg}$ and $4.0\,\mathrm{ms} = 0.0040\,\mathrm{s}$.

STEP 5/7

Substitution. First we calculate the change in momentum and then divide it by the time:

$$\begin{aligned} &m(v - v_0) \\ &= 0.058\,\mathrm{kg} \cdot (35 + 25)\,\tfrac{\mathrm{m}}{\mathrm{s}} \\ &= 3.48\,\mathrm{kg} \cdot \tfrac{\mathrm{m}}{\mathrm{s}} \end{aligned}$$

$$\begin{aligned} F_{\mathrm{avg}} &= \frac{3.48\,\mathrm{kg} \cdot \mathrm{m/s}}{0.0040\,\mathrm{s}} \\ &= 870\,\mathrm{N} \end{aligned}$$

STEP 6/7

Result. The racket pushes the ball with an average force of $8.7 \cdot 10^{2}\,\mathrm{N}$, to two significant figures like the mass $58\,\mathrm{g}$ and the time $4.0\,\mathrm{ms}$. The force is positive, that is, it points in the outgoing direction of the ball.

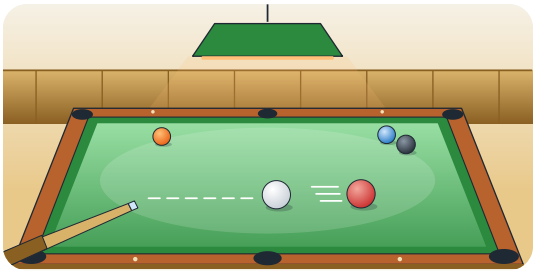
STEP 7/7

Reasonableness. The weight of the ball is $0.058\,\mathrm{kg} \cdot 9.81\,\mathrm{m/s^{2}} \approx 0.57\,\mathrm{N}$, about 1500 times smaller than the force of the racket, so neglecting the weight was justified. If the sign were forgotten and the velocities were simply subtracted, $35\,\mathrm{m/s} - 25\,\mathrm{m/s}$, the force would come out as only $145\,\mathrm{N}$, even though the racket must first stop the ball and then accelerate it in the opposite direction.

Conservation of momentum and one-dimensional collisions

Why can a cue ball stop dead when it hits a stationary ball head-on, while the other ball carries on at almost the same velocity?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy4-10.html`



IN THIS LESSON, YOU WILL LEARN TO

- explain, using Newton's third law, why the total momentum is conserved in a collision and in an explosion.
- calculate velocities in one-dimensional collisions and explosions with the law of conservation of momentum.
- distinguish an elastic collision from an inelastic one by means of kinetic energy.

PRIOR KNOWLEDGE

- The momentum of an object is $\overline{p} = m\overline{v}$, and the impulse of a constant net force changes it: $\Delta\overline{p} = \Sigma\overline{F}\,\Delta t$
  chapter *Momentum and impulse*
- The kinetic energy of an object is $E_\mathrm{k} = \frac{1}{2}mv^2$, and forces such as friction change mechanical energy into internal energy
  chapter *Conservation of mechanical energy and the work–energy principle*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 5.

Think about it

On a billiard table a red ball is at rest. A player hits the white cue ball straight towards it. The balls click, and in an instant the situation has changed: the cue ball has stopped almost where the red ball was, and the red ball speeds ahead at almost the same velocity as the cue ball had a moment ago. If the red ball were a heavy bowling ball, the cue ball would bounce back.

> **What determines how the motion is shared between colliding objects, and why does the cue ball stop exactly when it hits a similar ball?**

Theory

Key idea

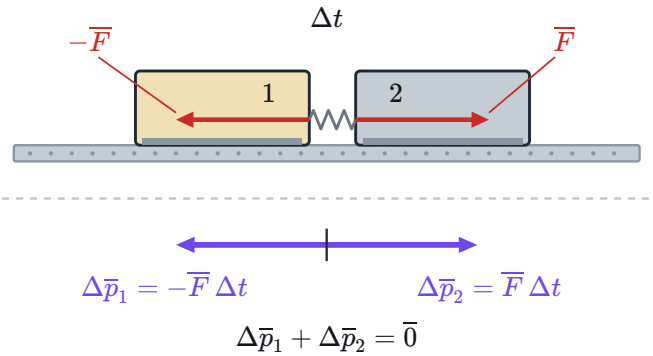
A system and its total momentum

In the chapter *Momentum and impulse* you learned that the momentum of an object is $\overline{p} = m\overline{v}$ and that the impulse of a constant net force changes it: $\Delta\overline{p} = \Sigma\overline{F}\,\Delta t$. In a collision, however, there are two objects that act on each other. So we look at them together as a **system**. The **total momentum** of a system is the vector sum of the momenta of its objects, $\overline{p}_{\text{tot}} = \overline{p}_1 + \overline{p}_2$.

The forces that the objects of the system exert on each other are **internal forces**, for example the forces of the bumpers of two carts on each other in a collision. Forces that come from outside the system are **external forces**, for example the weight of the carts, the normal force of the track and friction.

Why the total momentum is conserved

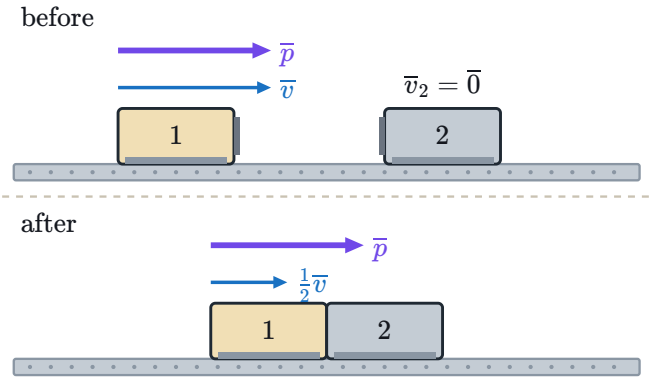
When cart 1 pushes cart 2 with a force $\overline{F}$, by Newton's third law cart 2 pushes cart 1 with an equally large force $-\overline{F}$ in the opposite direction. The forces begin and end at the same moment, so they act for the same time Δt. If the force is constant, the momentum of cart 2 changes by the impulse $\overline{F}\,\Delta t$, and the momentum of cart 1 changes by the same amount in the opposite direction. The momentum that one cart gains is exactly what the other loses. The total momentum does not change. The same holds even if the force changes during the collision, because at every moment the forces are equally large and opposite.



This reasoning works only when external forces do not change the momentum. On a horizontal air track the weight and the normal force cancel each other, and friction is very small. Colliding cars, on the other hand, do feel the friction of the road, but the collision lasts only a fraction of a second. The forces of the bumpers are then much larger than friction, so the impulse of friction during the collision remains negligibly small. This gives the **law of conservation of momentum**: when the sum of the external forces acting on a system is zero, or their impulse during the collision is negligibly small, the total momentum of the system is conserved.

Types of collisions

Momentum is conserved in every collision when the condition of the conservation law is met. Kinetic energy is not always conserved. A collision is **elastic** if the total kinetic energy of the objects after the collision is equal to that before it. Steel balls, billiard balls and carts with spring bumpers collide almost elastically. A collision is **inelastic** if some of the kinetic energy changes into other forms of energy. The objects then change shape, and kinetic energy becomes their internal energy and sound. When the objects stick together and move on together, the collision is **perfectly inelastic**, and the kinetic energy decreases the most: the largest possible part of it is transformed into other forms of energy.



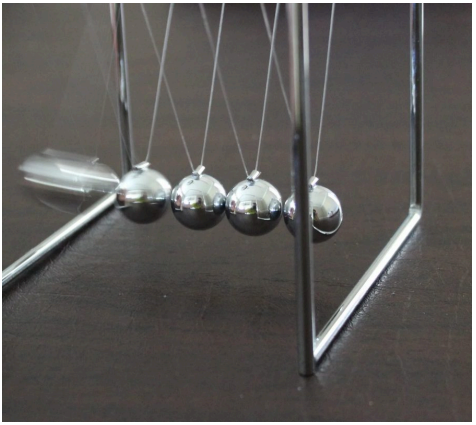
In the figure, carts of equal mass stick together with hook-and-loop tape. The total momentum is the same before and after, but now the same momentum is shared by twice the mass, so the common velocity is half of the original. Kinetic energy $\frac{1}{2}mv^2$ is not conserved: twice the mass and half the velocity give half of the original kinetic energy. The other half becomes internal energy of the tape and the carts.

An explosion

In an **explosion**, a system that is at rest or moving together breaks into parts because of internal forces: a spring is released between carts, skaters push each other away, or gunpowder pushes a bullet and a rifle apart. If the system is at rest at first, its total momentum is zero and stays zero. When a system at rest breaks into two parts, the parts get momenta of equal magnitude in opposite directions, and the lighter part gets the greater velocity. This is why a rifle kicks backwards when the bullet goes forwards. If there are more parts, the vector sum of their momenta is zero, but the momenta of the individual parts can differ in magnitude. Kinetic energy increases in an explosion, because energy of the spring, the muscles or the gunpowder is changed into kinetic energy.

The answer to the opening question

Billiard balls have equal masses, and their collision is almost elastic. When the velocity of the cue ball is v, conservation of momentum says that the sum of the velocities of the balls stays the same: $v_1' + v_2' = v$. Conservation of kinetic energy says that the sum of the squares of the velocities stays the same: $v_1'^2 + v_2'^2 = v^2$. Squaring the first condition and subtracting the second gives $2v_1'v_2' = 0$, so one of the balls is at rest after the collision. The cue ball cannot pass through the other ball, so it stops, and the other ball moves on with velocity v. In reality the balls also spin and the collision is not perfectly elastic, so the cue ball may keep moving a little.



In Newton's cradle, a ball swinging in from the left hits the balls at rest. The moving ball is blurred in the photo.

Photo: The Dean of Physics, CC BY-SA 4.0

Conservation of momentum

The conservation law with vectors and in one dimension

When the sum of the external forces acting on a system is zero, or their impulse during a collision or explosion is negligibly small, the total momentum is the same before and after:

$$\overline{p}_1 + \overline{p}_2 = \overline{p}'_1 + \overline{p}'_2$$

where $\overline{p}_1$ and $\overline{p}_2$ are the momenta of the objects before and $\overline{p}'_1$ and $\overline{p}'_2$ their momenta after ($\mathrm{kg \cdot m/s}$).

In a one-dimensional collision, all the velocities lie on the same line. We choose a positive direction, and then every velocity is a signed number: the velocity of an object moving in the opposite direction is negative. The conservation law becomes

$$
\begin{aligned}
&m_1 v_1 + m_2 v_2 \\
&= m_1 v_1' + m_2 v_2'
\end{aligned}
$$

where m_1 and m_2 are the masses of the objects (kg), v_1 and v_2 their velocities before and v_1' and v_2' their velocities after ($\mathrm{m/s}$).

Perfectly inelastic collision

When the objects stick together, they have a common velocity v' after the collision:

$$
\begin{aligned}
&m_1 v_1 + m_2 v_2 \\
&= (m_1 + m_2)\, v'
\end{aligned}
$$

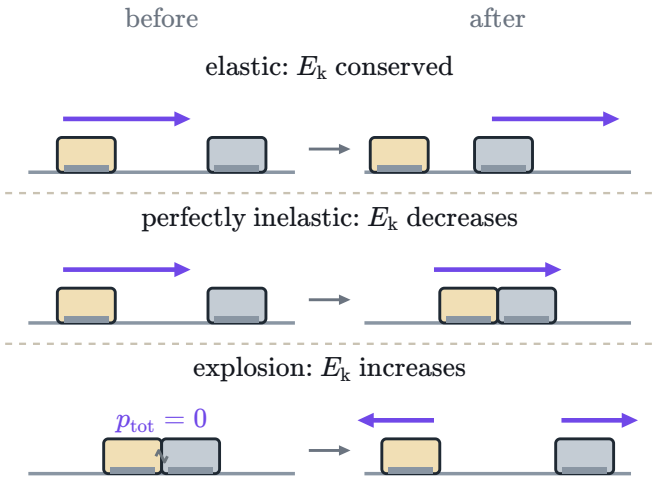
where v' is the common velocity of the objects after the collision ($\mathrm{m/s}$). For example, when a $0.40\ \mathrm{kg}$ cart at a velocity of $0.60\ \mathrm{m/s}$ sticks to a similar cart at rest, the common velocity is $0.30\ \mathrm{m/s}$, and the kinetic energy decreases from $0.072\ \mathrm{J}$ to $0.036\ \mathrm{J}$.

Elastic collision

In an elastic collision, in addition to momentum, the total kinetic energy of the objects is conserved:

$$
\begin{aligned}
&E_{\mathrm{k},1} + E_{\mathrm{k},2} \\
&= E'_{\mathrm{k},1} + E'_{\mathrm{k},2}
\end{aligned}
$$

where $E_{\mathrm{k}} = \tfrac{1}{2} m v^2$ is the kinetic energy of each object before and E'_{k} after (J). When the velocity of one object after the collision is known, the other follows from the law of conservation of momentum, and by comparing the kinetic energies we can see whether the collision was elastic. When an object hits an object of equal mass at rest elastically and head-on, the objects exchange velocities.



When external forces act, the law of conservation of momentum is used only across the collision or explosion itself, a short time from beginning to end. Before and after, external forces such as friction or weight can change the motion of the objects, and then the motion is studied with other laws, for example the work–energy principle.

Common misconception

Many people think that kinetic energy is conserved in a collision in the same way as momentum. Momentum is always conserved when the impulse of the external forces is negligibly small, but kinetic energy is conserved only in an elastic collision. If the velocity of carts that stick together is calculated from conservation of kinetic energy, the result is too large.

Another mistake is to forget the signs. When objects move towards each other, their momenta are in opposite directions, and one of them is negative. A third mistake is to think that momentum disappears when a clay ball dropped on the floor stops. The momentum of the ball passes to the Earth: the ball and the Earth form a system whose total momentum is conserved, but the mass of the Earth is so great that the change in its velocity cannot be noticed.

Summary

- The total momentum of a system is conserved when the sum of the external forces is zero or their impulse during the collision or explosion is negligibly small: $\overline{p}_1 + \overline{p}_2 = \overline{p}'_1 + \overline{p}'_2$.
- Conservation follows from Newton's third law: the internal forces in a collision are equally large and opposite and act for the same time, so the changes in the momenta of the objects cancel each other.

- In one dimension we choose a positive direction, and velocities are signed: $m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2'$.
- In an elastic collision, kinetic energy is also conserved. In an inelastic collision some of the kinetic energy changes into internal energy, and in a perfectly inelastic collision the objects move on together with a common velocity.
- In an explosion, the total momentum of a system at rest stays zero: when it splits into two parts, the parts get momenta of equal magnitude in opposite directions, and kinetic energy increases.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Railway wagons colliding

In a rail yard, freight car A, of mass $32\,000\ \mathrm{kg}$, moves at $1.5\ \mathrm{m/s}$ and catches up with car B, of mass $18\,000\ \mathrm{kg}$, moving in the same direction at $0.50\ \mathrm{m/s}$. The cars couple together and continue together. What is the common velocity of the cars immediately after coupling?

STEP 1/7

Given and find. $m_\mathrm{A} = 32\,000\ \mathrm{kg}$, $v_\mathrm{A} = 1.5\ \mathrm{m/s}$, $m_\mathrm{B} = 18\,000\ \mathrm{kg}$ and $v_\mathrm{B} = 0.50\ \mathrm{m/s}$. We are asked for the common velocity v' immediately after coupling.

STEP 2/7

Principle. We choose both cars as the system and the direction of motion of the cars as the positive direction. Coupling takes a moment, and during it the forces of the couplers are internal forces. The weight and the normal force of the rails cancel each other, and the impulse of rolling resistance is negligibly small during the short coupling, so the total momentum is conserved. The cars stick together, so the collision is perfectly inelastic.

STEP 3/7

Formula solved. The momentum before is equal to the momentum of the combined object after:

$$\begin{aligned} & m_{\mathrm{A}} v_{\mathrm{A}} + m_{\mathrm{B}} v_{\mathrm{B}} \\ & = \left(m_{\mathrm{A}} + m_{\mathrm{B}}\right) v' \end{aligned}$$

$$v' = \frac{m_{\mathrm{A}} v_{\mathrm{A}} + m_{\mathrm{B}} v_{\mathrm{B}}}{m_{\mathrm{A}} + m_{\mathrm{B}}}$$

STEP 4/7

Unit conversions. None are needed: the masses are in kilograms and the velocities in $\mathrm{m/s}$.

STEP 5/7

Substitution. First we calculate the momenta before coupling and then the common velocity:

$$\begin{aligned} & m_{\mathrm{A}} v_{\mathrm{A}} \\ & = 32\,000\,\mathrm{kg} \cdot 1.5\,\tfrac{\mathrm{m}}{\mathrm{s}} \\ & = 48\,000\,\mathrm{kg} \cdot \tfrac{\mathrm{m}}{\mathrm{s}} \end{aligned}$$

$$\begin{aligned} & m_{\mathrm{B}} v_{\mathrm{B}} \\ & = 18\,000\,\mathrm{kg} \cdot 0.50\,\tfrac{\mathrm{m}}{\mathrm{s}} \\ & = 9000\,\mathrm{kg} \cdot \tfrac{\mathrm{m}}{\mathrm{s}} \end{aligned}$$

$$\begin{aligned} & v' \\ & = \frac{48\,000 + 9000}{50\,000} \cdot \frac{\mathrm{kg} \cdot \mathrm{m/s}}{\mathrm{kg}} \\ & = 1.14\,\tfrac{\mathrm{m}}{\mathrm{s}} \end{aligned}$$

STEP 6/7

Result. The common velocity of the cars is $v' \approx 1.1\,\mathrm{m/s}$ in the original direction of motion of the cars. The result is given to two significant figures, like the velocities $1.5\,\mathrm{m/s}$ and $0.50\,\mathrm{m/s}$.

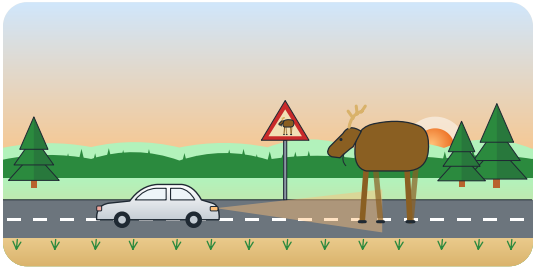
STEP 7/7

Reasonableness. The common velocity lies between the initial velocities of the cars, $0.50\ \mathrm{m/s}$ and $1.5\ \mathrm{m/s}$, and closer to the velocity of the heavier car A, as it should. The kinetic energy decreases from $38.3\ \mathrm{kJ}$ to $32.5\ \mathrm{kJ}$, that is, by about $15\ \%$, which fits a perfectly inelastic collision: kinetic energy changes into internal energy of the couplers and buffers, and into sound.

Road safety: braking, seat belts and airbags

Why is the stopping distance of a car travelling at $100\ \mathrm{km/h}$ almost three times that at $50\ \mathrm{km/h}$, and why does a seat belt protect you when the car cannot stop in time?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy4-11.html`



IN THIS LESSON, YOU WILL LEARN TO

- calculate reaction, braking and stopping distances and explain why braking distance grows with speed squared.
- estimate how speed, reaction time and road conditions affect stopping and the collision speed.
- explain with the impulse–momentum principle why the seat belt and the airbag reduce the force on a person in a collision.

PRIOR KNOWLEDGE

- Displacement is the area between the (t, v) graph and the time axis
  chapter *Acceleration and uniformly accelerated motion*
- Friction opposes sliding, and it is at most the product of the coefficient of friction and the normal force
  chapter *Weight, friction and the inclined plane*
- The change in momentum is the impulse: $\Delta\overline{p} = \overline{F}\,\Delta t$
  chapter *Momentum and impulse*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 6.

Think about it

Driving school teaches that even a small increase in speed lengthens the stopping distance a lot and that the seat belt is fastened even for a short trip. According to the Finnish Crash Data Institute (OTI), in the fatal accidents investigated in 2017–2021, 198 people died in cars and vans without a seat belt. The investigation teams estimated that 114 of them could have survived if the belt had been fastened.

> **Why does the stopping distance grow so steeply with speed, and how can a seat belt decide whether a person survives a collision?**

Theory

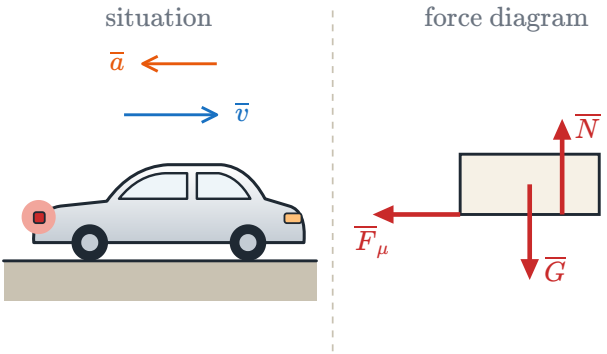
Key idea

Stopping starts with a reaction

When a driver notices danger, the car does not start to slow down at once. The driver must first notice the danger, decide to brake and move a foot to the brake pedal. This takes the **reaction time** t_r, usually about one second. For example, the crash speed calculator of Liikenneturva, the Finnish Road Safety Council, assumes a reaction time of $1\ \mathrm{s}$. Fatigue, alcohol and a glance at the phone make the reaction time longer. During the reaction time the car still moves at its original speed and travels the **reaction distance** s_r. Only then do the brakes begin to slow the car, and it travels the **braking distance** s_b before it stops. The reaction distance and the braking distance together are the **stopping distance** s_stop.

The road stops the car by friction

The brakes slow the rotation of the wheels, but it is the road that slows the car. On a horizontal road the friction between the tyres and the road is the only horizontal force, when air resistance is neglected. Friction points backwards, so by Newton's second law the acceleration also points backwards and the car slows down. In braking, the magnitude of the acceleration is called the **deceleration** a.

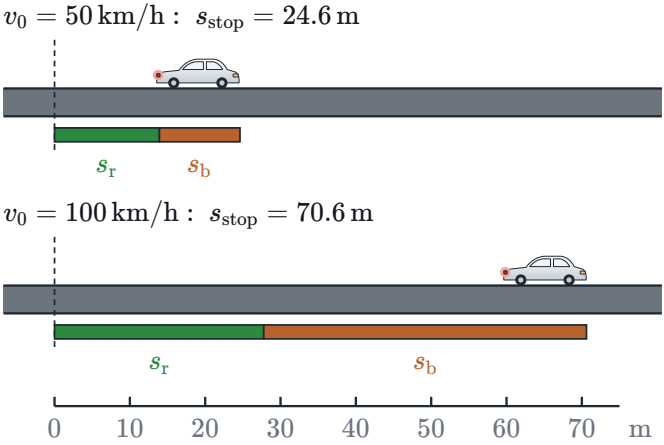


Friction has an upper limit that depends on the tyres and the road surface. On dry asphalt the deceleration can be about $9\ \mathrm{m/s^2}$, and the Liikenneturva calculator uses this value too. On a wet, snowy or icy road the friction and therefore the deceleration are smaller, on ice only a fraction of the value on a dry road. Anti-lock brakes (ABS) stop the wheels from locking. The surface of a rolling tyre does not

slide relative to the road, so static friction acts on the tyre. It can be greater than the kinetic friction of a sliding tyre, and the car stays steerable.

Speed makes the braking distance grow steeply

The reaction distance is directly proportional to the speed: at double the speed the car travels double the distance during the reaction time. The braking distance grows more steeply. With the same deceleration, stopping from double the speed takes twice as long, and for that whole time the car moves twice as fast. The braking distance is therefore four times as long. In the figure the reaction time is $1.0\,\mathrm{s}$ and the deceleration $9.0\,\mathrm{m/s^2}$ at both speeds.

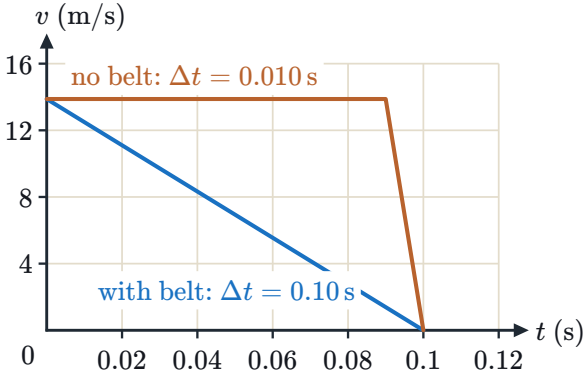


This gives the answer to the first part of the opening question. When the speed doubles from $50\,\mathrm{km/h}$ to $100\,\mathrm{km/h}$, the reaction distance doubles from $13.9\,\mathrm{m}$ to $27.8\,\mathrm{m}$, but the braking distance quadruples from $10.7\,\mathrm{m}$ to $42.9\,\mathrm{m}$. The stopping distance therefore grows from $24.6\,\mathrm{m}$ to $70.6\,\mathrm{m}$, almost three times as long.

Same change in momentum, different collision time

If the obstacle is closer than the stopping distance, the car collides with it. In a frontal collision the front of the car crumples and the car stops in about a tenth of a second. When the passenger stops with the car, their momentum $p = mv$ changes from mv to zero. For a passenger of the same mass the change in momentum is the same with and without a seat belt when they stop from the same initial velocity to the same final velocity: the initial and final velocities determine the change in momentum. In a real accident, however, the motion of a passenger

without a belt does not necessarily end when the car stops, because they may, for example, be thrown through the windscreen. When the change in momentum is the same, the only thing that can be changed is how long the **collision time** Δt is, that is, the time in which the passenger stops. In the chapter *Momentum and impulse* you learned that the change in momentum equals the impulse $\overline{I} = \overline{F}\,\Delta t$. The longer the collision time, the smaller the force needed for the same change in momentum.



Without a seat belt the passenger keeps moving at almost the original speed while the car slows down around them: by Newton's first law no force is yet slowing them down. They hit the steering wheel, the dashboard or the windscreen and stop in hundredths of a second. The **seat belt** fastens the passenger to the car, so they slow down together with the crumpling front of the car. The belt also stretches a little and spreads the force over the strong bones of the pelvis, chest and shoulder.



A 2014 Chevrolet Spark after a side impact test by the US National Highway Traffic Safety Administration (NHTSA). The side of the car has been pushed in.

Photo: Medical College of Wisconsin / NHTSA, public domain (PD-USGov-DOT)

The airbag inflates in hundredths of a second and catches the head and upper body. It lengthens the stopping time and spreads the force over a large area, so the pressure on the body is smaller. An airbag is designed to work together with the seat belt, not instead of it.

This is the answer to the second part of the opening question. The difference also shows in accident statistics. According to the Finnish Crash Data Institute (OTI), in the fatal accidents investigated by the road accident investigation teams in 2017–2021, 278 people wearing a seat belt and 198 people not wearing one died in cars and vans. The investigation teams estimated that 114 of those without a belt could have survived if the belt had been fastened. In Liikenneturva's monitoring, almost everyone in the front seat wears a belt.

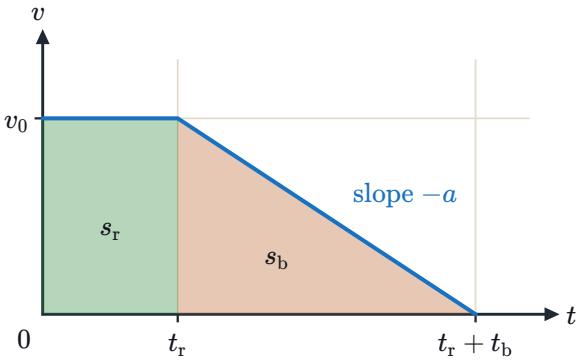
Stopping distance and the force of a collision

Reaction distance and braking distance

During the reaction time the car moves at a constant speed v_0, so the reaction distance is

$$s_\mathrm{r} = v_0 \, t_\mathrm{r}$$

where s_r is the reaction distance (m), v_0 is the speed of the car before braking ($\mathrm{m/s}$) and t_r is the reaction time (s). In braking the deceleration is assumed to be constant, so on the (t, v) graph the speed falls in a straight line to zero. As in the chapter *Acceleration and uniformly accelerated motion*, the distance travelled is the area between the graph and the time axis.



The speed falls by v_0 at deceleration a, so the braking time is

$$t_\mathrm{b} = \frac{v_0}{a}$$

where t_b is the braking time (s) and a is the deceleration, that is, the magnitude of the acceleration ($\mathrm{m/s^2}$). The braking distance is the area of the triangle, $\frac{1}{2}\, v_0\, t_\mathrm{b}$. Substituting the braking time gives

$$s_\mathrm{b} = \frac{v_0^2}{2a}$$

where s_b is the braking distance (m). The stopping distance is the sum of these:

$$s_\mathrm{stop} = v_0\, t_\mathrm{r} + \frac{v_0^2}{2a}$$

where s_stop is the stopping distance (m). The model assumes that the road is horizontal, the deceleration is constant during braking and air resistance is neglected. The speed is substituted in $\mathrm{m/s}$.

Deceleration from friction

On a horizontal road the car does not accelerate vertically, so the normal force equals the weight, $N = mg$. When braking as hard as the grip allows, the friction is $F_\mu = \mu N$, where μ is the coefficient of friction between the tyres and the road. By Newton's second law $\mu mg = ma$, and the mass cancels:

$$a = \mu g$$

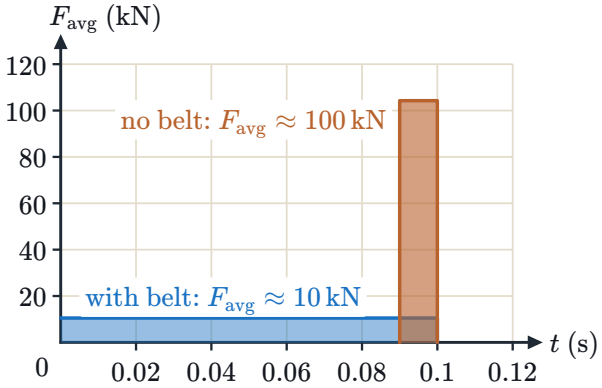
where μ is the coefficient of friction (no unit) and $g = 9.81\ \mathrm{m/s^2}$. The $9.0\ \mathrm{m/s^2}$ of dry asphalt corresponds to a coefficient of friction of 0.92. In this model the braking distance does not depend on the mass of the car. The braking distance of a heavy lorry is still often longer, because its brakes and tyres cannot always use all of the grip.

Average force in a collision

When the passenger's speed changes by Δv in a collision in time Δt, the impulse–momentum principle $\overline{F}_{\text{avg}}\,\Delta t = \Delta \bar{p}$ gives the magnitude of the average force acting on them:

$$F_{\text{avg}} = \frac{m\,\Delta v}{\Delta t}$$

where F_{avg} is the average force (N), m is the passenger's mass (kg), Δv is the magnitude of the change in speed ($\mathrm{m/s}$) and Δt is the collision time (s). When a $75\ \mathrm{kg}$ passenger stops from $50\ \mathrm{km/h}$, the change in momentum is about $1040\ \mathrm{kg\cdot m/s}$. With a belt it is spread over about $0.10\ \mathrm{s}$, and the average force is about $10\ \mathrm{kN}$. Without a belt the time is about $0.010\ \mathrm{s}$ and the force is ten times as large.



The areas of the rectangles in the graph are equal, because each is the same change in momentum $m\,\Delta v$. In reality the force varies during the collision, and the greatest force is larger than the average.

Common misconception

Many people think that double the speed means double the stopping distance. The reaction distance doubles, but the braking distance quadruples, because it depends on the square of the speed. Another mistake is to think that a seat belt reduces the change in momentum. When the passenger stops from the same speed, the momentum changes from mv to zero in both cases; the belt lengthens the time in which the change happens and therefore reduces the force. A third mistake is to believe that in a slow crash you can brace yourself with your hands. Even in a $50\ \text{km/h}$ collision the average force is about $10\ \text{kN}$ even with a belt, more than ten times a person's weight, and the muscle force of the hands is not enough against it.

Summary

- The stopping distance is the reaction distance $s_{\text{r}} = v_0\,t_{\text{r}}$ plus the braking distance $s_{\text{b}} = \frac{v_0^2}{2a}$, when the deceleration is constant.
- The reaction distance is proportional to the speed and the braking distance to the square of the speed: double the speed gives four times the braking distance.
- In braking, the friction between the tyres and the road slows the car. On a horizontal road $a = \mu g$, so a slippery road multiplies the braking distance.
- When the passenger stops from the same initial velocity to the same final velocity, their change in momentum is the same with and without a belt. The seat belt and the airbag lengthen the collision time, so the average force $F_{\text{avg}} = \frac{m\,\Delta v}{\Delta t}$ is smaller.

- The airbag spreads the force over a large area and works together with the seat belt, not instead of it.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: Braking and reaction time

A car is driving on dry asphalt at $80\,\mathrm{km/h}$. The driver glances at their phone, and their reaction time in the dangerous situation is $2.0\,\mathrm{s}$. The deceleration of the car in braking is $9.0\,\mathrm{m/s^2}$. How long is the stopping distance of the car?

STEP 1/7

Given and find. $v_0 = 80\,\mathrm{km/h}$, $t_\mathrm{r} = 2.0\,\mathrm{s}$ and $a = 9.0\,\mathrm{m/s^2}$. We are asked for the stopping distance s_stop.

STEP 2/7

Principle. During the reaction time the car moves at a constant speed, so the reaction distance is speed times time. After that the car slows down at a constant deceleration, when the road is horizontal and air resistance is neglected, so the braking distance is the area of the triangle on the (t, v) graph. The stopping distance is the sum of these.

STEP 3/7

Formula solved.

$$s_\mathrm{stop} = s_\mathrm{r} + s_\mathrm{b}$$

$$s_\mathrm{stop} = v_0\, t_\mathrm{r} + \frac{v_0^2}{2a}$$

STEP 4/7

Unit conversions. The speed is converted to $\mathrm{m/s}$:

$$\begin{aligned} 80\,\mathrm{km/h} &= \tfrac{80 \cdot 1000\,\mathrm{m}}{3600\,\mathrm{s}} \\ &\approx 22.22\,\mathrm{m/s} \end{aligned}$$

STEP 5/7

Substitution. First we calculate each partial distance and then the sum:

$$\begin{aligned} s_\mathrm{r} &= v_0\, t_\mathrm{r} \\ &= 22.22\,\tfrac{\mathrm{m}}{\mathrm{s}} \cdot 2.0\,\mathrm{s} \\ &= 44.44\,\mathrm{m} \\ s_\mathrm{b} &= \frac{(22.22\,\mathrm{m/s})^2}{2 \cdot 9.0\,\mathrm{m/s^2}} \\ &= 27.43\,\mathrm{m} \\ s_\mathrm{stop} &= (44.44 + 27.43)\,\mathrm{m} \\ &\approx 72\,\mathrm{m} \end{aligned}$$

STEP 6/7

Result. The stopping distance is about $72\,\mathrm{m}$, to two significant figures like the reaction time $2.0\,\mathrm{s}$ and the deceleration $9.0\,\mathrm{m/s^2}$. The reaction distance of $44\,\mathrm{m}$ is more than half of it.

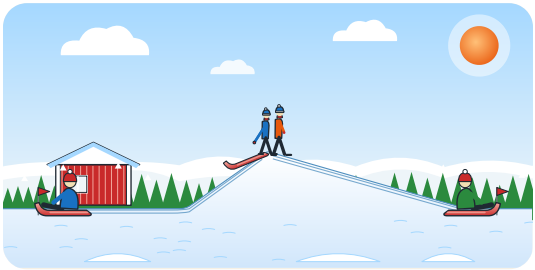
STEP 7/7

Reasonableness. With a reaction time of $1.0\,\mathrm{s}$ the reaction distance would be $22\,\mathrm{m}$ and the stopping distance about $50\,\mathrm{m}$. A glance at the phone therefore lengthens the stopping distance by about $22\,\mathrm{m}$, about five car lengths, even though the braking distance is the same. A speed of $22\,\mathrm{m/s}$ and a braking time of $22\,\mathrm{m/s} : 9\,\mathrm{m/s^2} \approx 2.5\,\mathrm{s}$ give a braking distance of about $\tfrac{1}{2} \cdot 22\,\mathrm{m/s} \cdot 2.5\,\mathrm{s} \approx 27\,\mathrm{m}$, as in the calculation.

Applied lesson: lift, parachute and sledding hill

Why can two sleds stop equally far from the top of a hill, although one goes down a steep slope and the other a gentle one?

Simulation, tasks and model solutions online:
`ruukkilab.fi/en/fy4-16.html`



IN THIS LESSON, YOU WILL LEARN TO

- choose, based on the free-body diagram, whether to solve a situation with the equation of motion or the work–energy principle.
- draw a free-body diagram and calculate the normal force or the acceleration in an accelerating lift.
- calculate the speed and sliding distance of a sled and the work done by air resistance with the work–energy principle.

PRIOR KNOWLEDGE

- The net force points in the direction of the acceleration, and $\Sigma \overline{F} = m\overline{a}$
  chapter *Newton's second law*
- On a slope the normal force is $mg\cos\alpha$ and the kinetic friction is μN
  chapter *Weight, friction and the inclined plane*
- The change in mechanical energy is the work done by forces other than the weight
  chapter *Conservation of mechanical energy and the work–energy principle*

Core path: follow the steps in order. Besides calculations, the core path includes investigations, explanations and graphs. Tasks marked *Optional* give extra practice. Optional tasks: 2.

Think about it

A physics group takes a bathroom scale into the lift of a shopping centre. When the lift starts up, the reading rises for a moment, and when the lift stops, it falls. Through the window you can see a skydiver who descends under a parachute at a steady speed, although the weight pulls the skydiver down all the time. On a sledding hill two children go down different sides of the same hill: one a steep, short slope, the other a gentle, long slope. The one who came down the steep side slides much farther on the flat, but

when the stopping points are measured, both are equally far from the top of the hill.

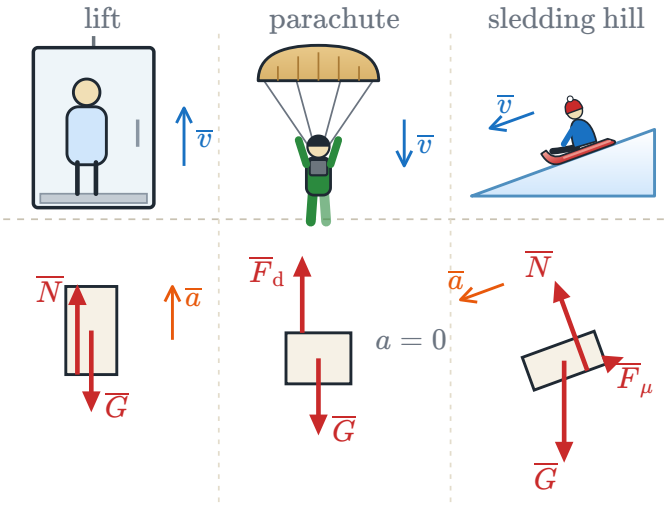
> **Familiar forces act in all three situations. When should you solve a situation with forces and when with energy, and why do the sleds stop equally far from the top?**

Theory

Key idea

First the free-body diagram, then the tool

The problems of this chapter are solved with two tools, and both start from the same place. The module has a third one too: for collisions and short impacts, the impulse principle and the conservation of momentum are used (chapter *Conservation of momentum and one-dimensional collisions*). First you choose the body and draw its free-body diagram, as in the chapter *Free-body diagrams and the combined effect of forces*. The diagram shows which forces act and in which directions. Then you choose the tool according to what is asked. When the question is about a force or an acceleration at some moment, you write the equation of motion (chapter *Newton's second law*). When the question is about the speed at some point or the change in energy over some distance, you use the work–energy principle (chapter *Conservation of mechanical energy and the work–energy principle*). Even then you need the free-body diagram, because it shows which forces do work.



Lift: a scale measures the normal force

A bathroom scale does not measure the weight but the normal force N with which it pushes the person standing on it upwards. The reading is shown in kilograms, because the scale divides the force by $9.81\ \mathrm{m/s^2}$. Only the weight and the normal force act on a person standing still in a lift. When the positive direction is up, the equation of motion is $N - mg = ma$, so $N = m(g + a)$. When the lift starts up and accelerates at $1.0\ \mathrm{m/s^2}$, the normal force on a $60.0\ \mathrm{kg}$ person is $649\ \mathrm{N}$, rounded $650\ \mathrm{N}$ as in the graph, and the scale shows $66\ \mathrm{kg}$. At a steady speed $a = 0$, and the scale shows the same as at home, whether the lift moves up or down. So the reading depends on the acceleration, not on the speed.

Parachute: at terminal velocity the mechanical energy decreases

A skydiver descending under a parachute soon falls at terminal velocity, that is, at a steady speed (chapter *Air resistance and falling through air*). By the equation of motion the acceleration is zero, so the air resistance is equal to the weight, $F_\mathrm{d} = mg$. In terms of energy, the kinetic energy does not change, but the potential energy decreases all the time. By the work–energy principle the change in mechanical energy is the work done by air resistance. It is negative, because air resistance points against the motion: over a height h, $W_\mathrm{d} = -F_\mathrm{d}h = -mgh$. When an $80\ \mathrm{kg}$ skydiver descends $100\ \mathrm{m}$, about $78\ \mathrm{kJ}$ of mechanical energy is

changed into motion of the air and finally into internal energy. Each tool tells the same thing in its own way.

Sledding hill: friction takes energy all the way

A sled slides down a straight slope. The free-body diagram shows that the component of the weight perpendicular to the slope and the normal force cancel each other, so $N = mg\cos\alpha$, as in the chapter *Weight, friction and the inclined plane*. The kinetic friction is then $F_\mu = \mu mg\cos\alpha$. The normal force does no work, because it is perpendicular to the motion. On the slope the weight does work mgh, and friction does negative work $-F_\mu L$, when the length of the slope is L. On a hill of the same height, the normal force and the friction are smaller on a steep slope and the slope is shorter, so friction takes less energy and the sled is faster at the foot. On the flat the sled slides until friction has turned all the kinetic energy into internal energy.

Equation of motion and work–energy principle

The first tool of an applied problem is Newton's second law:

$$\Sigma\overline{F} = m\overline{a}$$

where $\Sigma\overline{F}$ is the sum of the forces acting on the body, that is, the net force (N), m is the mass of the body (kg) and $\overline{a}$ its acceleration ($\mathrm{m/s^2}$). The law is written from the free-body diagram in components, $\Sigma F_x = ma_x$ and $\Sigma F_y = ma_y$, once the axes and the positive directions have been chosen. The second tool is the work–energy principle of mechanics:

$$E_{\mathrm{k},1} + E_{\mathrm{p},1} + W_{\mathrm{other}} = E_{\mathrm{k},2} + E_{\mathrm{p},2}$$

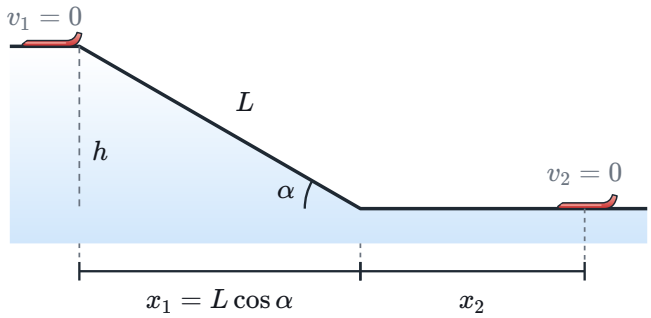
where the index 1 means the initial situation and 2 the final situation, $E_\mathrm{k} = \tfrac{1}{2}mv^2$ is the kinetic energy, $E_\mathrm{p} = mgh$ the potential energy from the chosen zero level and W_{other} the work done by forces other than the weight (J). The work–energy principle needs neither the acceleration nor the time, so it fits when the question is about the speed at a given point.

The work done by friction depends on the horizontal distance

On a straight slope with an angle of inclination α and a length L, the kinetic friction is $\mu mg\cos\alpha$ and the work it does is $-\mu mg\cos\alpha \cdot L$. The figure shows that $L\cos\alpha$ is the horizontal length x_1 of the slope. On the flat $N = mg$, and the work done by friction over a distance x_2 is $-\mu mgx_2$. In both cases the work done by kinetic friction is

$$W_\mu = -\mu mgx$$

where μ is the coefficient of kinetic friction, m the mass (kg), g the acceleration of free fall ($\mathrm{m/s^2}$) and x the horizontal length of the distance travelled (m). The result requires that the surface is straight, that the coefficient of friction is the same all the way and that only the weight and the normal force act perpendicular to the surface, so the sled is not, for example, pulled with a rope.



Why the sleds stop equally far away

A sled starts from rest at a height h, goes down the slope and stops on the flat. When the zero level is chosen at the flat, the work–energy principle gives $mgh - \mu mg(x_1 + x_2) = 0$. The mass cancels, and the whole horizontal distance of the sled from the top to the stopping point is

$$x_1 + x_2 = \frac{h}{\mu}$$

The angle of inclination does not appear in the result. A sled coming down the steep side travels a short horizontal distance on the slope, so it has more distance

left on the flat. A sled coming down the gentle side uses a larger part of the horizontal distance on the slope. This answers the question at the start. The model assumes that the coefficient of friction is the same all the way, that air resistance is small and that the bend at the foot of the slope is short. In a sharp bend the normal force and the friction grow for a moment, so a real sled stops a little earlier below a steep hill. In addition, the slope must be steep enough for the sled to slide: $\tan\alpha > \mu$.

Steps of the solution

1. Choose the body and draw its free-body diagram: for every force its cause and direction.
2. Decide the tool: the force or acceleration at a moment from the equation of motion, the speed at a given point from the work–energy principle.
3. For the equation of motion choose the axes and the positive directions. For the work–energy principle choose the zero level and write the work of the other forces with their signs: a force opposing the motion does negative work, and a force perpendicular to the motion does no work.
4. Solve the asked quantity in symbols, substitute only then, and check the result with a limiting case, for example without friction.

Common misconception

Many think that a scale in a lift shows more because the lift moves upwards. The reading depends on the acceleration, not on the speed: in a lift moving steadily up or down the scale shows the same as at home, and in a lift moving up that slows down to stop it shows less.

Another mistake is to think that at terminal velocity the mechanical energy is conserved, because the speed does not change. The kinetic energy stays the same, but the potential energy decreases, and the negative work done by air resistance changes it into other forms of energy. A third mistake is to write the friction force on a slope as μmg. On a slope the normal force is $mg\cos\alpha$, so the friction force is $\mu mg\cos\alpha$, and the work done by friction is $-\mu mg$ times the horizontal distance, not times the length of the slope.

Summary

- A mechanics problem always starts with choosing the body and drawing its free-body diagram. The force or acceleration at a moment comes from the equation of motion $\Sigma\overline{F} = m\overline{a}$, the speed at a given point from the work–energy principle.

- In a lift a scale measures the normal force $N = m(g + a)$, when the positive direction is up. The reading depends on the acceleration, not on the speed.
- At terminal velocity $F_{\mathrm{d}} = mg$, and the work done by air resistance is $-mgh$: the mechanical energy decreases although the speed does not change.
- On a straight slope $N = mg\cos\alpha$, and the work done by kinetic friction is $W_{\mu} = -\mu mgx$, where x is the horizontal distance. A sled that starts from rest at a height h stops at a horizontal distance h/μ from the top, when the coefficient of friction is the same all the way.
- When a body slides along a stationary surface, the normal force does no work, because it is perpendicular to the motion. In a lift the normal force of the floor is in the direction of the motion and does work.

Examples

Once you have studied the chapter's theory, move on to the guided examples. In the first you follow a worked solution step by step; in the next ones you make the choices yourself, and the hints help if you get stuck.

Example 1: A sled on a slope

A child and a sled, together $32.0\,\mathrm{kg}$, start from rest at the top of a straight slope. The length of the slope is $25.0\,\mathrm{m}$, the angle of inclination 20.0° and the coefficient of kinetic friction between the sled and the snow 0.200. How large is the speed of the sled at the foot of the slope? Air resistance is ignored.

STEP 1/8

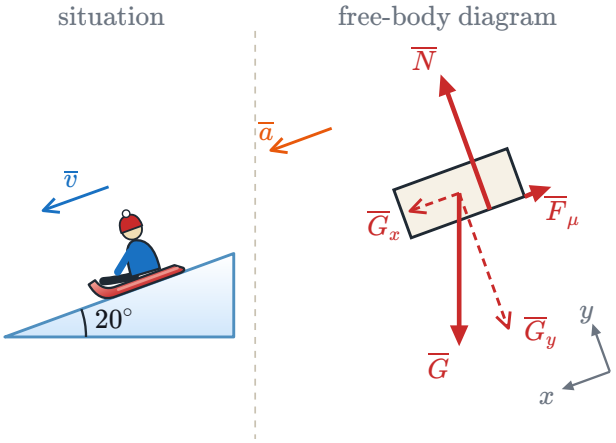
Given and asked. $m = 32.0\,\mathrm{kg}$, $L = 25.0\,\mathrm{m}$, $\alpha = 20.0^{\circ}$, $\mu = 0.200$, initial speed $v_1 = 0$ and $g = 9.81\,\mathrm{m/s^2}$. We want the speed v_2 at the foot of the slope.

STEP 2/8

Principle. The question is about the speed at a given point, so we use the work–energy principle. We choose the sled and the child together as the body. The weight does work, which the work–energy principle takes into account as a change in potential energy. The normal force does no work, because it is perpendicular to the motion. Kinetic friction points against the motion and does negative work $W_{\text{other}} = -F_\mu L$. The friction force comes from the free-body diagram.

STEP 3/8

Free-body diagram. The sled is drawn as a rectangle along the slope. The forces on it are from the Earth (the weight $\overline{G}$), from the snow (the normal force $\overline{N}$) and the kinetic friction $\overline{F}_\mu$ of the snow, which points up the slope, because the sled slides down. The axes are chosen so that x points down the slope and y perpendicularly out of the slope. In the y direction the sled does not accelerate, so $N - mg\cos\alpha = 0$ and $F_\mu = \mu N = \mu mg\cos\alpha$.



STEP 4/8

Formula solved. The zero level is chosen at the foot of the slope, so at the start $E_{\mathrm{p},1} = mgh$, where $h = L\sin\alpha$, and at the end $E_{\mathrm{p},2} = 0$. The speed is solved from the work–energy principle:

$$\begin{aligned} mgL\sin\alpha - \mu mg\cos\alpha \cdot L &= \tfrac{1}{2}mv_2^2 \\ v_2 &= \sqrt{2gL(\sin\alpha - \mu\cos\alpha)} \end{aligned}$$

STEP 5/8

Unit conversions. Not needed: the quantities are in SI units, the coefficient of friction has no unit and the angle is in degrees with the calculator in degree mode.

STEP 6/8

Substitution. First we calculate the expression in brackets and then the speed:

$$\begin{aligned} &\sin 20.0^\circ - 0.200 \cdot \cos 20.0^\circ \\ &\approx 0.3420 - 0.1879 = 0.1541 \\ v_2 &= \sqrt{2 \cdot 9.81 \cdot 25.0 \cdot 0.1541}\ \tfrac{\mathrm{m}}{\mathrm{s}} \\ &\approx 8.694\,\mathrm{m/s} \end{aligned}$$

STEP 7/8

Result. The speed of the sled at the foot of the slope is $8.69\,\mathrm{m/s}$, or about $31\,\mathrm{km/h}$, to three significant figures like the length of the slope $25.0\,\mathrm{m}$, the angle 20.0° and the coefficient of friction 0.200.

STEP 8/8

Reasonableness. Without friction the speed would be

$$\sqrt{2gh} = \sqrt{2 \cdot 9.81 \cdot 8.55}\,\tfrac{\mathrm{m}}{\mathrm{s}}$$
$$\approx 13.0\,\mathrm{m/s}$$

so friction reduces it, as it should. The same result comes from the equation of motion: in the x direction $mg\sin\alpha - \mu mg\cos\alpha = ma$, so $a \approx 1.51\,\mathrm{m/s^2}$, and from the result $aL = \tfrac{1}{2}v_2^2$ of the chapter *Conservation of mechanical energy and the work–energy principle* we get $v_2^2 \approx 75.6\,\mathrm{m^2/s^2}$ as above. The mass cancelled, so a heavier and a lighter sled are equally fast at the foot, when the coefficient of friction is the same.

Theory and worked examples from Ruukkilabra

ruukkilab.fi

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